Quantum Physics

At Any Cost

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Продкам маім
Якіх не абдыму,
Бо іх больш не мае,
И я ўжо іншы.

To the Great and Beautiful Nation
Which Gave Me Everything and More:
A Shelter, Opportunities and Inspiration,
And The Desire To Explore.

Contents

Personal Message

Dear Reader, thank you for picking up this book. Although we are not acquainted, I know a couple things about *You*. First, you have a beautiful name, and second, you are curious about science and quantum physics. This is awesome!

Today is a great time for learning quantum physics. On the one hand, the abundance of educational resources – both in printed and video format – is impressive and, at times, confusing. On the other hand, time is a valuable resource and must be used wisely. I kept these thoughts in mind, when writing this book.

Respecting your time and effort, I am asking you the following: If at any point you find the book too challenging, unintersting, or not helpful – *please stop and put it away!* But *do not give up* on learning quantum physics! Try other books.

There are so many great books and articles on quantum physics that simply listing them could fill the rest of the pages. Here is a short list of books on quantum physics that, in my opinion, would be great substitutes for mine:

1. *Quantum Mechanics For Beginners* by Suhail Zubairy.
2. *Quantum Mechanics: An Experimentalist's Approach* by Eugene Commins.

Personal Quirks

In this book you will encounter several examples of notation that might appear unusual. This is done on purpose. One example is when instead of the monstrous sigma (Σ) notation for the summation:

$$S = \sum_{i=1}^{i=N} a_i$$

we will use an archaic capital 'S'='$\int$':

$$S = \int_i a_i \quad i = \overline{1, N}.$$

Another example is the use of the modern notation for derivatives: $\partial_x f$. There is a couple more instances of such notational quirks. Please try being open-minded, flexible, and patient with me.

Acknowledgment

My sincere gratitude goes to all reviewers of the early drafts of the book for their valuable feedback. In particular, I want to thank Dr. Alex Rylyakov, Dr. Mikhail Makouski, Prof. Anton Kananovich, and Dr. Mohammad Teimourpour. To Dr. Teimourpour I must give separate thanks for numerous discussions, helpful suggestions on material presentation, and hospitality.

My friends, discussing the book with you was both illuminating and fun.

Finally, special acknowledgment must be given to my son, Daniel, for his help with fixing colors in many figures.

Yury Deshko
Weehawken, New Jersey
2024

Preface

This book is the result of lectures delivered to curious, motivated, and studious high schoolers. The lectures ran during the years 2019-2024 in various formats, but mostly in class during a three week summer school organized by Columbia University Pre-College Programs. Additionally, the same lectures were taught remotely to selected students of Ukrainian Physics and Mathematics Lyceum.

The material has been designed to be accessible to people with solid background in high-school algebra and physics (mostly mechanics). Several years of teaching to a relatively diverse set of students proved that nearly all material can be efficiently absorbed by most, provided diligent work is done on exercises and problem. The last fact confirms a well-known truism: *No real learning occurs without practice.*

Exercises are essential part of this book. They are carefully selected to help readers get better understanding of the material and they are also fully solved. The difficulty of the exercises varies from simple to quite challenging.

This book *is not a standard textbook.* It differs from many excellent introductions into Quantum Physics in that it lacks the breadth and rigor of the latter. However, this book serves a special purpose: It tries to act as the *bridge* between elementary and popular books and the more challenging college-level textbooks.

If a picture is worth a thousand words, then a formula is worth at least a hundred words. This book contains both pictures and formulas aplenty. Hopefully, the readers for whom this book is intended will enjoy both.

Some sections are marked with an asterisk, for example **Transposition***. Those sections contain material that is either optional or a bit more advanced that usual. These sections can be skipped without significant impact on the main message of the book.

The choice of the material, the emphasis in explanations, and philosophical preferences reveal author's clear bias towards empirical and operational viewpoint. While trying to be fair and open-minded, it is impossible to remain unaffected by the decades of working in laboratories trying to see formulas "in real life" and match concepts with actual numbers.

> ***Asher Peres On Quantum Phenomena***
> The approach is pragmatic and strictly instrumentalist. This attitude will undoubtedly antagonize some readers, but it has its own logic: quantum phenomena do not occur in a Hilbert space, they occur in a laboratory.
> Asher Peres, *Quantum Theory: Concepts and Methods*.

At Any Cost

The subtitle of this book has been inspired by the letter from Max Karl Ernst Ludwig Planck to an American physicist Robert Williams Wood. Describing his desperate attempts to explain the experimental results on the electromagnetic radiation from hot materials, Max Planck wrote[1] (italics are mine):

> ***Max Planck to Robert Wood***
> A theoretical interpretation therefore had to be found *at any cost*, no matter how high. It was clear to me that classical physics could offer no solution to this problem, and would have meant that all energy would

[1] Source!

> eventually transfer from matter to radiation. ...This approach was opened to me by maintaining the two laws of thermodynamics. The two laws, it seems to me, must be upheld under all circumstances. For the rest, I was ready to sacrifice every one of my previous convictions about physical laws. ...[One] finds that the continuous loss of energy into radiation can be prevented by assuming that energy is forced at the outset to remain together in certain quanta. This was purely a formal assumption and I really did not give it much thought except that *no matter what the cost, I must bring about a positive result.*

Trying to provide a theoretical explanation at any cost, Max Planck introduced the idea of energy quanta, initiating the development of quantum ideas and becoming "the father of quantum physics."

1. Introduction

This quantum business is so incredibly important and difficult that everyone should busy himself with it.

A. Einstein in a letter to his friend Jakob Laub in 1908, as quoted by A. Wheeler in "The Mystery and The Message Of The Quantum"

Abstract In this chapter.

Quantum physics is a century-old branch of physics. Its success is unparalleled and yet quantum physics is unfinished in one sense: There is no clear and widely adopted consensus on what some of quantum ideas "really mean."

1.1 What Is Quantum Physics?

There are many characterizations of quantum physics. In essence, quantum physics is the part of physics which focuses on *quantum systems* – physical systems showing *quantum behavior*. So, what effects or phenomena are quantum and which ones are non-quantum (also called *classical*)?

1.2 Who Needs Quantum Physics?

Whether you want to contribute to the progress of science and technology, or just make sense of the most advanced knowledge of the world – you will need familiarity with quantum physics.

Modern technologies are based on a very broad range of physical phenomena described both by classical and quantum physics. Anyone doing research or working in applied science and engineering would benefit from solid understanding of physical concepts and laws. The importance of quantum physics has grown steadily over the past hundred years and it is clear now: The future is quantum. Today the knowledge and skills in quantum physics are key to success in such areas as quantum chemistry, laser physics, and quantum cryptography, to name a few.

The knowledge of quantum physics is also important to anyone trying to simply appreciate modern understanding of the world. Quantum physics is the best picture of the univeres we have today. However, the message of quantum physics is so unusual and often subtle, that understanding it properly requires a careful study instead of a cursory glance through a popular literature.

1.3 Why is Quantum Physics Hard?

Quantum physics is not an easy subject. Mastering it requires learning new physical concepts and advanced mathematical tools. Additionally, it is necessary to examine critically some basic intuitive notions of classical physics and everyday experience. All this takes time and effort.

> ***Feynman on Quantum Physics***
> In his book *The Meaning of It All*, Richard Feynman – an American theoretical physicist and Nobel laureate – wrote:
> "Trying to understand the way nature works involves a most terrible test of human reasoning ability. It involves subtle trickery, beautiful tightropes of logic on which one has to walk in order not to make a mistake in predicting what will happen. The quantum mechanical and the relativity ideas are examples of this."
> *The Meaning of It All, Section I: The Uncertainty of Science.*

Let us examine those features of quantum physics that make it particularly challenging and discuss what can be done to make the learning easier.

1.3.1 Indirect Experience

Concepts of classical physics, especially when applied to macroscopic objects, can often be connected to our experiences. Velocity, heat, force, pressure, motion of projectiles, currents of fluids, propagation of sound and so on – all can be felt, seen, or otherwise perceived by humans. In contast, *we do not have a direct access to the quantum side of the world.* There are no experiments where one can say "Here, look at this entanglement!" or "Hey, touch this quantum of action!", or "Listen carefully, that is the decoherence!"

Analogy

Imagine you want to understand the life of inhabitants of extremely distant land. The environment and climate of that land makes it impossible for you to travel there and see everything with your own eyes.

Who knows, maybe they do not have such notions as "home", "road", "tomorrow", "force", and so on.

Now, if by "understand" we mean "explain and predict the behavior of aliens in our own terms", then the task might be hopeless. However, if by "understand" we mean "develop new concepts that describe and predict the behavior of aliens" then we might have a chance. Of course, the new concepts might be very foreign to us, at first. But the more we use them – the less our mind struggles.

The problem then becomes two-fold. First, we need give up our non-quantum intution. Remember, our intution and conceptual pictures are rooted in macroscopic reality. We interact with the latter using our macroscopic organs and instruments which are unable to capture subtle non-classical features of the world.

Second, we need to develop new intution and set of "pictures"/models which would be adequate for quantum world. While doing so, we must be careful not to "contaminate" this new way of thinking with old concepts.

1.3.2 Abstract Mathematics

The mathematics used in modern physics is becoming increasingly more abstract. It is the price to pay for the powerful and general tools which are indespensible for expressing most advanced results of physics, either classical or quantum.

Quantum physics does not use any special mathematical methods which would make it more challenging than classical physics. Of course, a student learning quantum theory might struggle with the heavy use of operators, complex vectors, Hilbert spaces, commutators, matrices and so on. The difficulty, however, is not due to the quantum nature of the subject, since nearly all these mathematical concepts and tools could be found in other – non-quantum – areas of mathematical physics.

Alfred Whitehead on Abstract Mathematics

> Nothing is more impressive than the fact that as mathematics withdrew increasingly into the upper regions of ever greater extremes of abstract thought, it returned back to earth with a corresponding growth of importance for the analysis of concrete fact. ...The paradox is now fully established that the utmost abstractions are the true weapons with which to control our thought of concrete fact
>
> *Mathematics as an Element in the History of Thought, Ch. 2, p. 46*

Surprisingly, current mathematical framework of quantum physics is the simplest possible. Quantum theory is *linear*, meaning that it is based on the *superposition principle* and uses the *simplest* kinds of operators – *linear operators*. It must be noted that the linearity is not exclusive to quantum theory, as, for example, classical electrodynamics is also a linear theory.

1.3.3 Inadequate Language

Perhaps the biggest barrier on the path to mastering quantum physics is... *ordinary language*. It has developed to communicate everyday experiences and emotions, but subtle natural phenomena are far removed from either of those.

> ***Bertrand Russell On Ordinary Language***
>
> Ordinary language is totally unsuited for expressing what physics really asserts, since the words of everyday life are not sufficiently abstract. Only mathematics and mathematical logic can say as little as the physicist means to say.
>
> *The Scientific Outlook (1931)*

> ***Werner Heisenberg On Ordinary Language***
>
> It is not surprising that our language should be incapable of describing the processes occurring within the atoms, for, as has been remarked, it was invented to describe the experiences of daily life, and these consist only of processes involving exceedingly large numbers of atoms. Furthermore, it is very difficult to modify our language so that it will be able to describe these atomic processes, for words can only describe things of which we can form mental pictures, and this ability, too, is a result of daily experience.

> *Principle of Quantum Theory (1930), Introductory, p. 11*

To capture deep and nuanced laws of nature, we must use a language of adequate power, built with new words and symbols completely foreign to common intuition and simple mental pictures. In other words, the adequate language has to be *abstract*, *purposefully developed*, and most likely *unintuitive* to an untrained mind.

R Constructing new languages is a standard task in the field of computer science. New languages are regularly proposed to better describe computational tasks in a specific domain. For example, a separate language exists for working with database requests.

In this field, a *domain specific language* is usually developed to address problems

A language is considered good if it does not allow constructing meaningless or harmful statements. For example, if a language allows definition of a set of sets which are not elements of themselves, the this language will be plagued with the famous Russell paradox.

Natural human language is notoriously bad for formulating precise and abstract results of science.

❗ *Particles*

Particles are excitations of quantum fields. They correspond to irreducible unitary representations of Lorentz group. Fields are operator-valued vector functions defined over Minkowski space and transformed simply under the action of the Poincare group.

Learning this language is the key to success.

1.4 Quantum Versus Classical

Quantum physics does what classical physics failed to do: It provides a consistent, relatively simple, and extremely powerful set of physical and mathematical tools that describe, explain, and predict behavior of subatomic, atomic, molecular, and even macroscopic phenomena.

To understand the limits of classical physics and the advantages of quantum, a solid understanding of both is required. Moreover, at time classical physics may appear decepticely similar to the quantum physics, since quite a lot of terminology, concepts, and mathematical tools are shared by two.

So when do we have to use quantum physics? Is there a border – and how sharp? – between quantum and classical phenomena? What are the main differences between the classical and quantum physics? What are the similarities?

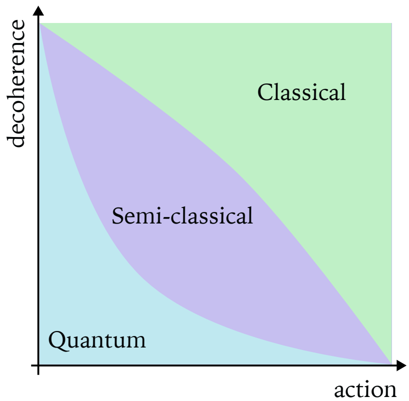


Fig. 1.1: Physical processes exhibit quantum effects when either *action* involved is on the order of 1-100 Planck constants or when *quantum phase* is not washed out by *decoherence* or both.

> ***Semiclassical Physics***
> A set of methods have been developed to solve problems where the deviations from classical behavior was not too strong, but not negligible.

> ***Correspondence Principle***
> Quantum effects do not dissapear abruptly. The transition from quantum to classical behavior is, in certain sense, continuous. This opens a way to approach some problems of quantum physics with the knowledge of classical physics.

1.5 Quantum Enigma

There are many adjectives that can be applied to quantum physics. It is tremendously successful, impactful, fascinating, and intellectually rewarding. It also seems mysterous and "least understood" part of physics.

> ***What is Quantum Mechanics?***
> In 1968 in a German town Lindau, 480 young scientists met with 20 Nobel Laureates in Physics – the founders and major contributors to quantum physics. By this time the theory of quantum physics was fully formulated and applied to many important problems. One of the paticipating Nobel Laureates – Willis E. Lamb Jr – wrote:
>
> "A remarkable feature of the 1968 conference of Nobel prize winners in physics at Lindau is that it was possible for me to ask such question in the presence of two of the founders of quantuam mechancis, Werner Heisenberg and P. A. M. Dirac, more than 30 years after the discovery, in a lecture attended by 400 students who had recently begun their study of the subject."

By 1968 most foundational papers and many great textbooks on quantum physics had been written, including the books by Dirac and Heisenberg. Didn't those works define what quantum mechanics was?

By now quantum physics has demonstrated even greater success, but the questions like "What is Quantum Mechanics?" are still routinely raised in many books and articles. We are still puzzled by this enigmatic omni-tool that we discovered.

> ***Frank Wilczek on Quantum Theory***
> Frank Wilczek, an American theoretical physicist, received his Nobel prize in 2004 "for the discovery of asymptotic freedom in the theory of the strong interaction".
>
> In a paper *What Is Quantum Theory?*, published in Physics Today, he wrote: "To summarize, I feel that after seventy-five years – and innumerable successful applications – we are still two big steps away from understanding quantum theory properly."

1.6 Brief Historical Context

The year 1900 is usually considered the birth year of quantum physics. On December 14 of 1900, at the meeting ????, the German physicist Max Karl Ernst Ludwig Planck presented his theoretical explanation of the *spectrum* of electromagnetic radiation emitted by hot bodies. In his work

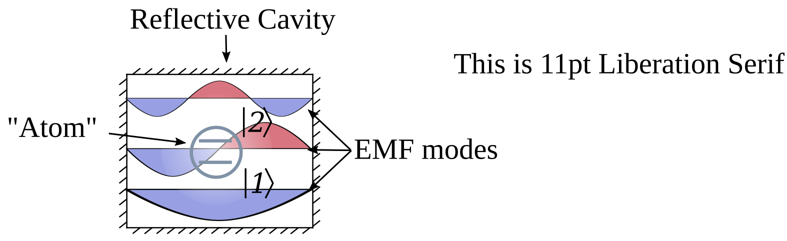


Fig. 1.2: Schematics can be used to represent functions, operators, their compositions and structure.

he introduced what is now known as *Planck's constant* h, which has a physical meaning of *elementary quantum of action.*

> ***Niels Bohr On*** h
> The Danish physicist Niels Bohr, one of the founders of quantum physics, wrote:
> *"A new epoch in physical science was inaugurated, however, by Planck's discovery of the elementary quantum of action, which revealed a feature of wholeness inherent in atomic processes, going far beyond the ancient idea of the limited divisibility of matter."*
> (Atomics Physics and Human Knowledge, Quantum Physics and Philosopy, Complimentarity and Causality.)

Let's take a look at what happened before and after Planck's discovery.

Pre-History: Spectroscopy and Molecular Theory

The evolution of physics in the century preceding quantum era is fascinating. It is important to know some of pre-history in order to appreciate the context in which quantum physics was born. We will highlight two big advancements, one in experimental, and the other in theoretical departments.

Spectroscopy

First, the methods of *spectroscopy*–study of energy distribution across various wavelengths of visible and invisible light–experienced significant development. Already in 1802 an English chemist and physicist William

Hyde Wollaston observed that[1] "If a beam of day-light be admitted into a dark room by a crevice 1/20 of inch broad[2], and received by the eye at the distance of 10 or 12 feet[3], through a prism of flint glass" one can see four colors (red, yellowish green, blue, and violet) separated by *dark lines.* Soon many more dark lines in the optical spectrum of the Sun were discovered and systematically studied by a German physicist Joseph Ritter von Fraunhofer. Today these *Fraunhofer lines* are understood to result from the *absorption* of radiation by different atoms in the atmospheres of the Sun and earth.

The energy absorbed by a substance can be *emitted.* The study of *emission spectra* from various bodies revealed a great deal of complexity and provided a lot of information about bodies' material composition and state.

There are four basic types of spectra one can observe, as illustrated in Figure 1.3. In a typical spectroscopic observation radiation from an excited object is sent through a *dispersive* component, such as a prism or a diffraction grating[4]. A beam of radiation is spread out according to colors and light of different color falls at different place of a detector (eye, photographic plate, or an electronic camera).

Light emitted by atoms consists of very definite colors which show up as *emission lines* in spectra, as shown in Figure 1.3(a). Exact locations of such lines tell which atom emits the radiation. Simpler atoms, such as hydrogen or helium, have simpler spectra.

Spectra of excited molecules contain many closely placed lines. Sometimes multiple lines coalesce into a *band*, as in Figure 1.3(b).

Radiation from large hot bodies, like Sun, reveals both discrete and continuous color distribution; see Figure 1.3(c). All colors are present in the spectrum, and dark Fraunhofer lines indicate that some of the radiation has been absorbed on its way to the detector by atoms and molecules.

Finally, an important radiation type corresponds to an *idealized* object, called *black body* – a theoretical material which does not relfect any incident radiation. In other words, *black body absorbs all incoming radiation.* Of course, black body also emits radiation, otherwise it would

[1]Wollaston William Hyde 1802 XII. *A method of examining refractive and dispersive powers, by prismatic reflection*, Phil. Trans. R. Soc. **92**: 365–380

[2]1/20 in = 1.27 mm.

[3]10-12 feet = 3-3.7 m.

[4]See Visual Glossary

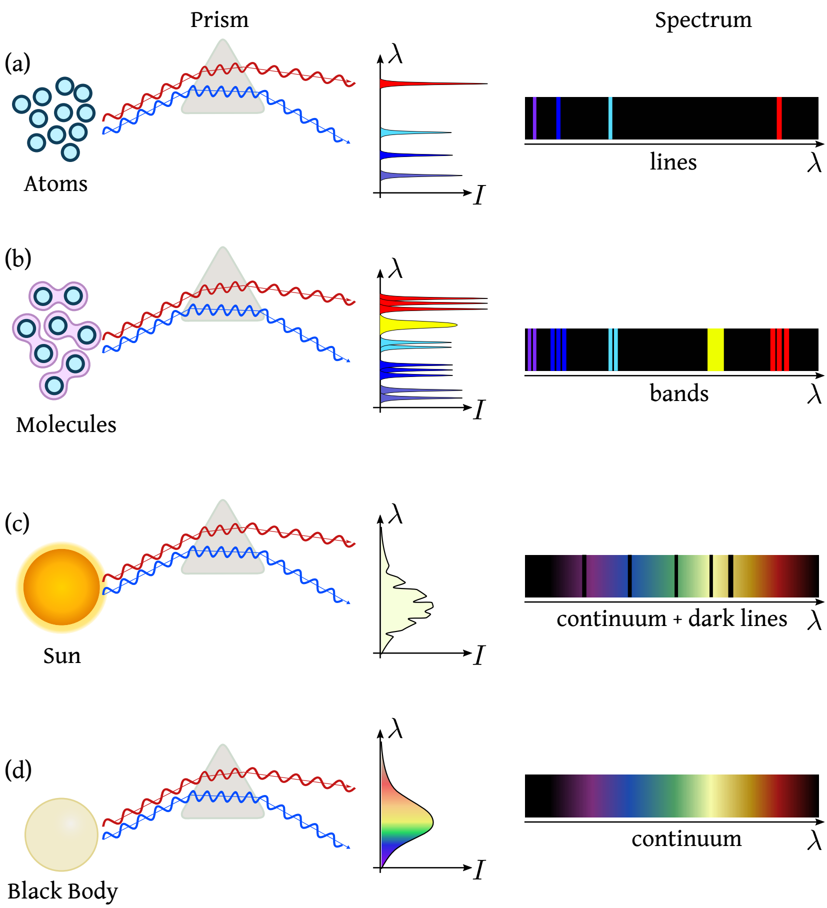


Fig. 1.3: Various types of spectra observed from different substances. (a); (b); (c); (d). See text for explanation.

never stop accumulating energy from incident light. Therefore, black body *is not truly black*, its perceived color depends on the temperature of the body. Emission spectrum of black-body, called *black body radiation* or *normal spectrum*, is of great interest, because it approximates emission spectrum from *any material* kept at a constant temperature. At the end of the 19th century, black body radiation was actively studied both experimentally and theoretically.

Max Planck And Black Body Radiation

Since black-body radiation spectrum does not depend on particular material, it has a universal nature. This fascinated Max Planck, as he wrote in his Scientific Autobiography[a]:

"Thus, this so-called Normal Spectral Energy Distribution represents something absolute, and since I had always regarded the search for the absolute as the lofties goal of all scientific activity, I eagerly set to work."

$$\frac{\delta E}{\delta \nu} \propto \frac{\nu^3}{e^{a\nu} - 1}\,, \qquad a = h/kT\,.$$

[a] Max Planck, *Scientific Autobiography and Other Papers*, Williams & Norgate, 1950, pp. 34-35.

The spectrum of black body radiation was the first type of spectrum to receive a theoretical explanation and an explicit formula. Spectra of atoms and molecules were properly studied only after the development of quantum mechanics.

Spectroscopy of the 19th century[5]

Molecular Theory

Second advancement of pre-quantum physics was connected with the hypothesis of atoms and molecules. Although atomistic ideas had been known for about two millenia, even in the 19th century far from every physicist was convinced that atoms and molecules were objects just as real as everyday things. There was no *direct evidence* for the existence of atoms and molecules, and the main support for the atomistic views came from *indirect evidence*, like the many useful results that followed from molecular theory. For example, the behavior of gases (diffusion,

[5] William McGucken, *Nineteenth-Century Spectroscopy*, The Johns Hopkins Press, 1969.

viscosity, laws connecting pressure and temperature, heat capacity, etc.) was especially well explained. Two major figures in this field were the Scottish physicis James Clerk Maxwell and the German physicist Ludwig Boltzmann.

In September of 1899, at the congress of the German Society of Natural Scientists and Physicians, Ludwig Boltzmann presented a review titled *"The Recent Development of Method In Theoretical Physics."*[6] He highlighted the rapid development of physics in the 19th century and discussed major experimental and theoretical results.

> ***Boltzmann's Prediction***
> *"I have to mention finally the relations which obtain according to the molecular theory between the principle of entropy and the calculus of probabilies, concerning the real significance of which there may be some difference of opinion, but which, no unprejudiced person will deny, are eminently qualify to extend our intellectual horizon and to suggest new combinations both of ideas and experiments."*

Boltzmann's anticipation that "the principle of entropy and the calculus of probabilities" will lead to "new combinations both of ideas and experiments" was fully confirmed by Max Planck in about a year!

As explained in the Section XX, Max Planck used nearly the same approach as Boltzmann to calculate entropy of a thermodynamic system. Only in his case, the system was not an ideal gas of molecules, but a collection of oscillating molecules that interact with electromagnetic radiation (in the form of heat).

1.6.1 1935 And 1965

The experimental and theoretical quantum physics were advancing fast. There were many milestones, two of which we would like to highlight, as they are strongly connected to the modern issues of quantum physics, technology, and even philosophy.

EPR Paper

In March of 1935 a paper titled *"Can Quantum-Mechanical Description of Physical Reality Be Considered Complete"* was published, authored by Albert Einstein, Nathan Rosen, and Boris Podolsky. The work is known

[6] *The Monist*, January, 1901, Vol. **11**, No. 2, pp. 226-257.

as *EPR paper*, and the issue raised in it is sometimes referred to as *EPR paradox*.

Schrodinger Cat

EPR paper stirred the community and prompted a couple of replies. Erwin Schrodinger wrote a paper *"The Current State of Quantum Theory"* (???) where he analyzed the EPR problem. He also introduced the concept of *entanglement* and now famous "dead-and-alive" cat.

Bohr Dense Paper

Niels Bohr – a giant of quantum physics – also replied to EPR paper in October of 1935. His main criticism was focused on how EPR group defined *physical reality*.

Bohr's reply is not an easy read.

Bell's Theorem

The debates about the intricacies of quantum theory never faded. But in 1965 they were taken to a new level, by John Stewart Bell, who proposed a way to *empirically* distinguish two modes of thinking about reality.

In essense, for the proposed experiment, classical physics predicted result A while quantum physics predicted the result $B \neq A$. Experimental tests pointed towards B.

1.6.2 Three Ages of Quantum

The evolution of quantum science and technology can be roughly divided into three stages: (a) "Old quantum physics" (b) modern quantum physics, and (c) information-age quantum physics (see Figure 1.4). Each stage is defined by the type of quantum systems in the focus of research and technology.

Old Quantum Physics

Observations of early quantum phenomena, such as atomic spectra or photoeffect, involved very large number of particles (atoms, photons, electrons; see Figure 1.4(a)). Addressing single particles seemed unrealistic in practice. This made the probabilistic nature of quantum theory even more similar to classical statistical mechanics.

Modern Quantum Physics

A new age of quantum physics began when it became possible to study behavior and interaction between individual particles: atoms and field excitations (e.g. photons). A whole new field, called cavity quantum

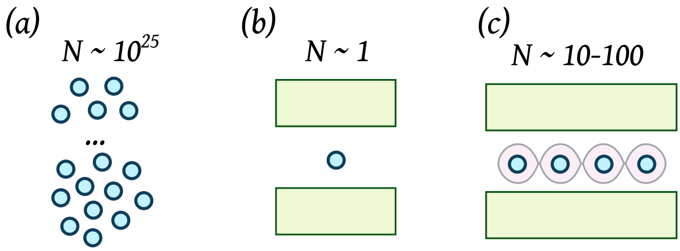


Fig. 1.4: Three stages of quantum science and technology: (a) Observation of large groups of objects (atoms, molecules); (b) Study of interaction between single particles (one atom + one photon); (c) Connecting tens and hundreds of quantum systems using *entanglement*.

electrodynamics (CQED) was born. Figure 1.4(b) shows a schematic of a cavity formed by two mirrors, with a single atom inside the cavity. During this era a number of beautiful experiments, demonstrating purely quantum effects which were previously discussed only theoretically, have been performed.

Information Age

The next step was studying quantum behavior and interaction of several particles (Figure 1.4(c)). The ability to produce and manipulate special *multipartite quantum states* opened the doors into the age of *quantum information science*. The role of *entanglement* as a *resource* for performing useful information processing tasks has been in the center of very active research.

The potential of quantum physics for sending and processing of information became apparent early. Quantum effects offer unique opportunities for secure communication and ultra-fast computation of certain algorithms. Despite many advances in these areas, the challenge is still great, especially in building a universal quantum computer.

> ***Slow Road to Quantum Computing***
> The idea of using quantum effects for computational task has been suggested by Richard Feynman in 1982 paper *Simulating Physics with Computers*.
>
> In an interview with Google's Director of Quantum Algorithms Ryan Babbush, Columbia university professor and the host of the World Science Festival Brian Green asked[a]: "What is the largest com-

> posite number that has been factored today with a quantum computer? " To which Ryan answered: "Oh, not a large one. [...] Something like 15 or 25 or something."
>
> [a] World Science Festival talk *What They Rarely Reveal About Quantum Computing: Without an Algorithm, It's Hype.* Available online.

The challenge is to create and maintain entanglement of a larger number of quantum systems. "In summary, in the span of less than two decades, photonic quantum information science has matured immensely. [...]. The number of photons simultaneously used in experiments has grown, from 2 to 4 up to 12" (Appl. Phys. Rev. 6, 041303 (2019))

Chapter Highlights

- *Natural evolution of mathematical objects from numbers, through vectors, leads to tensors.*
- *Each successive tier of mathematical object in the progression "numbers, vectors, tensors" is more abstract and more powerful.*
- *Numbers, vectors, and tensors are all conceptually connected.*

2. Physics

BEFORE STARTING THE JOURNEY TOWARDS QUANTUM PHYSICS IT IS worth taking a quick at physics in general as a branch of science and identify those basic assumptions, ideas, and tools which lie at its foundation. Quantum physics changed physicists' understanding of nature and our relationship with it. As Niels Bohr put it: *"we are faced with the necessity of a radical revision of the foundation for description and explanation of physical phenomena."*[1] To see and appreciate this necessity, we should get a better understanding of those foundational concepts that get affected most.

> ☑ ***Prerequisite Knowledge***
> To fully understand the material of this chapter, readers should be comfortable with the following concepts:
> - State
> - Dynamical equations

2.1 Goals and Methods

Physics is an activity of *human community* pursuing the following major goals: *Describe*, *explain*, and *predict* phenomena comprising the observed world. These goals might not seem lofty enough since they say nothing about the search for an absolute truth, the ultimate origins of the universe, or the true nature of reality. Admittedly, description, explanation, and prediction imply that someone is performing these activities and someone is benefiting from them. However, keeping in mind the human element

[1]Niels Bohr *On the Notions of Causality and Complementarity*, Science, January 20, 1950, Vol 111 p. 51

in physics is very important. The true nature of physical reality, if uncovered, will be communicated by physicists to other people in a human language, with the hope that we will appreciate it and find a use for it.

Description and explanation can be beautiful, satisfying, and valuable but the real power of physics comes from its predictive ability. Prediction is the ultimate test of any theory.

How do physicist achieve those goals?

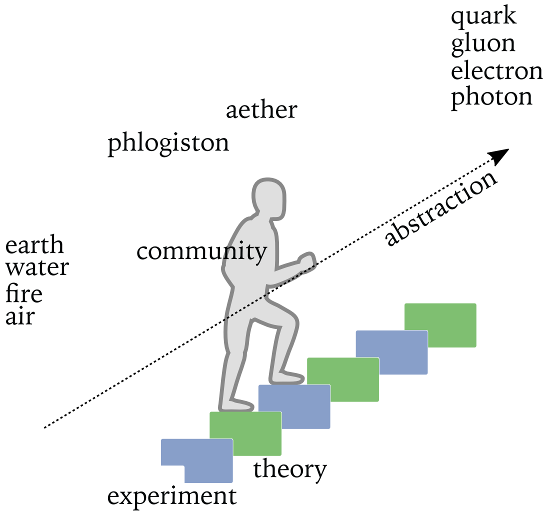


Fig. 2.1: Physics *community* uses experiment and theory to improve the picture of the world. Gradually, the experiments become more elaborate, more accurate, while theories become more abstract.

As history of science illustrates, we advance gradually, sometimes slower, sometimes more rapidly. As physicists explore new domains, they meet phenomena that challenge initial assumptions, sometimes leading to a realization that usual tools and concepts become *inadequate*. This is a *very exciting situation*, a time when the picture of the world gets clarified and updated. In recent centuries there were two major events in physics when our worldview was shaken: Discoveries of relativity theory and quantum mechanics.

> ***Feynman On Understanding Nature***
> *Trying to understand the way nature works involves a most terrible*

> *test of human reasoning ability. It involves subtle trickery, beautiful tightropes of logic on which one has to walk in order not to make a mistake in predicting what will happen. The quantum mechanical and the relativity ideas are examples of this.*

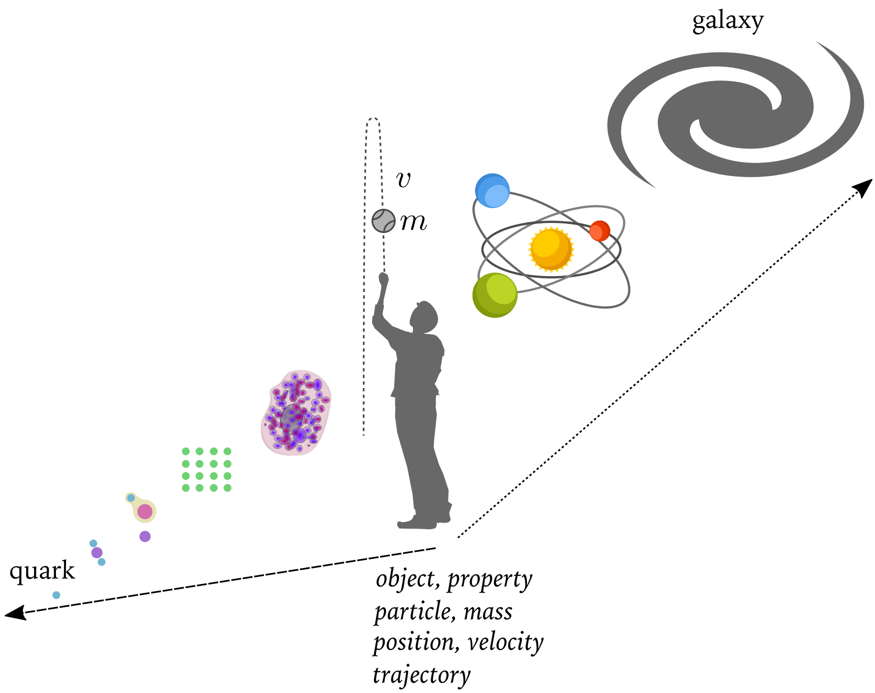


Fig. 2.2: Intuitive macroscopic concepts, such as *object, particle, property, velocity* etc. have their *domain of applicability*, outside of which there is no guarantee that they will work well.

2.2 Common Sense and Common Language

Common sense and basic logic play crucial role in physics.

Entitites, separability, properties, aspects.

2.2.1 Detached Observer

> ***Object and Properties***
> Is it possible to separate an object from its properties?

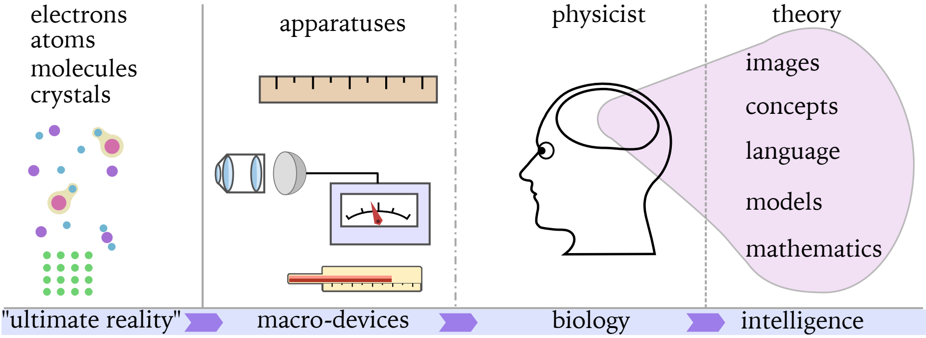


Fig. 2.3: Observers in classical view of the world are detached, separated from the "true" reality which they try to comprehend.

2.3 Classical and Quantum

Classical and quantum physics have many similarities. For example, both employ such basic ideas as: *system, state, dynamics, interaction, measurement, observer, information, probability*. These concepts do not describe any specific phenomena or intstruments, but rather represent elements of a framework used to *think the world*.

To a student, the similarities might make quantum physics seem less foreign, but they also may hinder the understanding of truly and purely quantum features. The more we learn and understand quantum physics, the clearer it becomes that the *meaning* of the basic concepts is often different.

The interconnection between the classical and quantum physics goes beyond shared concepts and mathematical tools. Some physicists pointed out that quantum physics *can't stand on its own*.

Lev Landau On Classical and Quantum

Deep connection between the classical and quantum physics is emphasized by Lev Landau in the third volume of theoretical physics "Quantum Mechanics":

" Usually a more general theory can be formulated in a way logically independent from the less general limiting case. Thus, relativistic mechanics can be formulated based on its own principles, without any reference to Newtonian mechanics. The basic assumptions of quantum mechanics, though, fundamentally can't be formulated without the reference to classical mechanics. "

Quantum Mechanics: Non-relativistic theory, Chapter 1, Section 1

2.4 System

A part of nature that can be clearly isolated and studied is called a *physical system.* An electron, an atom, a molecule, a crystal, a pendulum, a comet, a star – these are examples of physical systems of various degrees of complexity.

In quantum physics the concept of a system is expanded to include not only "physical objects" but also some *aspects of their behavior.* Thus, for the same physical entity (e.g. mode of electromagnetic field), we can focus on its polarization and treat it as a quantum system, disregarding its behavior in time and space. In other words, a separate *degree of freedom* can be viewed as quantum system. Another example is the spin of an electron trapped inside a tiny volume. Electrons have special degree of freedom called *spin.* If quantum experiment focuses on manipulating spin, then the latter can be treated as a distinct quantum system.

2.5 State

State is a very important concept in physics. It means *complete* but *minimal* knowledge about possible behavior of a given system. By *minimal*

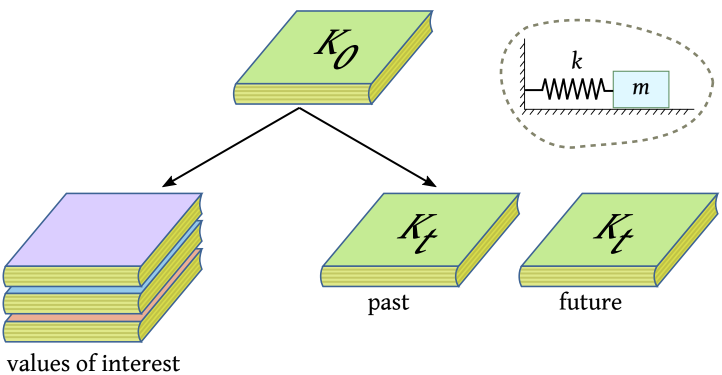


Fig. 2.4: State is a minimal and complete knowledge about a physical system.

knowledge we mean that if position of particle x is known, there is no need to know x^3 or any other one-to-one function of position. We only need to know and keep track of the *essential* information.

Completeness

How can we be sure that the information about a system is complete? Can there be some "hidden" information, unaccessible (may be yet, or even potentially forever) to us and yet affecting the behavior of a system?

The question of completeness is an important one. In the context of quantum physics it was raised for the first time by Albert Einstein, Boris Podolsky, and Nathan Rosen in 1935.

2.6 Deterministic Evolution

The completeness of a state is a very strong constraint. Not only it means "everything there is to know at a given moment", but also "know state now – know state always." The latter is an expression of *determinism*: the complete knowledge of a system is fully determined at all times once an initial state is known. However, state by itself is not sufficient to satisfy the latter requirement, it must be supplemented by the so called *dynamical equations*. These equations are specific to a physical system and encapsulate the laws that govern internal interactions. EXAMPLE?

Denoting the mathematical representation of the state as ξ, the evolution of the state between the moments of time $T = t$ and $T = t + \Delta t$ may be written as a functional dependence:

$$\xi_{t+\Delta t} = U_{t+\Delta t,t}\, \xi_t \,.$$

For $\Delta t > 0$ we determine the future state, while for $\Delta t < 0$ we determine the state in the past (relative to the moment t).

Example

For circular motion the state is the angle $\xi = \phi$, and the evolution is given by a simple formula

$$\phi_{t+\Delta t} = U_{t+\Delta t,t}\, \phi_t = \omega \Delta t + \phi_t \,.$$

Notice that in this case the evolution function depends on the time *difference* Δt and not on each moment of time separately:

$$U_{t+\Delta t,t} = U_{\Delta t} \,.$$

It must be emphasized again, that the final state $\xi_f = \xi_{t+\Delta t}$ is determined by two factors: the initial state $\xi_i = \xi_t$ and the laws of physics encoded in the evolution function $U_{t+\Delta t,t}$.

$$\xi_f \xleftarrow{U_{\Delta t}} \xi_i$$

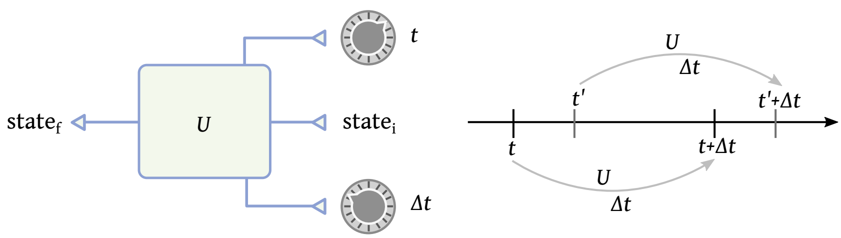


Fig. 2.5: Evolution operator transforms an initial state into the final state in time Δt.

The laws of physics are timeless[2], as illustrated by the Coulomb's law for the force between charges q and Q at a distance r apart: $F_C = kqQ/r^2$. The timeless nature of the physical laws requires that the same initial state ξ_i evolves into the same final state ξ_f regardless of when the evolution starts as long as the time interval between the beginning and the end of evolution is the same. Mathematically this is expressed as follows:

$$U_{\tau,t}\,\xi_i = U_{\tau',t'}\,\xi_i$$

for *any* initial state ξ_i, as long as $\tau' - t' = \tau - t = \Delta t$.

Thus, for *any* values of t and t', we have

$$U_{t+\Delta t,t} = U_{t'+\Delta t,t'}\,.$$

This equation says that the evolution function U becomes insensitive to the values t and t', and only depends on Δt – the time interval between the beginning and the end of evolution. Therefore, we can write the following connection between the states at different moments:

$$\xi_{t+\Delta t} = U_{\Delta t}\,\xi_t\,.$$

[2]Technical term is *time-translation invariant.*

This connection holds *for any moment of time* t and time interval Δt.

In physics the states are represented using numbers, vectors, functions, and similar mathematical objects. Common to all of these types of objects is a very basic property of "additivity" and "scalability". That is, one can – at least formally – add and subtract states, as well as multiply them by numbers. For example, for any two states ξ_1 and ξ_2, one can write equations like

$$\xi_3 = 2\xi_1 + 3\xi_2 \qquad \text{or} \qquad \Delta\xi = \xi_2 - \xi_1 \,.$$

Depending on a particular representation of the state, the evolution function U might be a "usual" function, an operator, or something else entirely. Regardless of what the exact *type* of U is, its job is always the same – map initial state ξ_i at time t into the final state ξ_f at time $t + \Delta t$.

For $\Delta t = 0$ the evolution function U must be a simple *identity* function:

$$U_0 = I \,.$$

Furthermore, for a continuous evolution, it is necessary for small changes in time δt to produce small changes in the state $\delta\xi$:

$$\xi_{t+\delta t} = U_{\delta t}\xi_t = \xi_t + \delta\xi \,.$$

For a continuous evolution of the state, the evolution function U must be continuous. This implies that for small time intervals it produces small changes:

$$U_{\delta t} \approx I + \delta U = I + G\delta t \,,$$

where G is called the *generator* of state evolution. The meaning of the generator is clear from its definition – it specifies how fast the state evolution happens: $G = \partial_t U$.

In terms of the generator, the evolution equation can be written using the relations

$$\delta\xi = \xi_{t+\delta t} - \xi_t = (I + G\delta t)\,\xi_t - \xi_t = G\delta t\xi_t \,.$$

Finally, dividing both sides by δt and using the ∂-notation, we arrive at the Schrodinger-type of equation for the *continuous* state evolution:

$$\partial_t\xi = G\,\xi_t \,. \tag{2.1}$$

The equation (2.1) is a general form of state evolution equations used in physics. It appears in many cases where the dynamics of a system is described as a *continuous deterministic evolution.*

Deterministic Evolution Equations

Equations similar to (2.1) can be found in many physical theories. In quantum theory it is Schrodinger equation, which can be written as follows:

$$\partial_t|\Psi\rangle = -i\hat{H}\,|\Psi\rangle\,.$$

2.7 Measurement

State of a system is closely related to the *knowledge* of its possible behavior. Where does that knowldege come from? In physics the primary source of quantative knowledge comes from experiments. Dutch experimental physicist Heike Kamerlingh Onnes, who received 1913 Nobel Prize in physics "for his investigations on the properties of matter at low temperatures which led, inter alia, to the production of liquid helium," put it as follows: *Door meten tot weten* or *Through measurement to knowing*.

Measurement is the source of our knowledge about the world. This is true for both classical and quantum physics. However, the role measurement plays in classical physics differs substantially from the role it plays in quantum physics. Let's examine the idea and process of measurement closer.

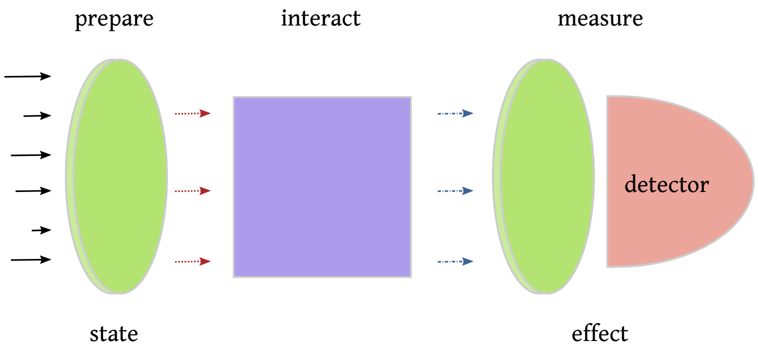


Fig. 2.6: Three stages of measurement process: Preparation of a system in a certain state, followed by the interaction of the system with external system, ending with the measurement which extract the information.

2.7.1 Joint Measurement

Imagine you put your left hand and right hand on different pads and discover that both hands move up and down randomly but in total sync.

2.8 Atoms

Classical physics, mainly classical mechanics and classical electrodynamics, describes atoms as unstable configuration of charges and predicts their decay in an "atomic heartbeat" (nanosecond). There should be no stable atoms according to classical physics, and the reason for all atoms having the same size is not clear at all.

What is observed is, firstly, stable configuration of positive and negative charges in every atom and, secondly, spontaneous decay of atoms from excited state into non-excited state. Quantum physics elegantly explains the latter.

2.9 Particles

Modern understanding of the particle concept is significantly more nuanced than simple "point-like object with certain physical attributes". Instead, particles are defined based on quantum behavior of corresponding *physical fields*. Quantum behavior of electromagnetic field corresponds to photons, and quantum behavior of electron-positron field corresponds to electrons (and their anti-particles).

> ❗ ***Particles***
> Particles are excitations of quantum fields. They correspond to irreducible unitary representations of Lorentz group. Fields are operator-valued vector functions defined over Minkowski space and transformed simply under the action of the Poincare group.

2.9.1 Photons And Electrons

Two types of fundamental elementary particles that will be discussed in this book are photons and electrons. Electronics, optics, and the physics of light-matter interaction cover a tremendous range of problems and provide an excellent playground for learning quantum physics.

2.10 Single Particle Detectors

Detecting a single particle is a challenging task. The first time humans distinctly observed a single particle was in 1908 in an experiment performed by Rutherford and Geiger.

2.10.1 Rutherford-Geiger Method

Rutherford and Geiger described their results in the paper *"An Electrical Method of Counting the Number of alpha-Particles from Radio-active Substances"*. The idea of their experiment, schematically illustrated in Figure 2.7, is as follows: Place a small amount of radioactive substance in vacuum and detect an effect of an extremely narrow "beam" or "ray" of radiation going in a specified direction. The effect Rutherford and Geiger used on to detect a single particle is based on *avalanche amplification* of electron currents in a specially designed detector.

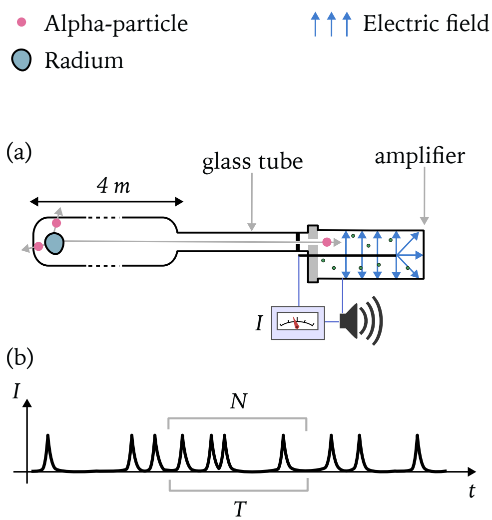


Fig. 2.7: (a) Schematic of the Rutherford-Geiger experiment for detecting a single subatomic particle. (b) Current from the amplifier at various moments of time.

> ***Amplification***
> The avalanche amplification of the effects caused by a single subatomic particle is just one illustration of an important feature of observations and measurements of quantum systems: *"the amplification of micro-*

scopic effects to the level of macroscopic observation,"[a] to borrow the phrase of an American theoretical physicist and Nobel Prize winner Julian Seymour Schwinger.

The amplification, in one form or another, is necessary if we want to perform and discuss any quantum experiment. As Niels Bohr put it:

"Here, it must above all be recognized that, however far quantum effects transcend the scope of classical physical analysis, the account of the experimental arrangement and the record of the observations must always be expressed in common language supplemented with the terminology of classical physics. This is a simple logical demand, since the word "experiment" can in essence be used only in referring to a situation where we can tell others what we have done and what we have learned."[b]

[a] *Quantum Kinematics and Dynamics*, Addendum *Non-selective measurements*, reproduced from the Proceedings of the National Academy of Sciences Vol 1. 45. pp. 1552-1553 (1959)

[b] *On the Notions of Causality and Complementarity*, Science, 1950, Vol. 111

Numbers

Rutherford and Geiger used radium as radioactive material. Radium atom contains 88 protons and 138 neutrons, with the total mass of the atom $m_{Ra} = 3.77 \times 10^{-22}$ gram. The sample of 1 gram would contain $N = 10^{22}/3.33 = 2.65 \times 10^{21}$ atoms. The most stable isotope of radium decays slowly, taking 1600 years for half of the atoms to decay. In one second a tiny fraction of the half of gram would decay, specifically:

$$N = \frac{2.65 \times 10^{21}}{2} \times \frac{1}{1600 \times 365 \times 24 \times 60 \times 60} = 2.65 \times 10^{10} \text{ atoms.}$$

From their measurements, Rutherford and Geiger estimated that one gram of radium emits $N \approx 3.2 \times 10^{10}$ alpha-particles.

If every atomic decay resulted in an electron going through a wire, then we would have a small current of

$$I = \frac{Q}{t} = \frac{Nq_e}{t} = \frac{3 \times 10^{10} \times 1.6 \times 10^{-19}}{1} = 4.8 \times 10^{-9}\,\mathrm{A} \approx 5\,\mathrm{nA}\,.$$

Such current is easily measureable by modern instruments. However, Rutherford and Geiger were using much smaller amount of radium (1/10000 of gram!) and had to amplify the current two thousand times, making it easier to detect and strong enough to make audible "clicks."

2.11 Polarization and Spin

Photons and electrons have certain characteristics which are not related to the "usual" motion through space with time. In technical jargon, they have *non-spatiotemporal degrees of freedom*. The exact meaning of this phrase will be clarified in Chapter 4.

Chapter Highlights

- *The power of mathematical concepts and methods increases with the level of abstraction.*
- *Learning new concepts often involves learning new terminology. The latter can create an artificial mental barrier.*
- *"Usual" numbers form a mathematical structure. The structure is revealed through various relations that exist between numbers.*
- *Relations between numbers are expressed using the concept of functions and operations (e.g., addition). Each operation is characterized by its arity – the number of arguments it accepts as an input.*

3. Mathematics

MATHEMATICAL CONCEPTS AND TOOLS USED IN QUANTUM THEORY ARE not significantly different from those used in classical physics. This fact, of course, does not make them any easier. However, it is important to remember that *there is no special mathematics that makes quantum physics extra challenging*.

In this chapter we develop mathematical tools needed for quantum theory. This is an essential investment that will pay off handsomely when we start discussing quantum phenomena. Mathematics plays a vital role in quantum physics since it becomes the only guide in the field where intuition and simple mental pictures fail.

☑ ***Prerequisite Knowledge***

To fully understand the material of this chapter, readers should be comfortable with the following concepts:

- Basic algebraic operations.
- Simple intuition about probabilities.
- Plots and axes.
- Familiarity with functions like $\sin x$, $\log x$.
- Use of subscripts, as in x_1.

Noncommutativity

Before we start learning specific mathematical tools, it is essential to understand and appreciate one extremely powerful concept that sets apart advanced mathematics used in quantum physics. This is the concept of *noncommutativity*.

Definition 3.1 ☞ *Noncommutativity*
An operation $\bullet$ is *noncommutative* if its result depends on the order of arguments:

$$A \bullet B \neq B \bullet A\,.$$

Example: $x^\wedge y \neq y^\wedge x$.

Noncommutativity brings variety and power. It is a natural ingredient of human language where various combinations of same letters have very different meaning: **post** ≠ **stop** ≠ **spot**. Furthermore, our actions and operations are often noncommutative, as we are reminded every time we dress up and go outside (in that order!) Obviously, the result of putting on socks and then shoes differs from doing it in reverse order. In short, noncommutativity is an essential feature of the world and our interactions with it. If mathematics is to adequately describe all that, it must have noncommutativity.

Usually, first introduction to algebra starts with addition, followed by multiplication. Both operations are *commutative*: $x + y = y + x$ and $x * y = y * x$. Very quickly students learn about subtraction, division, and exponentiation (x to the power of y) which lack commutativity. At even higher level, *function composition* (discussed in Section 3.3) demonstrates the power of noncommutativity. Thus, noncommutativity is present in the "usual" mathematics of classical physics but at a higher level. The situation in quantum physics is fundamentally different!

In quantum theory noncommutativity greets us at the door: even the basic quantities like position x and momentum p are now represented not by numbers, but by *special mathematical objects*, called *operators*, which, while extending and improving the concept of a number (see Fig. 3.1 and section 3.3), lose the comfortable property of commuting. To illustrate the point, consider the product xp, the physical meaning of which will be discussed later in section 4.4.1. In quantum theory we have

$$xp \neq px\,.$$

The difference $xp - px$ is closely related to the most fundamental physical quantity called *quantum of action*, discussed in section (XXX??).

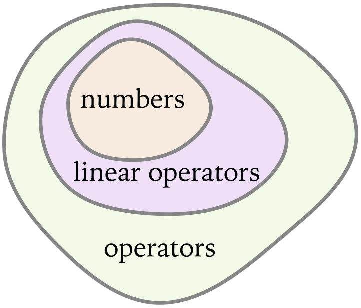


Fig. 3.1: Usual numbers are, in a certain sense, special case of *linear operators* – powerful mathematical objects explored later in the book.

Not only position x and momentum p are "upgraded" from numbers to operators, but any function of them, like their product xp, or kinetic energy $E_k = p^2/2m$ or potential energy $E_p = kx^2/2$ and so on[1], become operators too. In quantum theory any quantity that can be *observed* (measured) is described by an operator instead of a number.

> ***Dirac on Quantum versus Classical Mechanics***
> Paul Dirac, in §7 of his textbook *Principles of Quantum Mechanics* writes
>
> *"The different algebra for the dynamical variables is one of the most important ways in which quantum mechanics differs from classical mechanics."*
>
> In mechanics, position x and momentum p are called *dynamical variables*, since their change in time is governed by the *dynamical laws* – laws that involve forces (*dynamis* – "force" in Ancient Greek). The rules for manipulation of x and p , including addition and multiplication, are called *algebra*. Classical physics relies on a commutative algebra where x and p are usual numbers and $xp = px$. But classical physics fails to explain quantum phenomena.

To summarize: Noncommutative mathematics more accurately captures the intricate quantum features of nature. It is the essential feature of quantum theory where noncommutativity is placed at the very foun-

[1] We will learn how to understand a function of an operator later in the book.

dation.

3.1 Randomness

Randomness is the opposite of predictability, certainty, and *determinism*. Randomness (or *indeterminism*) plays an essential part in classical and quantum physics.

Mathematical description of randomness is based on the idea of *probability*. Probability quantifies or measures the notion of *likelyhood* of certain outcomes. A common example is the toss of a "fair" coin where the probability of tail is $1/2$. Or take another typical case: an ideal six-sided die. If we throw it 6 million times, and each side is equally likely to appear, we expect to see each value to come up 1 million times. Suppose we are interested in those cases when the face value is divisible by 3. That happes 2 million times, one million for 3 and another for 6. The chances that we will see such an event is therefore 2 out of 6 or 1/3.

Definition 3.2 ☞ ***Probability as Frequency***
Probability of an event E can be defined as the *relative frequency* or ratio of the number of times the event occured in a series of trials to the total number N of trials:

$$P_E = \frac{N_E}{N},$$

where P_E is the probability of the event E, and N_E is the number of times the event happened during an experiment.

Let's consider a real experiment with a six-sided die, illustrated in Figure 3.2. In this experiment a die has been first thrown 200 times, then again 200 times, and finally 1000 times. The histograms in the second column show how many time each face value appeared in a given series of throws. The plots in the first column correpond to the *running value* of the probabillity that the face value is divisable by 3 (i.e. equals 3 or 6) has appeared (event $E_{3|6}$). The running value means that we recalculate the ratio $N_{3|6}/N$ with each throw. This explains large oscillations between 0 and 1 at the very begginning, when we either get or don't get the event E in the first throw.

The histrograms reveal that in relatively short experiments ($N < 1000$) different face values might appear with significantly different probabilities, as if the die is not *fair*. This also results in the probability of

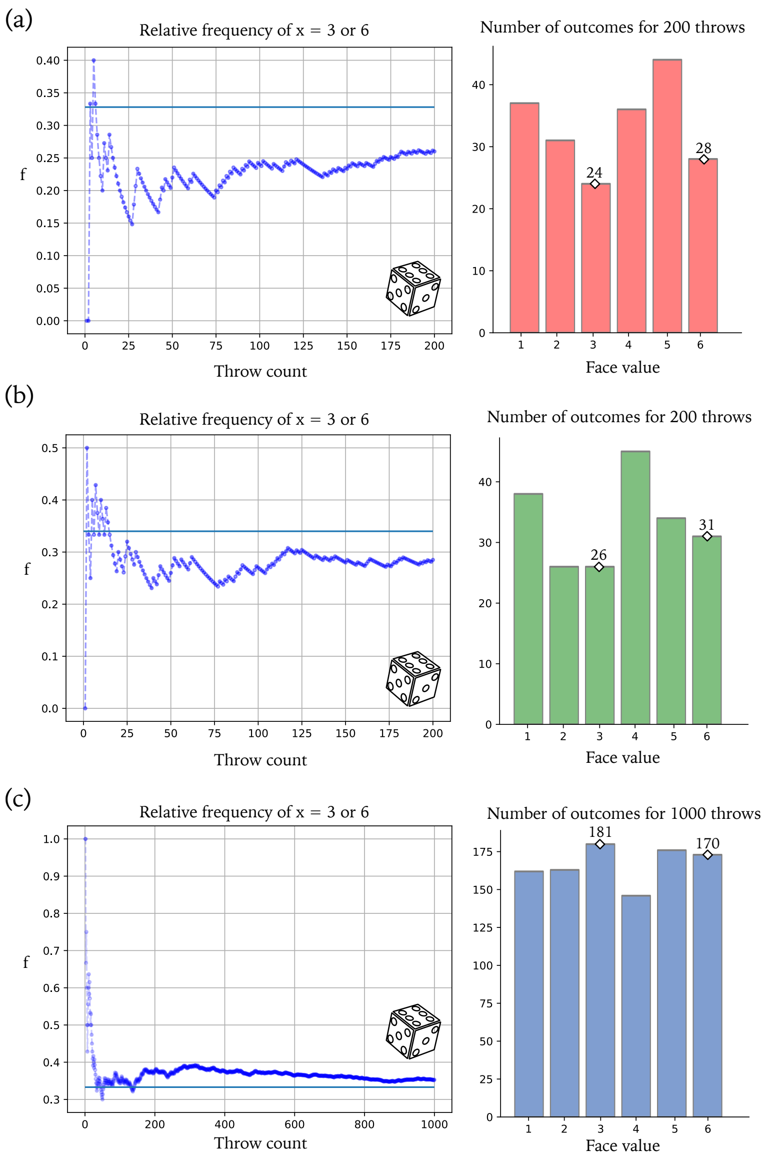


Fig. 3.2: Results of throwing a real six-sided die.

the event $E_{3|6}$ to differ appreciably from the "ideal" value of $1/3 \approx 0.33$, shown as a horizontal line in the plots. It requires a longer series of throws to see that the die is pretty fair and the probability of the event $E_{3|6}$ is close to the ideal.

❗ *Irreversibility*
The throw of a die is a good example of what is required in an actual quantum experiment. When a die is hitting the surface of a table, it might bounce off several times before settling on a certain face. Whatever energy (kinetic of potential) the die had is now *irreversibly* gone. If we had an ideal perfectly elastic materials, the die would endlessly keep bouncing up and down and we would not be able to complete a single observation.

Every observation in physics has a certain *finality*, caused by some *irreversible process* that transfers energy and information to the observing "device" (e.g. camera or human eye and brain). We will be returning to this important point in other chapters.

3.1.1 Ensembles and Series

In experiments with random outcomes we must deal with many trials. In each trial we observe a certain value $v_i\,,\ i = \overline{1, N}$. The sequence of values $v_1,\, v_2\,, v_3,\, \ldots,\, v_N$ is called an *statistical ensemble* or a *random series.*

3.1.2 Average and Variance

Two important characteristics of a random series are the *average value* and the *variance.* The usefulness of these parameters comes from their ability to concisely describe the main features of a very rich set of random values. Average can often be linked to the most likely outcome, while variance describes how large is the spread of values around the average.

The meaning of average or expectation value is quite simple. Imagine we throw a big party for M guests. Each guest can eat some number of candies, ranging from 0 to 10. Although we can't know this number in advance for each guest, we can still try to guess how many candies to purchase in order to satisfy all guest. Assuming that each guest "is the same" and can eat N candies, we would buy MN candies (perhaps more to be safe and guarantee that there will be enough candies for all). N is the *average* number of candies per guest.

Before we give formal definitions to these concepts, let's consider a specific example. Suppose we have M fair dice and toss them all at once. Denoting the sum total of face values as V, we can state that it can be anywhere from M to $6M$, although neither of these extreme values is likely. But which value will be most likely? The total value V is the sum of face value of each individual die:

$$V = v_1 + v_2 + v_3 + \ldots + v_M \,.$$

We can group terms according to their value and rewrite V as

$$V = 1 \cdot n_1 + 2 \cdot n_2 + 3 \cdot n_3 + \ldots + \ldots 6 \cdot n_6 \,,$$

where n_k is the number of dice with the face value k. For identical fair dice the ratio n_k/M is the same (1/6 in this particular case) and corresponds to the probability p_k of getting the face value k from a single die in a long series of throws. Therefore, the total face value of M dice can be written as follows:

$$V = 1 \cdot p_1 N + 2 \cdot p_2 N + 3 \cdot p_3 N + \ldots + \ldots 6 \cdot p_6 N = N\langle v \rangle \,,$$

where

$$\langle v \rangle = 1 \cdot p_1 + 2 \cdot p_2 + 3 \cdot p_3 + \ldots + \ldots 6 \cdot p_6$$

is called *average* of a face value. Another way to write average is

$$\langle v \rangle = \frac{V}{N} \,.$$

Definition 3.3 ☞ ***Average Value***

Given a random series $v_1,\ v_2\,, v_3,\ \ldots,\ v_M$, the average value is defined as

$$\langle v \rangle = \frac{v_1 + v_2 + v_3 + \ldots + v_M}{M} \,.$$

If each outcome v_k can take a value u_i with a probability p_i, then the average can be written as

$$\langle v \rangle = u_1 p_1 + u_2 p_2 + u_2 p_2 + \ldots + u_m p_m \,.$$

Along with *average value* the term *expectation value* is often used.

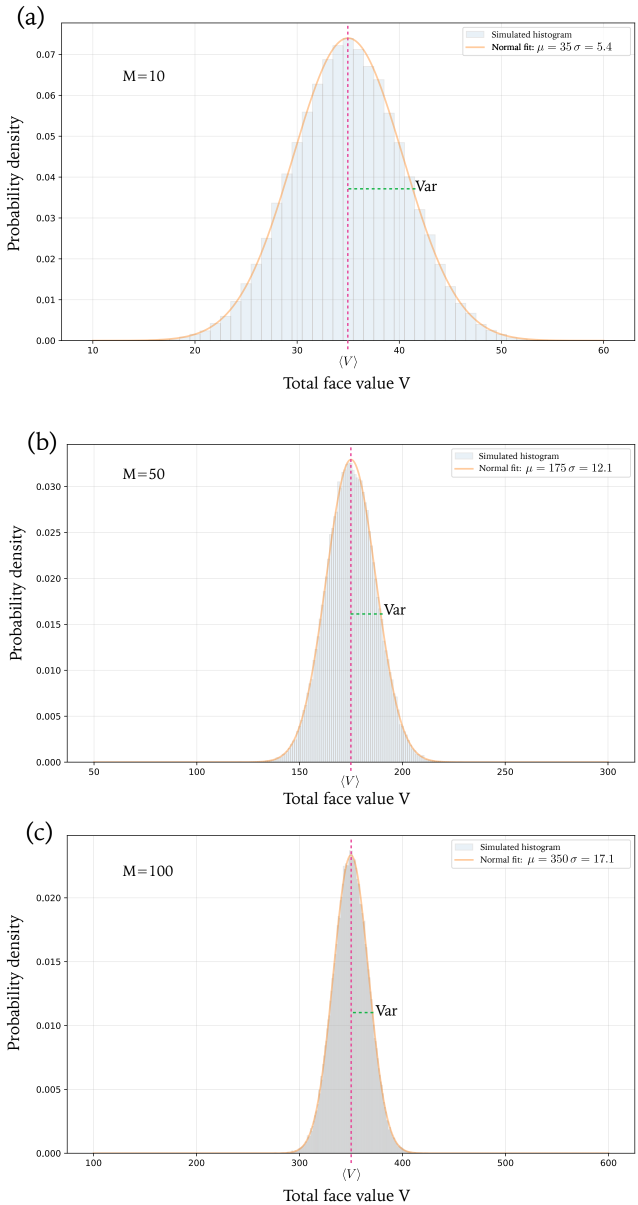


Fig. 3.3: Results of throwing M six-sided dice multiple times. The average and variance.

In Figure 3.3 the results of simulations of throwing $M = 10, 50, 100$ six-sided balanced dice is shown. The horizontal axis corresponds to the total face value $V = v_1 + \ldots + v_M$, while the vertical bars are proportional to the probability for a specific value to appear in a series of throws ($N = 100000$ in this example).

As immediately seen from the histograms, the extreme values $V_{min} = M$ and $V_{max} = 6M$ are stupendously unlikely. The average value $\langle V \rangle$ in this example lies in the middle between the minimum and maximum and also has the highest probability to appear. There are deviations from the average in each trial, but the distribution of V becomes seemingly narrower with increasing number of dice used. This is a curious feature of many processes that involve several random variables contributing to the final result.

To characterize the deviations from the average we can use *variance*. The idea is simple: Given a random series $V_1, V_2, V_3, \ldots, V_N$ we first calculate the average value $\langle V \rangle$. Then for each value V_k we calculate its deviation from the average $\Delta V_k = V_k - \langle V \rangle$. Next we want to find what the average of this deviation. But we must be careful, since positive and negative deviations appear equally likely and will cancel each other. To get around this, we first square the deviation and only then find the average $\langle (\Delta V)^2 \rangle$, obtaining *variance*. We can give the following definition:

Definition 3.4 ☞ ***Variance***
Given a random series $v_1, v_2, v_3, \ldots, v_M$, with the average $\langle v \rangle$, the variance of the series is the measure of its spread around the average value, given by

$$\operatorname{Var} v = \langle (v - \langle v \rangle)^2 \rangle \,.$$

Exercise 3.1 ✎
Show that

$$\operatorname{Var} v = \langle v^2 \rangle - (\langle v \rangle)^2 \,,$$

where $\langle v^2 \rangle$ is the average value of v^2:

$$\langle v^2 \rangle = \frac{v_1^2 + v_2^2 + v_3^2 + \ldots + v_M^2}{M} \,.$$

If v is measured in meters, then the units for variance would be in meters squared. It is often convenient to characterize the spread around the average in the same units as the average. This can be done using *deviation*:

Definition 3.5 ☞ ***Deviation***
Given a random series $v_1, v_2, v_3, \ldots, v_M$, with the average $\langle v \rangle$ and variance $\operatorname{Var} v$, the deviation of the series is the measure of its spread around the average value, given by

$$\Delta v = \sqrt{\operatorname{Var} v}\,.$$

R *Average* and *deviation* are important and convenient characteristics of a random series. They often (but not always!) concisely capture two basic aspects of the random series of values: the most likely result and the spread around it.

3.1.3 States And Hidden Variables

To make notation used in quantum physics feel less foreign we will introduce it early and in a more familiar setting. To this end, consider a scenario where a student, let's call him Alex, goes to a vending machine to buy a glass of juice. The machine offers two options: (1) A pure apple juice and (2) a pure orange juice, as shown in Fig. 3.4.

Suppose that each day for the whole year Alex chooses option 1. We can conclude almost with certainty that in future observations, Alex will choose option 1. We will say that Alex is in such a *state* that, if confronted with a choice at the vending machine, he will always go with the option 1. Symbolically we will write the *state* of Alex as $|1\rangle$.

Imagine another student, let's name her Bella, who sometimes chooses option 1 and other times – option 2 in an unpredictable (random) fashion. If both options are chosen equally frequently, we can say that Bella is in an *uncertain state* and write it symbolically as follows: $\frac{1}{2}|1\rangle + \frac{1}{2}|2\rangle$. All this mathematical expression states is our ability to predict the results of future observations, based on the observations performed in the past.

If students make up their minds right at the vending machine, being completely uncommitted prior, then calling the state of Bella *uncertain* is

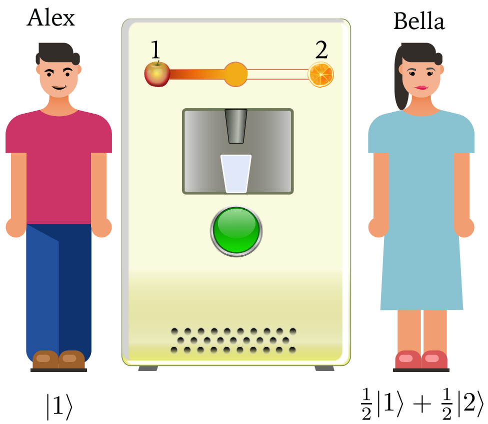


Fig. 3.4: Alex and Bella choose a juice at a vending machine.

justified. But another situation is possible: Bella approaches the machine already knowing what she wants that day and her mental state is quite certain in each transaction, however it is unknown to an outside observer. In other words, the factors that determine Bella's choice are *hidden* to the outside observer. In quantum theory such factors are called *hidden variables*.

Element Of Reality

It is not hard to accept the fact that, in the examples considered above, Bella is undecided before she interacts with the vending machine. But it is harder to imagine that Alex is undecided every time and yet *always* decides in favor of the apple juice. Such a certainty, the complete absence of randomness, leads us to think that there is *something in the reality* that determines Alex's choice.

Albert Einstein, Boris Podolsky, and Nathan Rosen suggested that "*If, without in any way disturbing a system, we can predict with certainty (i.e., with probability equal to unity) the value of a physical quantity, then there exists an element of physical reality corresponding to this physical quantity.* "

Thus, since we can, without asking Alex, predict with certainty the value of the option he chooses, there exists, according to Einstein, Podolsky, and Rosen an element of physical reality corresponding to Alex's choice.

R When we use the concept of the state, should we understand it as the state of the system, or our the state of "knowledge" about a system?

3.1.4 Joint Measurement

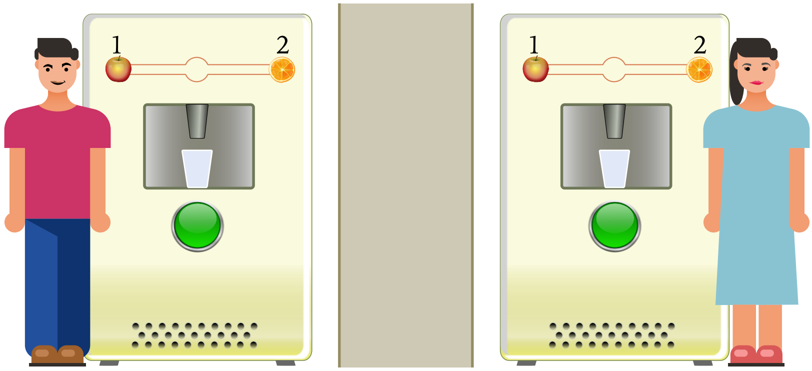


Fig. 3.5: Alex and Bella choose a juice at a vending machine.

When a student chooses a drink we consider it an *event*. We can speak about random events (random outcomes), their probabilities, expected values, and deviations. It is also possible to discuss two or more events in order to analyze possible connections between them. Suppose, for example, that Alex and Bella buy their drinks from different vending machines situated far apart, as illustrated in Fig. 3.5. Their choices are recorded and later examined, either separately or *jointly* (together).

If we consider Alex and Bella as a single team, then their combined (*joint*) choice can be considered a single event E. There are four different potential outcomes for each joint event: (1) Both Alex and Bella choose apple juice, $E = AA$, (2) Alex chooses apple juice, Bella chooses orange juice, $E = AO$, (3) Alex chooses orange juice, Bella – apple, $E = OA$, and finally (4) both choose orange juice, $E = OO$.

Imagine we have a long observational record, from which we can calculate relative frequency of each of the four possible outcomes: $AA \to p_1$, $AO \to p_2$, $OA \to p_3$, and $OO \to p_4$. Similar to the notation used to represent a state of a single person, we can write the general state of the "team" as follows:

$$p_1|11\rangle + p_2|12\rangle + p_3|21\rangle + p_4|22\rangle\,,$$

where $|11\rangle$ represents the situation when both Alex and Bell choose option 1 (apple juice) and p_1 is the probability of this outcome, $|12\rangle$ represents the situation when Alex chooses option 1 while Bell chooses option 2 with relative frequency p_2. Similarly for the remaining terms of the sum. For example, the case when Alex always chooses apple juice, while Bella vacillates between two equally appealing options would be described by the following state of the team (*joint* state):

$$|T_1\rangle = \frac{1}{2}|11\rangle + \frac{1}{2}|12\rangle\,.$$

Exercise 3.2

Suppose the joint state of two students is described by the state

$$|T_0\rangle = \frac{1}{2}|11\rangle + \frac{1}{2}|22\rangle\,.$$

What can we say about Alex? What can we say about Bella? How does this state compare to $|T_1\rangle$ given above? Can we distinguish between $|T_0\rangle$ and $|T_1\rangle$ observing Alex alone? Bella alone?

3.1.5 Correlations

Analyzing the state $|T_0\rangle$ we conclude that Alex randomly chooses any juice with equal probability. The same can be said about Bella. However, despite both Alex and Bella acting randomly, there is certain connection between their behavior, which results in total coincidence of their choices and complete absence of cases when they choose different juice. This is an example of *correlation* between random processes.

For the state $|T_0\rangle$ the correlation is very strong, complete, in fact. In other words, given $|T_0\rangle$ and observing only the behavior of Alex, we don't need to observe Bella, since her choices will "mirror" those of Alex as if she had no autonomy at all. If we observed Bella instead, we could predict the choices of Alex. Although random, the behaviors of Alex and Bella are *not independent*, they are *perfectly correlated.*

Exercise 3.3

> Consider the joint state given by
>
> $$|T\rangle = \frac{4}{10}|11\rangle + \frac{1}{10}|12\rangle + \frac{1}{10}|12\rangle + \frac{4}{10}|22\rangle\,.$$
>
> What can we say about the behavior of Alex? Of Bella? Is there any correlation in their behavior?

It is easy to create a perfect correlation between Alex and Bella if they can communicate. For instance, if they meet and agree to choose the same juice on each day. To make their choice random they flip a coin to determine the type of choice for that day. In this case we would observe the behavior described by the state $|T_0\rangle$.

3.1.6 Marginals

Sometimes, given the information about the "whole team", we want to deduce the information about its single member. For instance, given the joint state

$$|T\rangle = \frac{1}{12}|11\rangle + \frac{1}{6}|12\rangle + \frac{1}{4}|21\rangle + \frac{1}{2}|22\rangle\,,$$

we want to predict the behavior of Alex. To do so, we need to take a more "coarse-grained" view on the results of joint measurements (observations) and stop paying attention to what Bella is doing. In other words, we need to differentiate based only on the outcomes for Alex. Looking at $|T\rangle$, we see that Alex chooses apple juice in $N/12 + N/6 = N/4$ cases out of N. The second option appears in $N/4 + N/2 = 3N/4$ observations. Thus, we can describe the state of Alex as

$$|A\rangle = \frac{1}{4}|1\rangle + \frac{3}{4}|2\rangle\,.$$

Performing the same exercise for Bella, we obtain her state

$$|B\rangle = \frac{1}{3}|1\rangle + \frac{2}{3}|2\rangle\,.$$

The results we obtained for Alex or Bella are called *marginal distributions*. These are probabilities obtained from *joint distributions* by summing over terms that do not affect the outcome we want to focus on, like in the examples above.

Exercise 3.4
Using the state $|T_0\rangle$ defined above, find the states of Alex and Bella.

3.1.7 States Likeness

If the behavior of Alex is described as the state $|A_0\rangle = |1\rangle$, while the behavior of Bella as $|B_0\rangle = |2\rangle$, we can confidently say that there is a big difference between them. There is nothing in common in their choices, except their complete predictability.

Suppose we have $|A_1\rangle = 0.5|1\rangle + 0.5|2\rangle$ and $|B_1\rangle = 0.49|1\rangle + 0.51|2\rangle$. Although these states are not the same, they do seem similar. It is useful to have some mathematical way to *distinguish* such states, or, put differently, a quantitative measure of "likeness" of two states. Such a numeric function would have two inputs – states $|A\rangle$ and $|B\rangle$ – and it would produce a number representing how much in common the behaviors of Alex and Bell have. Given some such function, we could write using typical mathematical notation resembling a product($x * y$):

$$|A_0\rangle \Diamond |B_0\rangle = 0 \quad \text{and} \quad |A_1\rangle \Diamond |B_1\rangle \approx 1\,.$$

The first equations says that there is nothing in common between Alex and Bella – their preferences are completely different. The second equation states that the behaviors of both are nearly identical, with almost 100% likeness[2].

❗ *Likeness and Correlation*
We must not confuse correlation and similarity of states.

The choice of the "likeness" function is a matter of convention. It is dictated, foremost, by convenience which it brings into calculations. The only constraints for any candidate function should express "reasonable behavior." For instance, the measure of likeness ranges from zero to unity, and any state is completely similar to itself:

$$0 \le |A\rangle \Diamond |B\rangle \le 1 \quad \text{and} \quad |A\rangle \Diamond |A\rangle = 1\,.$$

We can also expect that the behavior of Alex resembles that of Bella's to the same degree as the behavior of latter resembles the behavior of the

[2] We can measure "likeness" on the scale from 0 to 1, or from 0 to 100%.

former. Symbolically: $|A\rangle\Diamond|B\rangle = |B\rangle\Diamond|A\rangle$.

The problem of *distinguishing* random behaviors (states) was discussed as early as 1923[3] by Sir Ronald Aylmer Fisher – a British mathematical statistician, working in the field of biology and genetics. Fisher's approach to represent random states using *angular coordinates* instead of more directly measurable probabilities was more advantageous for statistical calculations and was later analyzed by many researchers, including quantum physicist. For example, one of the founders of quantum information theory, William Kent Wootters, emphasized[4] the connection between Fisher's approach and the information which can be obtained during any experiment, not just quantum.

Fisher Circle, Wootter Semicircle, and Bloch Sphere

Here we"' briefly outline the motivation behind representing states using square roots of probabilities: $|A\rangle = \sqrt{p_1}|1\rangle + \sqrt{p_2}|2\rangle$ instead of a more intuitive form $|A\rangle = p_1|1\rangle + p_2|2\rangle$.

Fisher, focusing on statistical genetics, was analyzing probability p for a particular "value" of a gene to appear in a given population. In a simplest case the "value" was assumed binary so the probability for the other "value" of the gene is simply $q = 1 - p$. Instead of probability p Fisher introduced an angle variable θ, defined by $\cos\theta = 1 - 2p$. This is equivalent to noticing the similarity between the statistical and trigonometric identities: $p + q = 1$ and $\sin^2\phi + \cos^2\phi = 1$. Thus, introducing $p = \sin^2\frac{\theta}{2} = (1 - \cos\theta)/2$, Fisher was able to simplify many of his calculations.

The change of coordinates suggested by Fisher can be illustrated geometrically in Fig. 3.6. At first, each state written using probabilities $|A\rangle = p|1\rangle + q|2\rangle$, can be depicted by an arrow or a point in the upper right part of the (p, q) plane, as shown in Fig 3.6(a). Since $p + q = 1$, all states will belong to a straight segment connecting two "extreme" states $|1\rangle$ and $|2\rangle$. Using the angle θ, defined by $p = \sin^2\frac{\theta}{2}$, each state takes the form $|A\rangle = \sin\frac{\theta}{2}|1\rangle + \cos\frac{\theta}{2}|2\rangle$ and now corresponds to an arrow pointing to the semicircle in Fig. 3.6(b).

Construction similar to Fisher's was suggested by Wootters, with

[3] Fisher RA. XXI.—On the Dominance Ratio. Proceedings of the Royal Society of Edinburgh. 1923;42:321-341.

[4] Wootters, W.K. (2013). Optimal Information Transfer and Real-Vector-Space Quantum Theory. arXiv: Quantum Physics, 21-43.

the only difference in the range of values for the angle. William Wootters considered change of coordinates $(p, q) \to (\cos^2 \alpha, \sin^2 \alpha)$. As the result, each state can be written as $|A\rangle = \cos\alpha|1\rangle + \sin\alpha|2\rangle$ and represented using an arrow pointing to an arc of a circle in the top-right quadrant of Fig 3.6(c).

All three ways of representing states are equivalent and the choice is made based on convenience. Later we will learn one more powerful approach to represent states, based on *operators*. To close our discussion of geometrical representation of random states, we mention *Bloch sphere* which is used to represent states of quantum systems – known as *qubit* – with two distinct "levels". It is shown in Fig 3.6(d) and can be understood as an extension of Fisher semicircle. The extension from semicircle to a sphere is needed because in quantum physics a general state of a two-level system is described by *two angles* θ and ϕ, as we will later learn.

3.1.8 State Multiplication

The advantage of using square roots of probabilities when representing states becomes even more evident once we extend the mathematical operation of multiplication to states. The idea can be illustrated as follows. Consider a state $|A\rangle = \sqrt{p_1}|1\rangle + \sqrt{p_2}|2\rangle$. If we treat the "likeness" operation $\Diamond$ like a regular product, we obtain

$$|A\rangle\Diamond|A\rangle = p_1|1\rangle\Diamond|1\rangle + \sqrt{p_1 p_2}|1\rangle\Diamond|2\rangle + \sqrt{p_2 p_1}|2\rangle\Diamond|1\rangle + p_2|2\rangle\Diamond|2\rangle\,.$$

Now $|1\rangle\Diamond|1\rangle = 1 = |2\rangle\Diamond|2\rangle$, while states $|1\rangle$ and $|2\rangle$ are completely distinct: $|1\rangle\Diamond|2\rangle = 0 = |2\rangle\Diamond|1\rangle$. Therefore,

$$|A\rangle\Diamond|A\rangle = p_1 + p_2 = 1$$

as required.

This example is just a quick preview of what can be gained by using alternative coordinates (e.g., angles θ) together with *linear operations* such as product. We will have a more extensive introduction into these ideas once we get to the basics of linear algebra in the following sections.

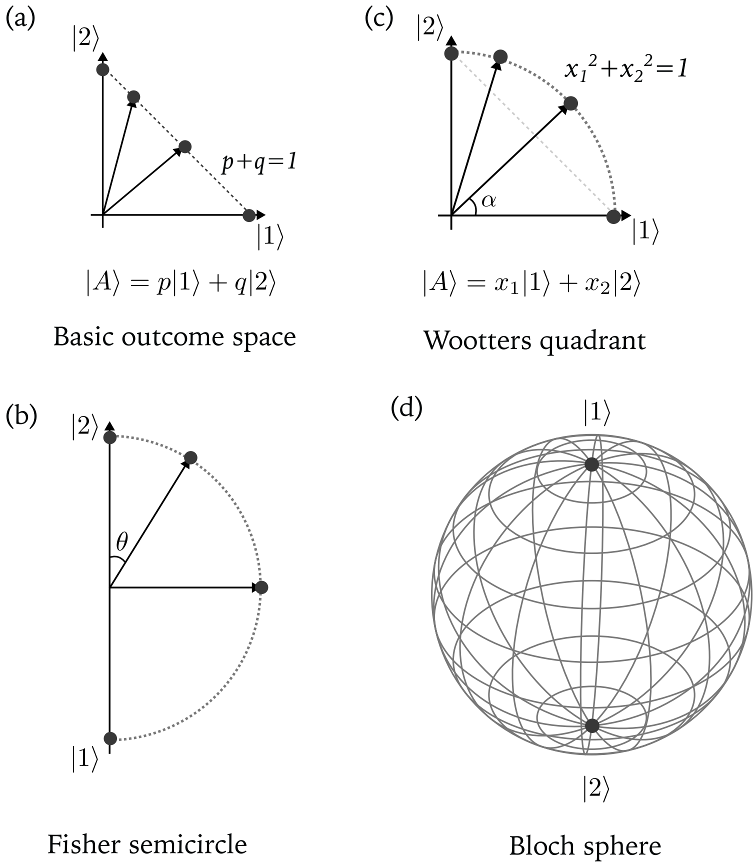


Fig. 3.6: Different approaches to represent states in probability space.

3.2 Functions

The idea of a function is a very basic one. Essentially, function is an unambiguous *rule*, an *algorithm*, which associates a certain value y (*result* or *output* of a function) with every meaningful *input* value x (*argument* of a function). As an example, consider the following function:

$$y = \frac{1}{2x^2 + 5}\,.$$

For any real number x we can compute the value y using only basic arithmetic operations.

Next consider a function **sqrt** which computes a square root of a number: $y = \mathbf{sqrt}\,x$. If we only work with real numbers, the range of meaningful inputs is reduced – only non-negative input values x are allowed. Another thing to notice is that now the function is simply given a name **sqrt** and shows no structure – it is not expressed in terms of other more basic operations. A few important functions are given names: $\sin$, $\cos$, $\exp$, $\log$, **abs**[5] and several more. Of course, the square root of a number is traditionally written with a special sign – called a *surd*: $\mathbf{sqrt}\,x = \sqrt{x}$.

3.2.1 Function Boxes

The input-output view of functions leads to a helpful picture where a given function is represented as a box with an input and an output (or multiple inputs and even multiple outputs), as illustrated in Figure 3.7.

3.2.2 Application Notation

The dominant rule for writing function f applied to an argument x uses parentheses around the argument, like so: $f(x)$. This rule, however, is *not absolute* and is abandoned as soon as one uses functions in linear algebra. For example, when an operator[6] $\widehat{f}$ is applied to a vector $|x\rangle$, we would write $\widehat{f}\ |x\rangle$, without the parentheses.

In this book we will use a uniform rule for function application: *Function and its argument are separated by a space. Only structured arguments are surrounded by parentheses. Simple arguments are written without*

[5]Absolute value of a number – its positive magnitude.

[6]The concept of an operator will be explained in section 3.7.

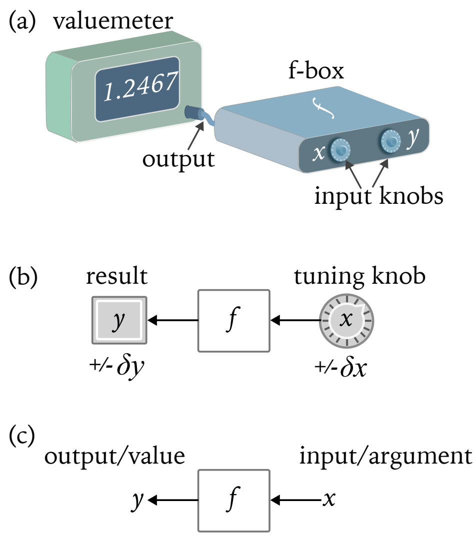


Fig. 3.7: Function can be viewed as a closed box with an input and an output.

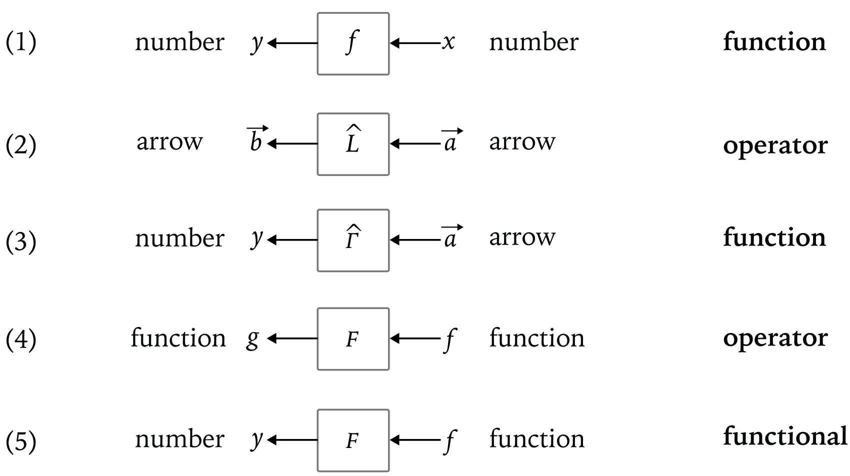


Fig. 3.8: Several types of functions: (1)"Usual" numeric function that computes a number for each numeric input value; (2)Function on "arrows" (or vectors), that computes an arrow for each arrow input value; (3) Function that computes a number (e.g., length) of an input arrow; (4) Function that transforms an input function into another function; see example in text. (5). Function that computes a single number for an input function. See section 3.8 for examples.

parentheses. Thus, we will always write

$$\sin x\,, \quad \exp x\,, \quad \mathrm{abs}\, x\,,$$

and so on. We can even write – without any confusion – expressions like

$$\cos\frac{\phi}{3}\,, \quad \log\sqrt{y}\,.$$

However, we will always use parentheses in the expressions similar to

$$\tan(\alpha+\beta)\,,$$

to avoid confusion with another valid expression: $\tan\alpha+\beta = (\tan\alpha)+\beta$.

3.2.3 Multi-Input Functions

Function of a single argument are the most familiar kind. However, functions with two inputs are also widely used. The simplest example is the function of two arguments (*binary function*):

$$\mathbf{add}\, x\, y = x + y\,.$$

On the left-hand side the function is written using so-called *prefix notation*, where the name precedes the arguments. On the right-hand side the same function is written using a more conventional *infix notation*, in which a special symbol is placed between two arguments. Other examples if binary functions can be given:

$$\mathbf{mul}\, x\, y = x * y = xy\,,$$

$$\mathbf{pow}\, x\, n = x^\wedge n = x^n\,,$$

$$\mathbf{max}\, x\, y = x \text{ if } x > y, \text{ otherwise } y\,.$$

Note that the infix notation works only for binary functions and special symbols (like "+") exist only for small number of them. In short, infix notation, although convenient, lacks generality needed for more powerful and abstract mathematics.

Any formula expressing a physical quantity can be viewed as a function with multiple inputs. Consider Newton's law for gravitational attraction between two point-like bodies. The force is given by

$$F = G\frac{Mm}{r^2}\,.$$

Assuming that the gravitational constant G is a fixed number, the expression for the force depends on three arguments – two masses and the distance between them.

Exercise 3.5
Consider a *ternary function* (function of three arguments):

$$f\,M\,m\,r = Mm/r^2\,.$$

Express it purely in terms of the binary function **mul** and the unary function **inv**.

3.2.4 Partial Application

The box-view of functions leads to a simple, yet powerful, concept of a *partially applied* function. A function of multiple arguments is called partially applied if not all its "inputs" are "filled" (i.e. assigned fixed values).

Consider, for instance, the ternary function from the Exercsie 3.5. Suppose we study how two specific bodies,with masses $M = 10$ and $m = 1$, interact gravitationally . Then the force F between these bodies is a unary function F_r of the distance r:

$$F_r = G * (f\,10\,1\,r) = 10G/r^2\,.$$

Here we partially applied the ternary function f to only two arguments, leaving the third argument r a free parameter. As the result of such partial application we obtained a unary function of the distance r between two boides.

Let's consider another example of partial application. This illustration might seem trivial, but it will help understand the role of partial application in the case of *dual* objects, such as ket and bra vectors of quantum theory. First, we write a product of two numbers using prefix notation: **mul** $x\,y$. Then, partially apply the binary function **mul** to some number, say 3. This results in a unary function **trp** which simply triples the value of its argument:

$$f_y = \mathbf{trp}\, y = 3y\,.$$

3.2.5 Linearity

Some functions are simpler than others. For example, a *symmetric* function has the same value for x and $-x$: $f\,x = f\,(-x)$. A general function does not have such a property, and in this sense symmetric functions are simpler than a general function.

Among the simplest kinds of functions, *linear functions* are of special importance. Such functions have the following properties:

$$f\,(x+y) = (f\,x) + (f\,y) \quad \text{and} \quad f\,(ax) = a(f\,x)\,.$$

These requirements are called *linearity conditions*.

Exercise 3.6
Check whether the function **trp** is a linear function.

For numeric functions, the linearity conditions are very restricting. Linear numeric functions all have the same form:

$$f_a\,x = a * x\,,$$

for some number a. Thus, for each number a there corresponds a linear numeric function f_a and its action on any argument x is a simple multiplication by the number a.

The simuation becomes less trivial when we consider linear functions whose arguments are not simple numbers (e.g. vectors, operators, or even functions).

3.3 Numberlikes

Physics without numbers is unimaginable. But simple numbers, like *natural numbers* 1, 2, 3, and so on are very limiting. As the range of application of mathematics increased, numbers evolved from natural numbers, to *whole* numbers, to *fractions*, to *real* numbers, and then to *complex* numbers and to *quaternions*[7]. The concept of a number became increasingly less intuitive, more abstract and powerful.

[7] Octanions are not used in physics widely enough to be discussed here.

Today mathematics offers several *mathematical objects* which behave essentially like numbers, but which also allow more powerful manipulations and thus can be used in wider range of problems. Examples of such *numberlikes* are *vectors*, *tensors*, and *operators*. We will explore all these objects in this chapter and will see that these three concepts are actually closely related to each other (e.g. vectors are tensors, and tensors are operators!)

Addition

The essential characteristic of numbers is the ability to *add* two of them to get another number:

$$x \boxplus y = z\,.$$

The addition operation satisfies two simple requirements

$$x \boxplus y = y \boxplus x \qquad \text{– commutativity}\,,$$

and

$$(x \boxplus y) \boxplus z = x \boxplus (y \boxplus z) \qquad \text{– associativity}\,.$$

Additivity is "contagious" – it propagates to other mathematical objects which operate on "usual" numbers. For example, it is easy to give a constructive meaning to the following expression:

$$f = \sin \boxplus \exp\ .$$

Here we add two numeric functions to create a new function f. To describe this function we must specify what it does to all possible arguments. In this case it is simply

$$f\,x = (\sin\,x) + (\exp\,x)\,.$$

Thus, the ability to add numbers leads to the ability to *add functions*. It must be emphasized, that in the expressions like $\sin \boxplus \exp$ we are not adding numerical values of the functions, we are adding functions – completely different mathematical objects. Adding functions becomes very useful for certain function types called *operators*, as explained in section 3.7.

Multiplication

Repeated addition of numbers leads to the idea of multiplication. For "normal" numbers mutiplication has two properties analogous to addition:

$$x * y = y * x \qquad \text{– commutativity},$$

and

$$(x * y) * z = x * (y * z) \qquad \text{– associativity}.$$

Furthermore, mutiplication and addition possess *distributivity*:

$$x * (y + z) = x * y + x * z .$$

The ability to add "normal" numbers allows one to introduce *multiplication of functions*. Indeed, we can give a constructive meaning to an expression

$$f = \sin \boxtimes \exp \ .$$

It must be emphasized again: this is not a multiplication of the numeric values of the functions $\sin$ and $\exp$, it is the mutliplication of the functions themselves. To find the value of a unary function f, we simply write

$$f\,x = (\sin\ x) * (\exp\ x) = y * z .$$

Now on both sides of this equality we have "normal" numbers: On the left-hand side we have the value $f\,x$ of a unary function f applied to the numeric argument x, on the right-hand side we have to numeric values $y = \sin\ x$ and $z = \exp\ x$ multiplied in a "usual" way.

The example given above can be generalized to an arbitrary pair of unary numeric functions f and h. It is not difficult to convince yourself that the "product" $f \boxtimes h$ is commutative, associative, and is also distributive with respect to the "addition":

$$f \boxtimes (h \boxplus g) = (f \boxtimes h) \boxplus (f \boxtimes g)$$

for three unary numeric functions f, h, and g. In other words, unary numeric functions can be made to behave like numbers. One can view functions and manipulate them as objects on their own, without referring to their arguments.

> **_Point-free Notation_**
> Manipulating functions without explicitely writing their arguments is

> known as *argument-free notation* or *point-free notation*. It is a useful practice and is common in quantum theory.

Composition

Functions, and their "brothers" operators, allow an additional way to combine two function in order to create another one. It is called *composition* or, sometimes, *sequencing* of two functions. Let's illustrate the idea using the familiar functions $\sin$ and $\exp$. We can "create" (define) a function f which acts on its input argument in the following way:

$$f\,x = \sin\left(\exp\,x\right).$$

We first apply the function $\exp$ to the input argument x to obtain a numeric value $y = \exp\,x$, and then apply the function $\sin$ to y. We applied two functions in sequence. A special notation exists for composition. We write $f = \sin\circ\exp$. Here is used an argument free notation, writing simply f instead of "f of x", similar how we write numbers simply as n instread of "n apples".

Unlike addition and multiplication, *composition is not commutative*:

$$\sin\circ\exp \neq \exp\circ\sin \quad \text{because} \quad \sin\left(\exp\,x\right) \neq \exp\left(\sin\,x\right).$$

However, *composition is associative.* Given three functions f, h, and g we can combined them in two different orders, specified by the parentheses:

$$\left(f\circ h\right)\circ g = f\circ\left(h\circ g\right).$$

Both sides of this equality represent the same value $f\left(h\left(g\,x\right)\right)$: We first evaluate $y = g\,x$, then feed it into h to find $z = h\,y$, and finally input it as the argument to f.

> **Exercise 3.7**
> Check whether composition is distributive with respect to an "addition" of functions.

> ***Composition In Quantum Physics***
> Composition is an very powerful way of creating new functions by combining a given pair of functions. Composition has special signifi-

cance for *linear operators* – linear functions operating on non-numeric arguments (e.g. on vectors).

In quantum theory different operators represent different measurement operations, such as measurement of position, momentum, energy, angular momentum, spin, and so on. Only *linear operators* are used in quantum theory. These operators act on special vectors, as we will later learn.

Like any two functions, two quantum operators $\widehat{A}$ and $\widehat{B}$ can be composed:

$$\widehat{C}=\widehat{A}\circ\widehat{B}\ .$$

It is customary to drop the infix composition sign and simply write $\widehat{C}=\widehat{A}\widehat{B}$.

An operator can be composed with itself. In this case a special notation exists to avoid long and clumsy expressions:

$$\widehat{A}\circ\widehat{A}=\widehat{A}\widehat{A}=\widehat{A}^2\ .$$

In general, the expression $\widehat{A}^n$ represents the operator $\widehat{A}$ composed with itself n times.

3.4 Kalcoolus

Quantum theory uses many mathematical tools, including linear algebra and calculus. In this book the full power of the latter won't be need. Instead, we will use "kids menu calculus" or "kalcoolus" – a small set of simple calculus-related tools sufficient to express most of the equations of quantum theory. We will begin by discussing kalcoolus variants of *derivative* and *integral*.

3.4.1 $\Delta - \delta - \partial$ Notation

Imagine we are tracking time from the moment we wake up at 6 a.m. to the moment we go to bed 16 hours later. Every event E during the day is assigned a certain time t_E. We can ask how much time elapsed between two events. If the first event happens right after we wake up and the second at noon, then we write the time interval as

$$\Delta t = t_2 - t_1\ .$$

This is an example of Δ-notation for *sizable change* of any variable, in this case it is time.

Now consider an eye-blink, where we close the eyes at t_1 and open them at t_2. In this case we write

$$\delta t = t_2 - t_1 \,,$$

using δ-notation to emphasize that the time interval is *tiny*.

Suppose we are monitoring the temperature outside by looking at a thermometer. For each moment of time t we can specify the measured temperature f_t. The temperature in the morning and the temperature at noon can be quite different, so we use Δ-notation:

$$\Delta f = f_{t_2} - f_{t_1} = f(t_2) - f(t_1) = f(t_1 + \Delta t) - f(t_1) \,.$$

For the tiny time interval the change of temperature is also expected to be tiny. Physically this means that there are no extreme events causing rapid jumps in temperature. Mathematically, this means that the function f is *continuous*. In such cases we use δ-notation:

$$\delta f = f(t_1 + \delta t) - f(t_1) \,.$$

Such relation can be written for any moment of time t:

$$\delta f = f(t + \delta t) - f(t) \,.$$

Often it is important to know *how quickly* a value is changing. For example, we can speak of the *rate of change* of temperature with respect to time:

$$\frac{\delta f}{\delta t} = \frac{f(t + \delta t) - f(t)}{\delta t} \,.$$

Instead of the fractions, we will use ∂-notation:

$$\partial_t f = \frac{\delta f}{\delta t} \,.$$

The notation just introduced can be applied to *any* function of *any* variable or *any* number of variables. It is general, relatively simple, and efficient.

Exercise 3.8
Given the kinetic energy of a moving body is $E = mv^2/2$, write out fully the meaning of the following expressions: $\partial_m E$ and $\partial_v E$.

Derivative
The rigorous mathematical notion of derivative is used when we want to be absolutely sure that we are dealing with a unique value for the rate of change. To arrive at this unique value, one would analyze the result of making the change δx of the variable x gradually ever smaller, and prove that the ratio

$$\frac{f(x+\delta x) - f(x)}{\delta x}$$

is approaching a unique value.

For simplicity and practical purposes, we will use the rate of change $\partial_x f$, remembering that we are making a small error when compared to a rigorous derivative:

$$\text{derivative} = \partial_x f + \text{small error}\,.$$

3.4.2 Bernoulli Sums

A Swiss mathematician Jacob Bernoulli, who worked at the end of the 17th century, made many important contributions to mathematics. In his 1713 book "The Art of Conjecturing" he presented formulas for the "sum of powers":

$$\int n = \frac{1}{2}nn + \frac{1}{2}n\,,$$

$$\int nn = \frac{1}{3}n^3 + \frac{1}{2}nn + \frac{1}{6}n\,,$$

and so on up to the power of n^{10}, and gave a general expression for $\int n^c$.

The summation sign used by Bernoulli was introduced by his contemporary – a German mathematician Gottfried Leibniz. The sign is just the first letter of the word *'Sum'*, written in accordance with the rules of that time. For example, a 1661 astronomy reference book *"Astronomia Carolina"*, contains a section titled *Of the ſecond Inequality of the Moon.*

History aside, notation is a matter of convention. In this book we will use Leibniz summation sign for "regular" sums, like Bernoulli, as well as for "special" sums used to express *integrals*.

> ***Example***
> Using Leibniz summation sign, we can say that "sizable change is the sum of many tiny changes" by writing the following expression:
>
> $$\Delta x = \int \delta x$$
>
> The right-hand side stands for a long expression $\delta x + \delta x + \ldots + \delta x$.

3.5 Arrows

Before we introduce the concepts of *vectors*, *vector spaces*, and *operators* acting on vectors, we will examine an introductory model for vectors based on *arrows* – directed line segments, as illustrated in Figure 3.9.

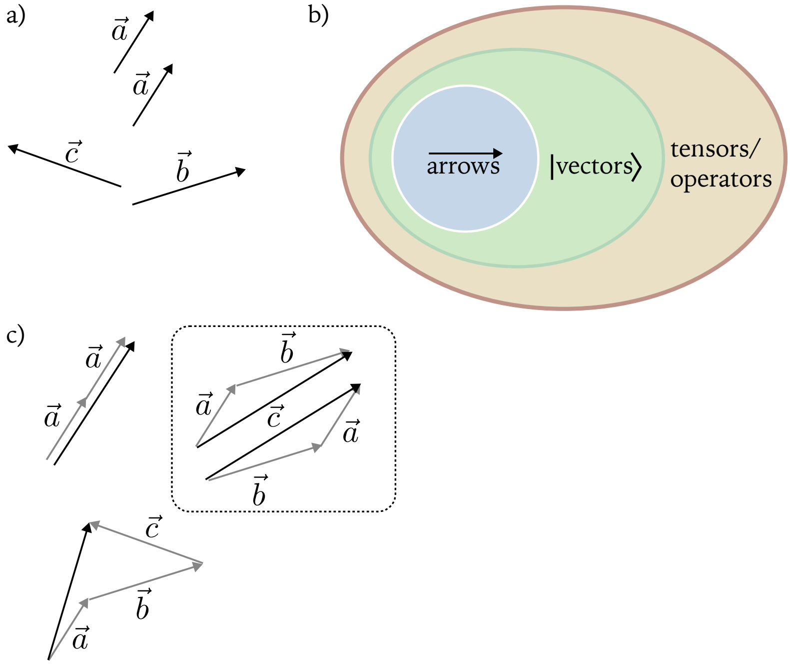


Fig. 3.9: Arrows provide a simple geometric example of vector quantities. The idea of vectors, however, is more powerful and extends beyond this simple representation as directed line segments.

A very common notation for vectors uses arrows over letters:

$$\vec{a}\,,\vec{b}\,,\vec{c}\,,\dots,\vec{\alpha}\,,\vec{\beta}\,.$$

However, not all vectors behave like arrows, so this suggestive notation is not universal. Furthermore, to introduce the basic notation of quantum theory as early as possible, we will switch to what is known as *Dirac notation* for vectors. We already met with this notation when discussing probability in section 3.1.3 and now is the time to extend the range of its application.

3.5.1 Dirac Notation

Instead of placing an arrow on top of a letter, we enclose the letter between a pair of symbols as follows:

$$\vec{a} \xrightarrow{\text{Dirac}} |a\rangle\,.$$

The advantages of this notation accumulate and become more apparent the longer we study quantum physics.

3.5.2 Arrow Algebra

Arrows are *numberlike* – they can be added and multiplied, and the rules for addition and multiplication resemble those of regular numbers. However, arrows allow a richer set of operations compared to numbers. For example, there are three different types of multiplication for any pair of arrows, depending on the type of the final result: 1) a number; 2) another arrow; 3) more complex geometric figures like a piece of a plane. We will mainly focus on the first type, called *scalar product*, and the thrid one, called *tensor product*.

3.5.3 Bases

Consider a pair of non-parallel arrows $|a\rangle$ and $|b\rangle$, arranged for convenience tail-to-tail. Their sum yields the third arrow $|c\rangle$:

$$|c\rangle = |a\rangle + |b\rangle = 1|a\rangle + 1|b\rangle\,.$$

The numbers in front of $|a\rangle$ and $|b\rangle$ are called *components* of the arrow $|c\rangle$ *relative* to $|a\rangle$ and $|b\rangle$. By varying these numbers, while keeping the arrows $|a\rangle$ and $|b\rangle$ fixed, we can *obtain any arrow in a plane*. In other words, any arrow $|v\rangle$ can be written as

$$|v\rangle = v_a|a\rangle + v_b|b\rangle \,.$$

Here, again, the pair of numbers (v_a, v_b) represent components of $|v\rangle$ relative to $|a\rangle$ and $|b\rangle$.

Non-parallel arrows $|a\rangle$ and $|b\rangle$ are *independent*, in the sense that neither can be expressed in terms of the other. In contrast, the arrow $|v\rangle = v_a|a\rangle + v_b|b\rangle$ is not independent from $|a\rangle$ and $|b\rangle$. Thus, the set of arrows $\{|a\rangle, |b\rangle, |v\rangle\}$ is not independent. For arrows in a plane, the number of independent arrows can not be more than two – matching the dimensionality of the plane.

The set of n independent arrows is called *basis* for the n-dimensional space in consideration. Any set of $m < n$ independent arrows will be *incomplete* and insufficient to serve as a basis. In essense, basis is a set of "building blocks" for all possible arrows. The set of "building blocks" must be rich enough to build up anything else (*complete set*), and it does not need any redundant elements (*independent set*).

Basis in not unique, as we can easily show. Given basis $|a\rangle$ and $|b\rangle$, we can rotate and scale each arrow separately, to obtain arrows $|\alpha\rangle$ and $|\beta\rangle$:

$$|a\rangle \to |\alpha\rangle \text{ and } |b\rangle \to |\beta\rangle \,.$$

As long as $|\alpha\rangle \nparallel |\beta\rangle$, they can be used as basis.

Exercise 3.9
Consider $|+\rangle = |a\rangle + |b\rangle$ and $|-\rangle = |a\rangle - |b\rangle$. Prove that if $|+\rangle = \lambda|-\rangle$ then $|b\rangle = \mu|a\rangle$. Find μ.

As the previous exercise shows, the arrows $|+\rangle$ and $|-\rangle$ are independent and can be used as basis.

Exercise 3.10
Given $|v\rangle = |a\rangle + 2|b\rangle$, express it in the $\{|+\rangle, |-\rangle\}$ basis:

$$|v\rangle = v_+|+\rangle + v_-|-\rangle \,.$$

Find v_+ and v_-.

3.5.4 Normalized Bases

Vectors are sometimes defined as mathematical quantities with *magnitude* and *direction*. This implies that a meaningful notion of *length* (magnitude) can be assinged to vectors. For arrows this is obvious, while for other types of vector-like quantities might be not so obvious.

The length of a vector is called its *norm*. Vectors with unit length are called *normalized*. Any vector can be "scaled" to have a unit length. Indeed, given a vector $|a\rangle$ with length a, we get a normalized vector as follows:

$$|u\rangle = \frac{|a\rangle}{a} \,.$$

When doing calculations, it is convenient to normalize all basis vectors. This results in *normalized basis*.

3.6 Scalar Product

The first way to multiply two vectors that we will study is called *scalar product*. In this operation two vectors are combined to yield a number (a.k.a *scalar*):

$$|a\rangle \cdot |b\rangle = x \,.$$

Here we followed traditional *infix notation* using "$\cdot$" as an analogue of "$*$" for numbers. Soon we will learn the usefulness of *prefix notation* for scalar product and its relation to operators, but for now we will keep writing scalar product of vectors similar to numbers.

Guided by simplicity, we expect scalar product to satisfy two basic algebraic laws:

$$|a\rangle \cdot |b\rangle = |b\rangle \cdot |a\rangle \ \text{ – commutativity} \,.$$

$$(|a\rangle + |b\rangle) \cdot |c\rangle = |a\rangle \cdot |c\rangle + |b\rangle \cdot |c\rangle \ \text{ – distributivity} \,.$$

Let's apply the last requirement to a vector $2|a\rangle$, representing it as $|a\rangle + |a\rangle$:

$$(2|a\rangle) \cdot |b\rangle = (|a\rangle + |a\rangle) \cdot |b\rangle = |a\rangle \cdot |b\rangle + |a\rangle \cdot |b\rangle = 2(|a\rangle \cdot |b\rangle) \,.$$

Thus, we see that one more requirement is both natural and helpful:

$$(x|a\rangle) \cdot |b\rangle = x(|a\rangle \cdot |b\rangle) \,.$$

R We can formulate this last result as a rule: *"numbers can be pulled outside of scalar product."*

3.6.1 Products of Basis Vectors

Combining scalar product with the representation of vectors in a *normalized* basis $\{|u_1\rangle, |u_2\rangle\}$ leads to important insights. Consider two vectors:

$$|a\rangle = a_1|u_1\rangle + a_2|u_2\rangle \text{ and } |b\rangle = b_1|u_1\rangle + b_2|u_2\rangle .$$

Scalar product $|a\rangle \cdot |b\rangle$ can be calculated step-by-step, first expanding $|a\rangle$:

$$|a\rangle \cdot |b\rangle = a_1|u_1\rangle \cdot |b\rangle + a_2|u_2\rangle \cdot |b\rangle .$$

Then, expanding $|b\rangle$, we obtain:

$$|a\rangle \cdot |b\rangle = a_1b_1|u_1\rangle \cdot |u_1\rangle + a_1b_2|u_1\rangle \cdot |u_2\rangle + a_2b_1|u_2\rangle \cdot |u_1\rangle + a_2b_2|u_2\rangle \cdot |u_2\rangle .$$

Using the commutativity $|u_1\rangle \cdot |u_2\rangle = |u_2\rangle \cdot |u_1\rangle$, we can slightly simplify the last expression:

$$|a\rangle \cdot |b\rangle = a_1b_1|u_1\rangle \cdot |u_1\rangle + (a_1b_2 + a_2b_1)|u_1\rangle \cdot |u_2\rangle + a_2b_2|u_2\rangle \cdot |u_2\rangle .$$

If the scalar products of all basis vectors are known, we can calculate scalar product of any pair of vectors. We reduced the problem to defining the scalar product of a pair of unit-length vectors:

$$|u_1\rangle \cdot |u_1\rangle \, , |u_2\rangle \cdot |u_2\rangle \, , \text{ and } |u_1\rangle \cdot |u_2\rangle .$$

The only difference between two unit-length vectors $|u_1\rangle$ and $|u_2\rangle$ is their orientation. Since there is no preferred direction in the plane, we can only speak of the *mutual orientation* of the pair of vectors. A convenient measure of the mutual orientation is the *angle between* corresponding arrows. We are, therefore, led to the following candidate for the scalar product:

$$|u_1\rangle \cdot |u_2\rangle = f_\theta ,$$

where θ is the angle between $|u_1\rangle$ and $|u_2\rangle$, measured counterclockwise from $|u_1\rangle$ to $|u_2\rangle$. At this point, the only thing we can say about the function f is that it is continuous and periodic in its argument θ. Let's examine two simple candidates – trigonometric functions $\cos\theta$ and $\sin\theta$.

3.6.2 Scalar Product Meaning

Once the scalar product of two unit vectors is defined, we can ask what intuitive interpretation can be given to a scalar product in general. To this end, let's take a pair of normalized vectors $|u_1\rangle$ and $|u_2\rangle$ and scale them to an arbitrary lengths:

$$|u_1\rangle \rightarrow |a\rangle = a|u_1\rangle \text{ and } |u_2\rangle \rightarrow |b\rangle = b|u_2\rangle\,.$$

Scalar product of $|a\rangle$ and $|b\rangle$ is then simply

$$|a\rangle \cdot |b\rangle = abf_\theta\,.$$

The factor ab – the product of two lengths – suggests that the scalar product is related to area, as illustrated in Figure 3.10.

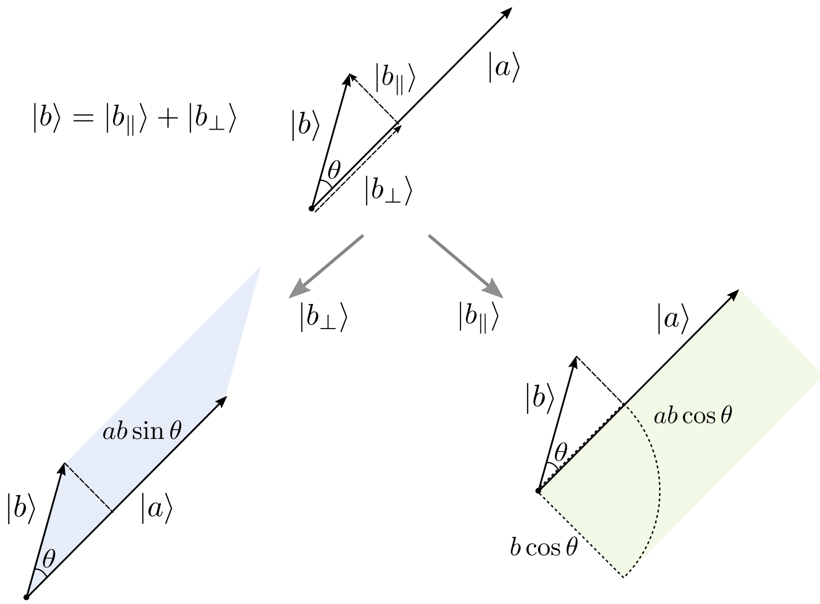


Fig. 3.10: Two way to interpret scalar product of two vectors.

If we note that the vector $|b\rangle$ can be written as the sum of two parts:

$$|b\rangle = |b_{||}\rangle + |b_\perp\rangle\,,$$

one parallel to the vector $|a\rangle$ and the other perpendicular to it, then two different interpretations can be given. First, if $f = \sin$, then the scalar product

$$|a\rangle \cdot |b\rangle = ab \sin\theta$$

would correspond to the area of a parallelogram built on two vectors $|a\rangle$ and $|b\rangle$ or the area of a rectangle built on two vectors $|a\rangle$ and $|b_\perp\rangle$. This approach is fruitful and it is used in a more advanced version of the product of two vectors, related to *tensor product* as discussed later.

The second simple option corresponds to $f = \cos$ and the scalar product

$$|a\rangle \cdot |b\rangle = ab \sin\theta \,.$$

It is the area of the rectangle built on two vectors $|a\rangle$ and $|b_\parallel\rangle$.

> ***Fidelity, Alignment, and Overlap***
> Scalar product of unit vectors can be viewed as the measure of their "alignment" or "overlap". This interpretation is useful in quantum theory, where vectors are used to represent states of quantum systems. Comparing two vectors means asking how similar or different two states are.

3.6.3 Orthonormal Bases

The choice of basis vectors is dictated by their usefulness in a given problem. In many cases it is convenient to use basis vectors which have unit length and which are *mutually orthogonal*:

$$|u_1\rangle \cdot |u_1\rangle = 1 = |u_2\rangle \cdot |u_2\rangle \text{ and } |u_1\rangle \cdot |u_2\rangle = 0 \,.$$

In this case scalar product of any two vectors

$$|a\rangle = a_1|u_1\rangle + a_2|u_2\rangle \text{ and } |b\rangle = b_1|u_1\rangle + b_2|u_2\rangle \,.$$

takes on a very simple form:

$$|a\rangle \cdot |b\rangle = a_1 b_1 + a_2 b_2 \,.$$

Although we arrived at this result analysing only two-dimensional case, it is easily generalized to any number of dimensions. For an n-dimensional case we would have the following expression for the scalar product

$$|a\rangle \cdot |b\rangle = \int a_i b_i \,, \quad i = \overline{1, n} \,.$$

3.7 Operators

Operators are functions of a particular kind. A "usual" function, like cos, *transforms* a number into another number, while an operator $\widehat{T}$ transforms a vector into another vector:

$$\widehat{T}\ |a\rangle = |b\rangle\,.$$

It is helpful to consider examples.

Example Operators

It is easy to come up with examples of operators:

- Unit operator (or *identity* operator), such that

$$\widehat{I}\ |a\rangle = |a\rangle\,.$$

- "Zeroing" operator that maps every vector into a zero vector:

$$\widehat{0}\ |a\rangle = |0\rangle\,.$$

- "Flipping" operator that changes a vector into its opposite: vector:

$$\widehat{F}\ |a\rangle = -|a\rangle\,.$$

- "Orthogonal" operator that makes any vector perpendicular to its original direction. In other words, it rotates the original vector by 90 degrees counter-clockwise, leaving the length the same:

$$\widehat{J}\ |a\rangle = |b\rangle\,,\ \text{such that}\ b = a\ \text{and}\ |a\rangle\cdot|b\rangle = 0\,.$$

  Here a and b are the lengths of the vectors.
- "Normalizing" operator that turns an arbitrary vector into a normalized vector with the same direction:

$$\widehat{N}\ |a\rangle = |a\rangle/a\,.$$

- Rotation operators that perform a rotation of a vector by a specified angle θ:

$$\widehat{R}_\theta\ |a\rangle = |b\rangle\,,\ \text{such that}\ a = b\,,\ \text{and}\ |a\rangle \cdot |b\rangle = a^2 \cos\theta\,.$$

Special cases of these operators are $\widehat{F}$=$\widehat{R}_\pi$, $\widehat{J}$=$\widehat{R}_{\pi/2}$ and $\widehat{I}$=$\widehat{R}_0$.

- Scaling operators that change the length of a given vector by a specified factor:

$$\widehat{S}_x\ |a\rangle = x|a\rangle\,.$$

Special cases of these operators are $\widehat{F}$=$\widehat{S}_{-1}$, $\widehat{I}$=$\widehat{S}_1$, and $\widehat{0}$=$\widehat{S}_0$.

3.7.1 Operator Algebra

Since operators are functions, we can manipulate them like functions. In particular, we can use them as objects in their own right, which can be added or composed. Indeed, if we can add vectors, we can give useful meaning to the sum of operators:

$$\widehat{T}=\widehat{I} + \widehat{J} + \widehat{S}_2\ \ .$$

To fully describe an operator, we must find how it acts *on any* vector. In this example we get

$$\widehat{T}\ |a\rangle = \widehat{I}\ |a\rangle + \widehat{J}\ |a\rangle + \widehat{S}_2\ |a\rangle = |a\rangle + |b\rangle + 2|a\rangle = 3|a\rangle + |b\rangle\,,$$

where the vector $|b\rangle$ is orthogonal to $|a\rangle$, while having the same length.

Operators can be multiplied by "usual" numbers, and they can be composed.

3.7.2 Linear Operators

Among the simplest non-trivial operators are *linear operators*. As mentioned earlier in section 3.2.5, linear functions satisfy two conditions:

$$\widehat{T}\ (|a\rangle + |b\rangle) = \widehat{T}\ |a\rangle + \widehat{T}\ |b\rangle\,,$$

and

$$\widehat{T}\ (x|a\rangle) = x\,\widehat{T}\ |a\rangle\,.$$

R Thus, operator action distributes over a sum, and numbers can be pulled outside.

To define a linear operator, we need only to describe how it affects basis vectors. Indeed, if an arbitrary vector $|a\rangle$ is represented in some basis, we can write

$$\widehat{T}\ |a\rangle = \widehat{T}\ (a_1|u_1\rangle + a_2|u_2\rangle) = \widehat{T}\ (a_1|u_1\rangle) + \widehat{T}\ (a_2|u_2\rangle)\,,$$

and then pull numbers outside to obtain:

$$\widehat{T}\ |a\rangle = a_1\,\widehat{T}\ |u_1\rangle + a_2\,\widehat{T}\ |u_2\rangle\,.$$

We only need to know $|t_1\rangle = \widehat{T}\ |u_1\rangle$ and $|t_2\rangle = \widehat{T}\ |u_2\rangle$, so that

$$\widehat{T}\ |a\rangle = a_1|t_1\rangle + a_2|t_2\rangle\,, \quad \widehat{T}\ |b\rangle = b_1|t_1\rangle + b_2|t_2\rangle\,, \text{ and so on.}$$

Now both $|t_1\rangle$ and $|t_2\rangle$ are vectors and can be represented in the same basis as $|a\rangle$:

$$|t_1\rangle = T_{11}|u_1\rangle + T_{12}|u_2\rangle\,,$$

and

$$|t_2\rangle = T_{21}|u_1\rangle + T_{22}|u_2\rangle\,.$$

The four numbers T_{11}, T_{12}, T_{21}, and T_{22} are called *components of an operator* in a given basis. It is crucial to remember that these components are specific to a basis. If we decide to change to a different basis, components of all vectors and operators will change.

Matrix

Components of a linear operator in a given basis are often written as square table, called *matrix*:

$$\widehat{T} = \begin{pmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{pmatrix}\,.$$

We emphasize again: If we change to a different basis, components of the operators will change, and the matrix will look different. Sometimes matrices are convenient for computations, but they are rarely used in this book.

■ **Example 3.1** Consider an operator that transforms the state $|0\rangle$ into a linear combination $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2})$ and the state $|1\rangle$ into a linear combination $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2})$. It is called *Hadamard* operator and has the following matrix representation.

$$\widehat{H} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}.$$

■

Numbers On Steroids

Operators can be used to solve problems that do not have solutions in terms of real numbers. For example:

$$\widehat{A} + \widehat{B} = 6\,\widehat{I} \quad \text{and} \quad \widehat{A}\widehat{B} = 36\,\widehat{I}\,.$$

To simplify this problem we can first re-scale the operators, introducing

$$\widehat{a} = \widehat{A}\,/6 \quad \text{and} \quad \widehat{b} = \widehat{B}\,/6\,.$$

Now the problem becomes a bit simpler:

$$\widehat{a} + \widehat{b} = \widehat{I} \quad \text{and} \quad \widehat{a}\widehat{b} = \widehat{I}\,.$$

We are looking for two operations with arrows in a plane such that two successive operations do not change any arrow (the second equation). Since an action of any operator on arrow rotates it and changes the length of the latter, the operator $\widehat{a}$ should cancel rotation and scaling done by $\widehat{b}$. For example, if $\widehat{b}$ scales arrow by a number b and rotates by an angle β, the operator $\widehat{a}$ scales arrows by $a = 1/b$ and performs rotation by $-\beta$ – same magnitude but in the opposite direction.

Exercise 3.11

Show that $\widehat{a}$ and $\widehat{b}$ only rotate any arrow by an angle of $60°$.

Dirac Equation

Paul Dirac discovered an equation that describes electron moving at any speed $v < c$ where c is the speed of light. To make his equation work, he had to perform "linearization" of an older equation from quantum mechanics – an operation that made his equation rely on linear operators instead of numeric coefficients. In short, his equation looked like this:

$$(\widehat{A}\, p + \widehat{B}\, mc)|\psi\rangle = |\phi\rangle\,.$$

Here p is the momentum of an electron, m is its mass. The exact meaning of $|\psi\rangle$ and $|\phi\rangle$ will be explained once we get to quantum theory in section 5.3. For now it is important to appreciate the fact that Dirac's approached worked successfully once he switched from simple numbers to operators $a \to \widehat{A}$ and $b \to \widehat{B}$.

3.7.3 Super-operators*

An idea of a function is quite general, it implies mapping one value to another, or calculating result given a certain input. Given a number, a function can yield another number, or given a vector a function can produce another vector. In the latter case we call the function an *operator*.

In the mathematical tool-set of both classical and quantum physics there are functions of *higher order* in the sense that they can act on functions or operators. For example, a *super-operator* is a mapping from any operator $\widehat{T}$ into another operator. Although it sounds abstract, it is not a complicated idea. Indeed, all we need is a rule that finds some operator $\widehat{B}$ given an operator $\widehat{A}$. One non-trivial rule can be stated as follows: Given an operator $\widehat{A}$, apply it twice:

$$\widehat{A} \quad \xrightarrow{\mathcal{F}} \quad \widehat{B} = \widehat{A} \circ \widehat{A}\ .$$

Since the composition of two operators is again an operator, this is a satisfactory definition.

3.8 Functionals

Another important type of function is called *functional*. A functional maps a function into a number. Let's consider several examples.

Total Mass

Suppose an astrophysicist is trying to model a spherically symmetric star and calculates *density* of the star as the function of distance from its center: $r \to \rho_r$. The total mass of the star can then be evaluated as the sum of masses of all spherical shells with thickness δr:

$$M = \int \delta V \rho_r = \int 4\pi r^2 \delta r \rho_r\,.$$

For a given function ρ_r this summation will result in a number – star's total mass. Such mapping $\rho_r \to M$ is an example of a functional.

Total Fuel

Consider a car moving on a straight highway between two points A and B. The amount of fuel the engine consumes at a given moment depends on the speed of the car at that moment and can be described by the function μ_v. Suppose the position of the car as the function of time x_t is known and are looking for the total fuel consumed during the travel. This can be done in three steps.

First, we find the speed of the car as the function of time by applying the operator ∂_t to x_t: $v_t = \partial_t x$. Second, we find the fuel consuption rate μ as the function of time by plugging v_t into μ_v: $f_t = \mu(v_t)$. Finally, we can find the total amount of consumed fuel as the sum

$$F = \int f_t \delta t \,.$$

Combining all three steps into a single mathematical expression will result in a more cumbersome formula:

$$F = \int \delta t \mu(\partial_t x) \,.$$

This formula encodes a recipe for mapping any function x_t into a number F – an example of a functional.

Total Action

A body in a "free fall" is moving with constant acceleration due to the force of gravity. Its speed increases as the body approaches the ground. If the body starts at rest at height H, its position along the vertical y axis depends on time as $y_t = H - gt^2/2$ and the velocity changes according to the equation $v = -gt$.

The potenital energy $E_p = mgy$ of the body decreases, while its kinetic energy $E_k = mv^2/2$ grows. The total mechanical energy $E = E_p + E_k$ remains fixed according to the law of energy conservation. Thus, the potential energy of the body is transformed into the kinetic energy.

Another physical quantity is often important – the *imbalance* of kinetic energy over the potential energy:

$$L = E_k - E_p \,.$$

It does not remain constant, and for the case of a free fall we can easily find its time dependence:

$$L_t = mg^2t^2 - mgH\,.$$

Given L_t, we can calculate a fundamental physical quantity – total *action* of the process:

$$A = \int \delta t L_t\,.$$

The summation extends to the moment $t = T$ when the body reaches the ground ($y = 0$). This happens at $T = \sqrt{2H/g}$.

Performing the summation requires evaluation of two familiar sums:

$$\int t^2\delta t = \frac{T^3}{3} \quad \text{and} \quad \int \delta t = T\,.$$

Substituting the values of T and simplifying, the expression for the total action takes the form

$$A = mgT(\frac{gT^2}{3} - H) = -\frac{mH}{3}\sqrt{2gH} = -\frac{mv_mH}{3}\,.$$

Here we used $v_m = gT = \sqrt{2Hg}$ – the maximal speed of the body at the end of the free fall process. Finally, denoting the maximum momentum of the body as $p_m = mv_m$, we obtain $A = -p_mH/3$. Note that the action can be expressed as the product of momentum and distance.

Action is a physical quantit of fundamental importance. It plays a prominent role in both classical mechanics (the principle of *stationary action*) and in quantum physics (the principle of *action quantization*). Both principles will be explored in details later in the book.

Exercise 3.12
Calculate the total action of a free fall process for an electron falling from the height 0.1 meter.

Assorted Examples

Examples of functionals given above involve evaluation of sums in order to find *total quantities* of various kinds:

$$Q = \int \delta x f_x\,.$$

The total quantity Q depends on the behavior of the input function f_x over an extended range of x values. Simpler forms of functionals can also be used. For example:

$$\mathcal{M}\, f = f_0$$

returns the value of the input function f_x at zero. This functional, despite its trivial look, is very useful and widely used in physics and mathematics. Its rigorous mathematical form is called *Dirac delta function* .

Dirac Delta Function

The idea of delta function is simple: it describes the density of mass (or charge, probability, and so on) for a point-like particle. Formally, such density can be written as δ_x.

Since the total mass (charge, probability) is finite, the summation of the density over the region where the particle might be must be a fixed number:

$$m = \int \delta x \delta_x \,.$$

Another example of a simple functional is the maximum of a function:

$$\mathcal{X}\, f = \max f_x \,.$$

Finally, one can map any function f_x into a number like so:

$$\mathcal{R}\, f = \frac{f_1}{1!} + \frac{f_{1/2}}{2!} + \frac{f_{1/3}}{3!} + \ldots + \frac{f_{1/n}}{n!} + \ldots \,.$$

For $f = \sin$ we obtain $\mathcal{R}\,\sin \approx 1.1479$.

Exercise 3.13

For the functionals $\mathcal{M}$, $\mathcal{X}$, and $\mathcal{R}$ check whether they are *linear.*

3.9 Spaces

In mathematics and physics the concept of *space* becomes more abstract, in comparison with the intuitive view of space as a "container for things." Space in mathematical sense is more akin to how people understand space in expressions like "space of ideas", "space of solutions", "color space", "design space", "parameter space", and so on.

First of all, space is a set of mathematical objects of similar nature. For example, all numbers, or all arrows considered as one single entity, can be viewed as space. Second, unlike a simple set, space has some *structure*, some *relations* between its elements. For example, there might some sense of direction, or distance, or simply the notion of continuity. Advanced spaces, used in mathematical physics, have quite a complex structure, with distances, angles, algebraic operations, and even with operations required for calculus.

We will be mainly using *vector spaces*, understood as rich collections of all vectors, endowed with addition and multiplication operations, as defined earlier.

3.10 Duality

Many mathematical concepts have very natural "companions", related in the spirit of mirror images or "yin-yang." In this kind of relationship both "companions" are on equal footing, neither concept is preferred, neither is basic while the other is derived. This situation is called *duality*.

Scalar product of vectors leads to two interesting and useful dualities. The first duality deals with vectors and results in *dual vectors*, while the second duality deals with operators and gives the notion of *adjoint operators*.

3.10.1 Dual Vectors

To arrive at dual vectors we can start with scalar product, written using a *prefix notation*:

$$[\cdot]\,|a\rangle\,|b\rangle = x\,.$$

Here we introduced a binary operator (function of two vector arguments) that maps a pair of vectors into a number.

The next step is to consider the operator $[\cdot]$ *partially applied* to the first argument only:

$$[\cdot]\,|a\rangle \bullet\ .$$

Although here we denoted "an empy slot" with a grey circle $\bullet$, it is unnecessary and can be dropped. Furthermore, we will use Dirac notation for the mathematical object that results from the partial application of scalar product, writing it as follows:

$$\langle a| = [\cdot]\,|a\rangle\,.$$

Such an object can be written for *every* vector:

$$\langle b| = [\cdot]\,|b\rangle\,, \quad \langle c| = [\cdot]\,|c\rangle\,, \qquad \text{and so on.}$$

A beautiful thing about this whole construction is that it gives us *new kinds of vectors.* Also, it teaches us a lesson that vectors can be viewed as functions, revealing a connection between the two concepts.

It is easy to demonstrate that $\langle a|$ is a linear function. Recall that a function with two arguments, when partially applied, becomes a function of a single argument. Therefore, since $[\cdot]$ is the function of two vector arguments, $\langle a|$ must be the function of just one vector argument. Moreover, since $[\cdot]$ is linear in both arguments, $\langle a|$ must be linear in its single argument. More explicitely:

$$\langle a|\,(|b\rangle + |c\rangle) = \langle a||b\rangle + \langle a||c\rangle\,.$$

To check this equality, we first replace $\langle a|$ with its definition, then switch to the infix notation, and use the distributivity of scalar product.

Exercise 3.14

Prove the equalities:

$$\langle a|\,(|b\rangle + |c\rangle) = \langle a||b\rangle + \langle a||c\rangle\,,$$

and

$$\langle a|\,(x|b\rangle) = x\langle a||b\rangle\,.$$

Bra, Ket, and Braket

In Dirac notation the application of $\langle a|$ to $|b\rangle$ is written in a less noisy way:

$$\langle a|b\rangle\,.$$

The whole expression is called *bracket*, whereas $\langle a|$ is called *bra vector*, and $|b\rangle$ is called *ket vector.*

Now how can we show that the collection of objects $\langle a|$, $\langle b|$, $\langle c|$ and so on forms a vector space? To be a vector an object must behave like one, demonstrating all properties that fully define vector. We will examine whether bra-vectors can be added, and whether they can be expanded in some basis.

Can $\langle a|$ and $\langle b|$ be added? Of course they can, because they are functions over "addable things" (over vectors), and the meaning of $\langle v| = \langle a| + \langle b|$ is simple:

$$\langle v|c\rangle = \langle a|c\rangle + \langle b|c\rangle \,.$$

For similar reasons, any $\langle a|$ can be multiplied by "usual" numbers, so that the expressions like

$$\langle w| = x\langle a| + y\langle b|$$

have a straightforward interpretation.

Since there is exists a basis in the ket-vector space, there is one in the bra-space. Indeed, suppose we have $|a\rangle = a_1|u_1\rangle + a_2|u_2\rangle$ and we use the linearity of the scalar product in its first argument:

$$\langle a| = [\cdot]\,(a_1|u_1\rangle + a_2|u_2\rangle) = a_1([\cdot]\,|u_1\rangle) + a_2([\cdot]\,|u_2\rangle) \,.$$

Since

$$[\cdot]\,|u_1\rangle = \langle u_1| \,, \qquad \text{and} \qquad [\cdot]\,|u_2\rangle = \langle u_2| \,,$$

we conclude that any bra-vector can be expanded in terms of some other bra-vectors, just like ket-vectors are expanded in some basis:

$$\langle a| = a_1\langle u_1| + a_2\langle u_2| \,.$$

In summary, if we need a bra-basis, we can use ket-basis and partially apply the scalar product operator to each ket-basis vector.

One last step is needed to rigirously prove that both ket-vectors and bra-vectors are indeed vectors: we must investigate how their components change when we change bases. The only challenge this task presents is its tediousness[8].

From now on we will recognize that to every ket-vector $|a\rangle$ there corresponds a *dual* bra-vector $\langle a|$. We must remember that this duality is generated by the operation of scalar product. Once a vector space has scalar product defined, we get a dual vector space "for free." Such dual vector space turns out extremely useful in quantum theory, as we will soon see.

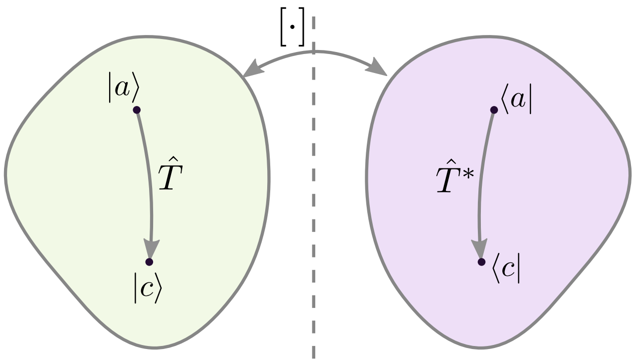


Fig. 3.11: Duality generated by the scalar product applies both the vector spaces and operators acting on them.

3.10.2 Dual Operators

We have shown that each vector $|a\rangle$ has dual $\langle a|$, and the whole space of ket-vectors has a dual counterpart, as shown in Figure 3.11. Given an operator $\widehat{T}$ acting on ket-space, we can define an operator $\widehat{T}^*$ acting on bra-space. The action of $\widehat{T}^*$ on an arbitrary bra-vector $\langle a|$ is defined as follows: we first transform bra-vector $\langle a|$ into its dual ket-vector $|a\rangle$ and find $\widehat{T}\,|a\rangle = |c\rangle$, and finally find its dual $\langle c|$, by partially applying scalar product:

$$\langle c| = (\widehat{T}\,|a\rangle) \cdot \bullet \;.$$

This procedure maps any $\langle a|$ into some $\langle c|$, as required. Let's examine how this bra-vector $\langle c|$ acts on an arbitrary ket-vector $|b\rangle$:

$$\langle c|b\rangle = (\widehat{T}\,|a\rangle) \cdot |b\rangle \;.$$

Postfix Notation

The action of an operator $\widehat{T}$ on a ket-vector $|a\rangle$ is nearly always written using *prefix notation*:

$$\widehat{T}\,|a\rangle = |c\rangle \;.$$

To emphasize the distinction between dual vector spaces, the application of the dual operator $\widehat{T}^*$ on the dual vector $\langle a|$ follows *postfix*

[8] See, for instance, *Tensors For Inquiring Minds* by Yury Deshko.

notation:

$$\langle a|\, \widehat{T}^* = \langle c|\, .$$

The use of prefix-postfix notations highlights the "mirror-like" relationship between the dual objects:

$$\langle a|\, \widehat{T}^* \;\Big|\; \widehat{T}\, |a\rangle$$

Conjugation

Switching between dual objects – vectors or operators – is an important and often used operation in quantum theory. This operation is called *conjugation* and is often denoted by an asterisk:

$$|a\rangle \quad \xrightarrow{*} \quad \langle a|\, ,$$

$$\widehat{T} \quad \xrightarrow{*} \quad \widehat{T}^*\, ,$$

$$\widehat{T}\, |a\rangle \quad \xrightarrow{*} \quad \langle a|\, \widehat{T}^* \quad .$$

Another, more traditional, way to express these relations is as follows:

$$\langle a| = (|a\rangle)^*\, , \qquad \text{and} \qquad \langle a|\, \widehat{T}^* = (\widehat{T}\, |a\rangle)^*\, .$$

3.10.3 Adjoint Operators

The notion of an *adjoint* operator is very important in quantum theory. Adjoint operators replace the usual real numbers used as the values of physical quantities, like position, momentum, energy, angular momentum, to name a few. Before give a definition of an adjoint operator, we'll explore the context in which adjoint operator is used.

To begin, recall that a linear operator transforms an input vector into an output vector. For some operators such transformation is always *reversible*: There exist another operator that can "cancel" the action of the first, as illustrated in Figure 3.12(a), where the operator $\widehat{U}$ acts as the *inverse* of the operator $\widehat{T}$:

$$\widehat{U} \circ \widehat{T} = \widehat{I} \quad .$$

The inverse of an operator, if there exists one, is denoted using the negative power: $\widehat{U}$=$\widehat{T}^{-1}$. Not every operator has an inverse, with $\widehat{0}$ ("zeroing" operator) and $\widehat{N}$ (normalizing operator) providing simple examples.

Scalar product leads to more sophisticated relations between operators. One such relation, shown in Figure 3.12(b), amounts to one operator being "inverse relative to the second input" of the scalar product. In simpler terms: *Acting on both inputs changes nothing*. This can be expressed in the following equation:

$$|a\rangle \cdot |b\rangle = (\widehat{T}\,|a\rangle) \cdot (\widehat{U}\,|b\rangle)\,.$$

If this equality holds true for *any* pair of vectors $|a\rangle$ and $|b\rangle$, then we will call the operator $\widehat{T}$ *scalar-inverse* of the operator $\widehat{U}$:

$$\widehat{U} \overset{[\cdot]}{=\!=} \widehat{T}^{-1}\ .$$

All rotation operators $\widehat{R}_\theta$ are scalar-inverse of themselves:

$$|a\rangle \cdot |b\rangle = (\widehat{R}_\theta\,|a\rangle) \cdot (\widehat{R}_\theta\,|b\rangle)\,.$$

Indeed, rotating both vectors by the same angle, does not change their mutual orientation and does not affect their lengths, keeping the product $ab\cos\theta$ constant.

Exercise 3.15
Find the scalar-inverse of the scaling operator $\widehat{S}_x$.

One more useful relation is generated by the scalar product. As illustrated in Figure 3.12(c), this relation means that one operator, say $\widehat{T}$, achieves the same affect as the other operator (e.g., $\widehat{U}$), while acting on the second input of the binary scalar product operator $[\cdot]$. Symbolically we can write:

$$|a\rangle \cdot (\widehat{U}\,|b\rangle) = (\widehat{T}\,|a\rangle) \cdot |b\rangle\,.$$

If this is true for all possible vectors, then we will call the operator $\widehat{T}$ *scalar-equivalent* to the operator $\widehat{U}$.

$$\widehat{U} \overset{[\cdot]}{=\!=} \widehat{T}\ .$$

A conventional name for the scalar-equivalent operator is *adjoint*. From the definition given above it is clear that both the original operator

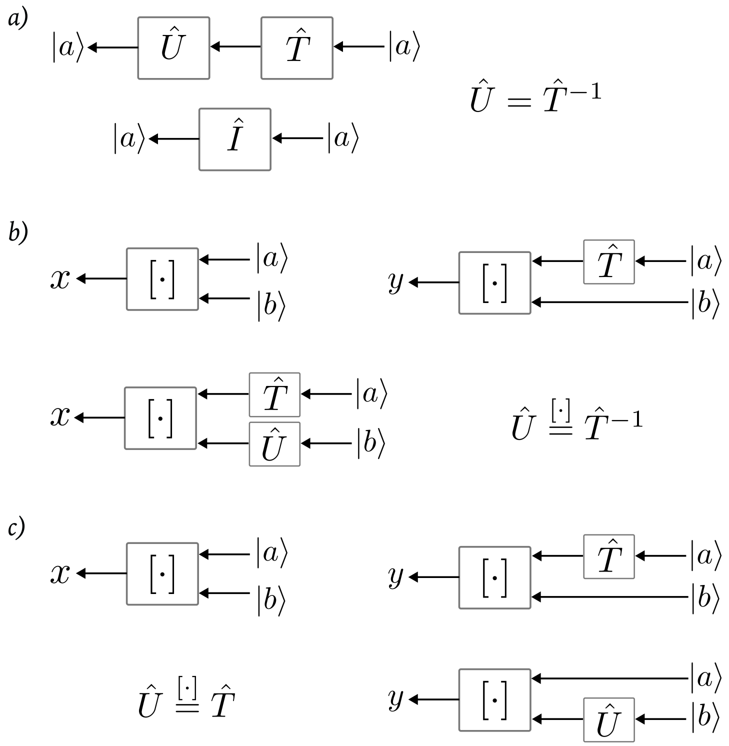


Fig. 3.12: Some operators have "relatives" which have certain interesting properties.

$\widehat{U}$ and its adjoint $\widehat{T}$ act on the same ket-vector space. The adjoint of the operator $\widehat{U}$ is often denoted using "*dagger*"-notation:

$$\widehat{T}=\widehat{U}^{\dagger} \;.$$

Definition 3.6 ☞ ***Adjoint Operator***
An adjoint of the operator $\widehat{U}$ acting on ket-space, is another operator, denoted as $\widehat{U}^{\dagger}$, acting on the same ket-space, and satisfying the following requirement:

$$|a\rangle \cdot (\widehat{U}\,|b\rangle) = (\widehat{U}^{\dagger}\,|a\rangle) \cdot |b\rangle$$

for all pairs of ket-vectors.

All scaling operators $\widehat{S}_x$ are scalar-equivalent to themselves:

$$|a\rangle \cdot (\widehat{S}_x\,|b\rangle) = (\widehat{S}_x\,|a\rangle) \cdot |b\rangle \,.$$

Indeed, scaling the first vector by some factor x has the same effect on the product $ab\cos\theta$ as scaling the second vector by the same factor.

Exercise 3.16
Find the scalar-equivalent of the rotation operator $\widehat{R}_\theta$.

Interestingly, finding the adjoint of a given operator is simpler than finding its inverse. This is most easily done using components:

$$U_{ij} = |u_i\rangle \cdot (\widehat{U}\,|u_j\rangle) = (\widehat{U}^{\dagger}\,|u_i\rangle) \cdot |u_j\rangle \,,$$

now we can use the commutativity of the scalar product

$$U_{ij} = |u_j\rangle \cdot (\widehat{U}^{\dagger}\,|u_i\rangle) = U^{\dagger}_{ji} \,.$$

Thus, the components of the adjoint $U^{\dagger}_{ji}$ are obtained from the components of the operator U_{ij} by "swapping" indices. For example:

$$U^{\dagger}_{11} = U_{11}\,, U^{\dagger}_{22} = U_{22}\,, \qquad \text{and so on, while}$$

$$U_{12}^{\dagger} = U_{21}\,, U_{23}^{\dagger} = U_{32}\,, \qquad \text{and so on.}$$

Transposition

The swapping of components $T_{ij} \to T_{ji}$ is called *transposition*, and the resultant operator whose components are T_{ji} is called *transposed* relative to the original operator $\widehat{T}$.

The concepts of conjugation and adjoint operator, discussed above, have interesting and powerful representations in terms of *tensor product* – the topic of the next section.

3.11 Tensor Product

A pair of vectors can be combined in a variety of ways, yielding results of different kinds, see Figure 3.13. Two familiar operations of vector addition and scalar product are close analogs of numeric addition and multiplication, respectively. Addition "glues" two vectors into the third vector, scalar product "fuses" two vectors into a number. There exists one more method of "bundling" vectors which results not in a number or yet another vector, but in a more advanced mathematical objects called *tensor*. Tensors are closely related to operators and we will focus on this aspect of tensor product.

We start with a pair of ket-vectors $|a\rangle$ and $|b\rangle$ and try "bundling" them in a way that resembles multiplication. Denoting this new operation $\otimes$, we want to have

$$|a\rangle \otimes (|b\rangle + |c\rangle) = |a\rangle \otimes |b\rangle + |a\rangle \otimes |c\rangle\,,$$

and

$$(x|a\rangle) \otimes (y|b\rangle) = xy|a\rangle \otimes |b\rangle\,.$$

The last requirement indicates that this new type of product has the nature of an area, since it is proportional to the product of the lengths of two vectors. Thus, it can't be another vector, but must be a construction of different kind that we will call *tensor*, and the product $|a\rangle \otimes |b\rangle$ – a *tensor product.*

Using the arrow model of vectors, we can consider as a possible candidate a "blade" – a piece of plane swept by "sliding" the vector $|a\rangle$ along the vector $|b\rangle$, as illustrated in Figure 3.13(c). This geometric object

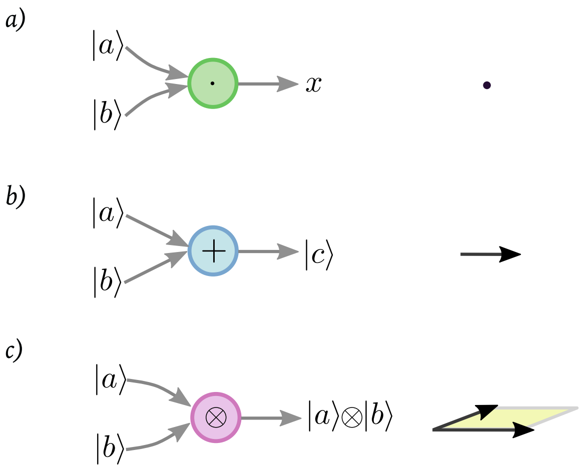


Fig. 3.13: Three basic operation with vectors: a) Scalar product; b) vector sum; c) tensor product.

has "magnitude" or size, defined by the area of the blade, and it has direction or orientation. This approach is used in geometric algebra, where the *outer product* is defined along these lines.

The geometric representation of product $|a\rangle \otimes |b\rangle$ might be appealing, but not at all required. Furthermore, it might even be unhelpful when we would like to generalize the operation to the product of other vector types:

$$|a\rangle \otimes \langle b| \,, \quad \langle a| \otimes |b\rangle \,, \quad \text{or} \quad \langle a| \otimes \langle a| \,.$$

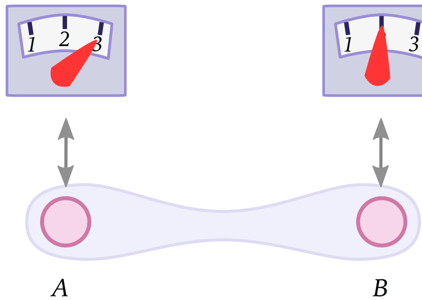
In other words, we should not constrain ourselves by reliance on a geometric intuition. How then do we interpret these various products? Well, one simple *operational* criterion helps: A mathematical object is defined by *what it does* and *how it is used.* Let's study the two most important cases of tensor products.

3.11.1 Ket-Ket

The first tensor product is made of two ket-vectors:

$$\widehat{P} = |a\rangle \otimes |b\rangle \,.$$

In quantum theory, where ket-vectors "store" the information of all possible measurement (*state* of a quantum system), the product is used to describe a *joint system*, in which two distinct parts, say A and B, can be identified and measured by different measuring devices. The situation is similar to the one described earlier in section 3.1.4.



$$|a\rangle = a_1|1\rangle + a_2|2\rangle + a_3|3\rangle \qquad |b\rangle = b_1|1\rangle + b_2|2\rangle + b_3|3\rangle$$

Fig. 3.14: Joint measurement of two systems.

Let's study an example, shown in Figure 3.14. Imagine that we measure the value of some discrete quantity (quantized energy of an atom, for instance) using identical devices in different locations, corresponding to the systems A and B. Suppose that in each trial the value of energy is measured to be either 1, or 2, or 3 *randomly*. Then, after many trials of measuring the energy of the system A, we may summarize the statistics as follows:

$$|a\rangle = a_1|1\rangle + a_2|2\rangle + a_3|3\rangle\,,$$

where $a_i = N_i/N$ is the *probability* to obtain the result i, expressed as the ratio of the number of times N_i the value i appeared in a series of N measurements. Similarly, after many trials of measuring the energy of the system B, we may summarize the statistics as an expression:

$$|b\rangle = b_1|1\rangle + b_2|2\rangle + b_3|3\rangle\,,$$

with the similar meaning of the coefficients b_i.

Now we can write the summary of observation of two systems joined together:

$$\widehat{P}= |a\rangle \otimes |b\rangle = a_1 b_1|1\rangle \otimes |1\rangle + a_1 b_2|1\rangle \otimes |2\rangle + \ldots a_3 b_3|3\rangle \otimes |3\rangle\,.$$

From the probability theory it is known (see section 3.1.4) that the numbers $p_{ij} = a_i b_j$ give the probability to measure the value $|i\rangle$ in the experiment on the system A, while getting the value $|j\rangle$ in the measurement performed on B.

It might seem like unnecessary complication to use the tensor product expressions like

$$\widehat{P} = \int p_{ij} |i\rangle \otimes |j\rangle$$

instead of keeping separate vectors $|a\rangle$ and $|b\rangle$. But if we try to use individual vectors $|a\rangle$ and $|b\rangle$ when the results of joint measurements have the following statistics:

$$\widehat{P}_0 = \frac{1}{2}|1\rangle \otimes |1\rangle + \frac{1}{2}|2\rangle \otimes |2\rangle \,,$$

we discover that no $|a\rangle$ and $|b\rangle$ satisfy the requirement $\widehat{P}_0 = |a\rangle \otimes |b\rangle$.

Exercise 3.17
Prove that the assumption

$$\widehat{P}_0 = |a\rangle \otimes |b\rangle$$

leads to contraditions and, therefore, can not be true.

The information encoded in the tensor product $\widehat{P}_0$ is an example of *strongly correlated* behavior. In this example, it says that whenever system A is found (randomly!) with the energy 1, the same energy is measured on the system B; similarly for the energy 2. Effects like these are of great interest in quantum theory, and the operation of tensor product is indispensable for describing such phenomena.

3.11.2 Ket-Bra

The second type of tensor product involves one ket-vector and one bra-vector, "bundled" into a new object which behaves like an *operator*, unlike braket used in scalar product. Given vectors $|a\rangle$ and $\langle b|$, we can form a tensor product

$$\widehat{P} = |a\rangle \otimes \langle b| \,.$$

Economical Notation
Instead of writing $|a\rangle \otimes |b\rangle$ we will often write $|a\rangle|b\rangle$. Instead of $|a\rangle \otimes \langle b|$ we will often write $|a\rangle\langle b|$.

What does $\widehat{P}$ do? It can be used as an operator, acting on ket-vectors as follows:

$$\widehat{P}\,|c\rangle = |a\rangle \otimes \langle b||c\rangle = |a\rangle\langle b|c\rangle = x|a\rangle\,,$$

where $x = \langle b|c\rangle$ is a number. The operator $|a\rangle \otimes \langle b|$ transforms any input vector into a vector parallel to $|a\rangle$. In other words, it *projects* all vectors onto the direction specified by $|a\rangle$. Operators of this kind are sometimes called *projectors*, but we will reserve this name for the operators of special form $|a\rangle \otimes \langle a|$.

Exercise 3.18
Suppose $|a\rangle \perp |b\rangle$ and $\widehat{P}= |a\rangle \otimes \langle b|$. Find the result of $\widehat{P}\,|a\rangle$.

Definition 3.7 ***Projector***
A tensor product $|a\rangle \otimes \langle a|$ is called *projector*. It projects all vectors onto the direction specified by the vectors $|a\rangle$, while only scaling $|a\rangle$:

$$|a\rangle \otimes \langle a||a\rangle = a^2|a\rangle\,.$$

In quantum theory ket-vectors represent the state of a quantum system – the information about all possible measurement results. Projectors like $|a\rangle \otimes \langle b|$ describe the change of state $|b\rangle \to |a\rangle$, or *transition* between states.

Another application of tensor products in quantum theory is representation of *operators* in a choses basis. To see this, recall that every linear operator is fully specified once we know how it transforms basis vectors:

$$\widehat{T}\,|a\rangle = \widehat{T}\,(a_1|u_1\rangle + a_2|u_2\rangle) = a_1(\widehat{T}\,|u_1\rangle) + a_2(\widehat{T}\,|u_2\rangle)\,.$$

If $\widehat{T}\,|u_1\rangle = |t_1\rangle$, then we can express it using tensor product $\widehat{P}_1= |t_1\rangle \otimes \langle u_1|$. The transormation of $|u_2\rangle$ can be written similarly: $\widehat{P}_2= |t_2\rangle \otimes \langle u_2|$. Since $\widehat{P}_1$ and $\widehat{P}_2$ are both operators, they can be multiplied by numbers and added to obtain another operator. The action of $\widehat{T}$ then becomes expressed as

$$\widehat{T}\ |a\rangle = a_1\ \widehat{P}_1\ |u_1\rangle + a_2\ \widehat{P}_2\ |u_2\rangle = \widehat{P}_1\ (a_1|u_1\rangle) + \widehat{P}_2\ (a_2|u_2\rangle)\,.$$

Now since

$$\widehat{P}_1\ (a_1|u_1\rangle) = \widehat{P}_1\ (a_1|u_1\rangle + a_2|u_2\rangle) = \widehat{P}_1\ |a\rangle\,,$$

and

$$\widehat{P}_2\ (a_2|u_2\rangle) = \widehat{P}_2\ (a_1|u_1\rangle + a_2|u_2\rangle) = \widehat{P}_2\ |a\rangle\,,$$

because of the orthogonality of the basis. For example:

$$\widehat{P}_1\ (a_2|u_2\rangle) = a_2|t_1\rangle \otimes \langle u_1||u_2\rangle = 0\,.$$

The action of the operator $\widehat{T}$ becomes

$$\widehat{T}\ |a\rangle = \widehat{P}_1\ |a\rangle + \widehat{P}_2\ |a\rangle = \left(\widehat{P}_1 + \widehat{P}_2\right)|a\rangle\,.$$

Now we can drop the input ket-vector, writing the operator in terms of the ket-bra tensor products:

$$\widehat{T} = |t_1\rangle \otimes \langle u_1| + |t_2\rangle \otimes \langle u_2|\,.$$

We can leave the expression as is or convert it into tensor product of only basis vectors. Recalling that

$$|t_1\rangle = T_{11}|u_1\rangle + T_{12}|u_2\rangle\,, \qquad \text{and} \qquad |t_2\rangle = T_{21}|u_1\rangle + T_{22}|u_2\rangle\,,$$

and using the linearity of the tensor product, we arrive at the following exression:

$$\widehat{T} = T_{11}|u_1\rangle\langle u_1| + T_{12}|u_1\rangle\langle u_2| + T_{21}|u_2\rangle\langle u_1| + T_{22}|u_2\rangle\langle u_2|\,.$$

Here we used the *economical notation* for tensor product. It can be further compactified and generalized to any number of dimensions:

$$\widehat{T} = \int T_{ij}|u_i\rangle\langle u_j|\,, \quad i,j = \overline{1,n}\,.$$

3.11.3 Useful Ket-Bras

A unit operator $\widehat{I}$ plays an important role in many calculations. It can be represented as

$$\widehat{I} = \int |u_i\rangle\langle u_i|\,, \quad i = \overline{1,n}\,.$$

It is easy to check this for $n = 2$ explicitely:

$$(|u_1\rangle\langle u_1| + |u_1\rangle\langle u_1|)|a\rangle = |u_1\rangle\langle u_1|a\rangle + |u_2\rangle\langle u_2|a\rangle\,.$$

The right-hand side is clearly the expansion of the ket-vector $|a\rangle$ in the basis $|u_i\rangle$:

$$|u_1\rangle\langle u_1|a\rangle + |u_2\rangle\langle u_2|a\rangle = a_1|u_1\rangle + a_2|u_2\rangle = |a\rangle\,.$$

Completeness

The expression

$$\widehat{I} = \int |u_i\rangle\langle u_i|\,, \quad i = \overline{1,n}$$

is called the *completeness* statement of the set of basis vectors. It says that the set of orthonormal vectors is *not missing* any vector necessary to construct all other vectors in the space. Indeed, if we omit even a single term $|u_k\rangle\langle u_k|$ in the sum $\int |u_i\rangle\langle u_i|$ then the term $a_k|u_k\rangle$ will be missing in the expansion

$$\left(\int |u_i\rangle\langle u_i|\right)|a\rangle = a_1|u_1\rangle + \ldots + a_{k-1}|u_{k-1}\rangle + a_{k+1}|u_{k+1}\rangle + \ldots a_n|u_n\rangle\,.$$

The operator $\widehat{J}$ that rotates vectors in a plane by 90 degrees counter-clockwise takes the basis ket-vector $|u_1\rangle$ into $|u_2\rangle$ and $|u_2\rangle$ into $(-|u_1\rangle)$. Using tensor products $\widehat{J}$ can be written as follows:

$$\widehat{J} = |u_2\rangle\langle u_1| - |u_1\rangle\langle u_2|\,.$$

The following exercise demonstrates how easy it is to find and adjoint of an operator in ket-bra form.

Exercise 3.19

Given an operator $\widehat{T} = |c\rangle\langle d|$, show that its adjoint is $\widehat{T}^\dagger = |d\rangle\langle c|$. Thus, there is a simple rule:

$$(|c\rangle\langle d|)^\dagger = |d\rangle\langle c| .$$

From the previous exercise follows that every projector operator $\widehat{\Pi} = |a\rangle\langle a|$ is its own adjoint:

$$\widehat{\Pi}^\dagger = \widehat{\Pi} .$$

Such operators are called *self-adjoint*, they play an important role in quantum theory, representing measurements of various physical variables, such as position, momentum, energy, or angular momentum and others.

Hermitian Operators and Polynomials

In quantum theory self-adjoint operators are called *Hermitian operators*, after French mathematician Charles Hermite (1822 – 1901). Thus, in quantum theory, *Hermitian operator* has the property

$$\widehat{T} = \widehat{T}^\dagger .$$

Among several mathematical concepts named after Hermite, there are *Hermite polynomials*, which satisfy a simple recurrent relation:

$$\mathrm{H}_{n+1} = 2x\mathrm{H}_n - 2n\mathrm{H}_{n-1} ,$$

with the first two polynomials being $H_0 = 1$, $H_1 = 2x$. Hermite polynomials appear in the analysis of *quantum harmonic oscillator.*

Although Charles Hermit wrote about these polynomials in 1864, they were properly studied five years earlier by a Russian mathematician Pafnuty Chebyshev (or Tchebycheff), who also has polynomials named after him.

Exercise 3.20

Write Hadamard operator in terms of the tensor product of vectors $|0\rangle$, $|1\rangle$, $|+\rangle$ and $|-\rangle$.

$$\widehat{H} = |+\rangle\langle 0| + |-\rangle\langle 1| \,.$$

R Conjugation rule:

$$|a\rangle\langle b| \overset{\dagger}{\longleftrightarrow} |b\rangle\langle a| \,.$$

3.11.4 Tensor Product of Operators

The idea of "bundling" can be extended to operators. Thus, given the operators $\widehat{A}$ and $\widehat{B}$, we can write their tensor product:

$$\widehat{C} = \widehat{A} \otimes \widehat{B} \;\; .$$

Such an operator expresses an action (measurement or interaction) on two parts of a *composite* quantum system, e.g. two atoms in a molecule or two "beams of light" in a single experiment. To illustrate further, imagine we are studying two atoms, described by the state vectors $|a\rangle$ and $|b\rangle$, respectively. As explained above, the system of two independent atoms as one whole can be described using tensor product:

$$|\psi\rangle = |a\rangle \otimes |b\rangle \,.$$

Now if we perform some operation $\widehat{A}$ on the first atom, and some operation $\widehat{B}$ on the second atom in parallel, we can express this as follows:

$$\widehat{C}\,|\psi\rangle = \left(\widehat{A} \otimes \widehat{B}\right)\left(|a\rangle \otimes |b\rangle\right) \,.$$

Since the atoms are assumed independent in this example, with the joint state being a tensor product, and the actions on each system is performed independently of each other, we will have the following final states of each system:

$$\widehat{a} \quad \to \quad \widehat{A}\,|a\rangle \,,$$

and

$$\widehat{b} \quad \to \quad \widehat{B}\,|b\rangle \,.$$

Therefore, the final state of the composite system becomes:

$$|\phi\rangle = \left(\widehat{A}\,|a\rangle\right) \otimes \left(\widehat{B}\,|b\rangle\right) \,.$$

We can summarize it all as a rule:

$$\widehat{A} \otimes \widehat{B}\ (|a\rangle \otimes |b\rangle) = \widehat{A}\ |a\rangle \otimes\ \widehat{B}\ |b\rangle\ .$$

To keep things simple we analyzed the case of two operators and two vectors, but the ideas can be generalized to any number of factors. We will do that when necessary.

3.12 Functions As Vectors

Perhaps surprisingly, certain classes of functions, which are very useful in physics, have all characteristic of vectors. As we showed, familiar numeric functions can be treated like numbers: they can be added and multiplied yielding new function. For example, we can write

$$f = 2\cos + \frac{1}{3}\sin\ .$$

(Reminder: This is an argument-free notation. When applied to a specific value x, this functions would return $f_x = 2\cos x + \frac{1}{3}\sin x$.)

If trigonometric functions $\cos$ and $\sin$ acted as "basis", then this expression could be viewed as a vector:

$$|f\rangle = 2|\cos\rangle + \frac{1}{3}|\sin\rangle\ .$$

More generally, to convince ourselves that functions (at least some classes of functions) can be viewed as vectors, we need to find some such "basis" and find a way to express functions in the form:

$$|f\rangle = c_1|b_1\rangle + c_2|b_2\rangle + \ldots + c_n|b_n\rangle\ .$$

Turns out, in many cases it is possible. We will consider a couple of most important and relatively simple examples: polynomials and trigonometric functions. To illustrate the ideas we will use a moderately complicated function

$$f_x = (1 + x)e^{-x^3 \sin x}\ ,$$

built from product and composition of polynomials, trigonometric, and exponential functions. The graph of f_x is plotted in Figure 3.15 using open circles. Two different ways to approximate this funciton in the range $[-2, 2]$ is demonstrated in parts (a) and (b). Let's take a closer look.

3.12.1 Taylor Series

In a finite range of interest, it possible to represent a function using polynomials. The function f_x defined above, behaves almost like the following polynomial of the 13th degree:

$$\begin{aligned} P_x \quad &= 1 + x + 0.107x^2 + 0.156x^3 - 1.565x^4 - 1.669x^5 + \\ &+1.27x^6 + 1.362x^7 - 0.444x^8 - 0.484x^9 + 0.74x^{10} + \\ &\qquad +0.082x^{11} - 0.005x^{12} - 0.005x^{13}\,. \end{aligned}$$

Polynomial approximation works well for all functions which are "well-behaved" (smooth and not blowing up to infinity) in a finite range of interest. The smaller the range, the simpler the required polynomial becomes. This idea is illustrated in Figure 3.16. At the highest zoom level any function looks like a piece of straight line, described by a first degree polynomial $a_0 + a_1 x$. Expanding the range (zooming out), the function first starts to resemble a parabola, then cubic, and so on.

It can be shown that given a function f_x, it can be approximated around some point of interest $x = a$ using a simple series:

$$f_x \approx c_0 + c_1(x-a) + c_2(x-a)^2 + \ldots + c_n(x-a)^n$$

or more compactly

$$f_x \approx \int c_i(x-a)^i\,, \quad i = \overline{1,n}\,.$$

This series, first introduced by an English mathematician Brook Taylor in 1715, is called *Taylor series*.

As the result, the information about a function f becomes "encoded" in a list of coefficients of the polynomial:

$$f \quad \leftrightarrow \quad (c_0, c_1, c_2, \ldots, c_n)\,.$$

The function can then be written similar to a vector with components c_i and some basis $|u_i\rangle$ (polynomials in this case):

$$|f\rangle = \int c_i |u_i\rangle\,, \quad i = \overline{1,n}\,.$$

Like in the case of arrow-vectors, the basis is not unique. Let's examine

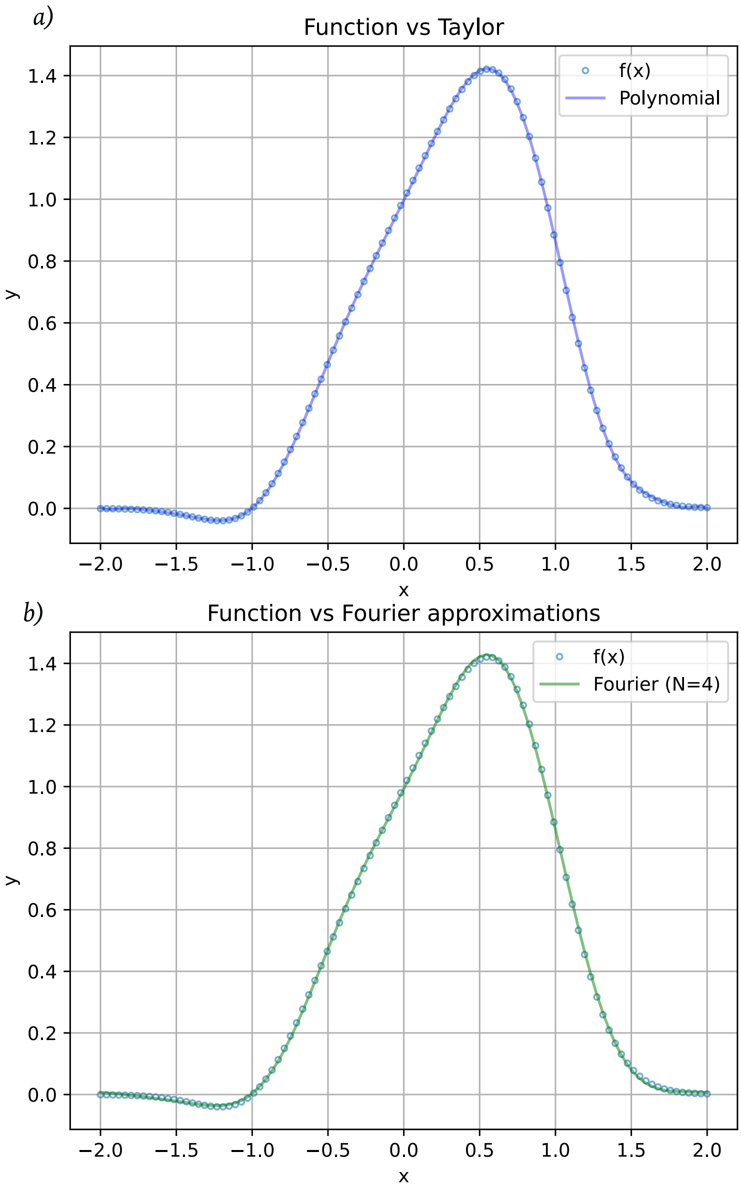


Fig. 3.15: A function $f_x = (1 + x)e^{-x^3 \sin x}$ can approximated using polynomials or trigonometric functions.

$$f(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n + \ldots$$

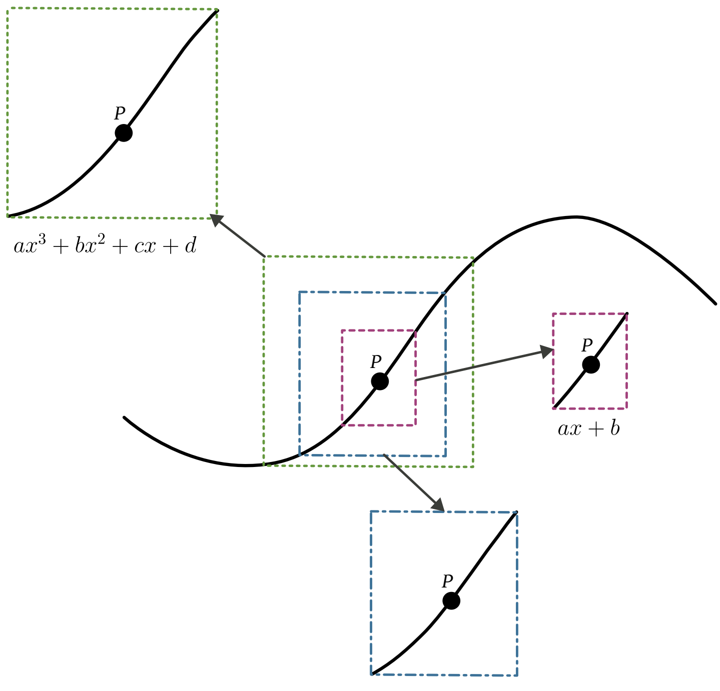


Fig. 3.16: Many "well-behaved" functions can be described using polynomials when "zoomed in" on a finite range. The smaller the range of interest – the lower the degree of the polynomial needed for accurate fit.

another way to represent a function using $\sin$ and $\cos$.

> ***Analytic Functions***
> In order to absolutely precisely represent a well-behaved function f using polynomials, one needs an *infinite* number of coefficients:
>
> $$f \quad \leftrightarrow \quad (c_0, c_1, c_2, \ldots, c_n, \ldots)\,.$$
>
> If a function allows such representatation, it is called *analytic function*. Analytic functions are of great value in mathematical physics, where they often describe very smooth physical behavior without jumps or infinities.

3.12.2 Fourier Series

As shown in Figure 3.15(b), the same function $f_x = (1+x)e^{-x^3 \sin x}$ can be well approximated using the following series:

$$\begin{aligned} F_x = & \frac{954}{1000} + \frac{571}{1000}\cos x + \frac{4}{10}\sin x + \frac{34}{1000}\cos(2x) + \frac{212}{1000}\sin(2x) - \\ & - \frac{78}{1000}\cos(3x) - \frac{28}{1000}\sin(3x) - \frac{11}{1000}\cos(4x) - \frac{32}{1000}\sin(4x)\,. \end{aligned}$$

The similarity between f_x and F_x is only approximate and works for a limited range.

A French mathematician and physicist Joseph Fourier introduced these simple trigonometric series in his 1822 work *"The Analytical Theory of Heat"*. They are known as *Fourier series*, and are used to represent *periodic* functions as an infinite sum

$$F_x = \int \left[a_n \cos(nx) + b_n \sin(nx)\right], \quad n = \overline{0, \infty}\,.$$

Non-periodic functions, such as f_x, can be represented either only approximately, or using $\cos$ and $\sin$ with summation over *non-countable/continuous* index n.

> ***Fourier Transform***
> The operation of representing functions with either countable or continuous summation over trigonometric functions is called *Fourier transform*. It is a very powerful mathematical tool, widely used in science

and engineering.

$$|f\rangle = \int \left[a_n |\cos_n\rangle + b_n |\sin_n\rangle \right] , \quad n = \overline{0, \infty} .$$

Again we have a situation when a function can be described compactly with a set of coefficients:

$$f \quad \leftrightarrow \quad (a_0, a_1, b_1, a_2, b_2, \ldots, a_n, b_n) .$$

3.12.3 Hilbert Spaces

We saw indications that functions, being "numberlikes", also have vector properties and can be expanded in certain basis. A basis can be, in theory, *infinite* (in numerical practice it is always finite):

$$|f\rangle = \int c_i |u_i\rangle , \quad i = \overline{1, \infty} .$$

In physical applications, including quantum theory, the coefficients and the basis have well-defined meaning. For example, coefficients can represent *amplitudes* of oscillations of a body, or powers in certain *"communication channels"*, or some other physical property. These interpretations suggest, firstly, that each coefficient is finite, and secondly that there is some *limit on their total sum.* For example, if the coefficients represent the amplitudes of oscillations with various frequencies, then the square of a coefficient c_i would be related to the energy of an oscillation (See section YYY) and we would have

$$\int |c_i|^2 = N < \infty$$

as a mathematical statement that the total energy of an oscillating system is finite.

The finite sum of the coefficients (or their squares) has an important consequence: it can be used to define a *magnitude* (a "length") of a function as equal to the sum. This leads to a possibility of a *scalar product* for a pair of functions and everything that the scalar product enables (e.g. dual spaces).

Although not all functions have the nice properties described above, a great number of functions used in quantum theory do. These functions, taken as one whole set, form a *space* with length and scalar product operation defined. A special feature of this space is that, in general, basis must be *infinite*. Of course, in practice, and even in many special theoretical problems, a finite number of spaces is used.

Vector spaces, endowed with scalar product, but requiring infinite bases, is called *Hilbert space*, in honor of a German mathematician David Hilbert. Hilbert spaces are very important mathematical tools in quantum theory.

3.13 Application: Circular Motion

Let us examine how the concepts and tools discussed above can be applied to a simple case of circular motion. Consider a particle moving in a circle

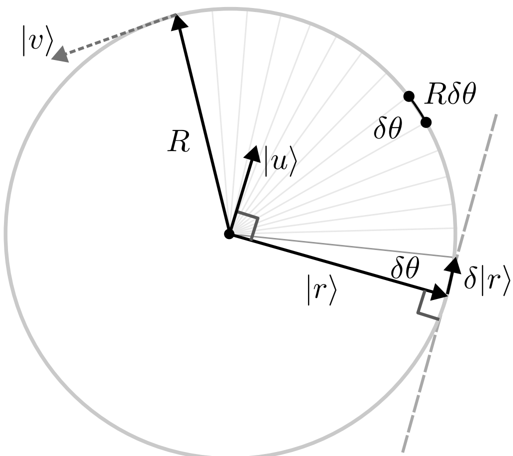


Fig. 3.17: Circular motion.

with the radius R, as shown in Figure 3.17. If we choose the center of the circle as the reference point, we can specify the position of the particle using an arrow $|r\rangle$. During motion the direction of this arrow is constantly changing, but its length R remains the same.

After a short time interval δt, the position of the particle changes by $\delta|r\rangle$:

$$|r_t\rangle \quad \rightarrow \quad |r_{t+\delta t}\rangle = |r_t\rangle + \delta|r\rangle\ .$$

The length of the path covered by the particle during the time interval δt can be approximated by the length of the arc $\delta L = R\delta\theta = v\delta t$. The arrow $\delta|r\rangle$ can be written as $\delta L|u\rangle$ where $|u\rangle$ is the unit length vector pointing in the direction of motion. This unit vector can be constructed from $|r\rangle$ by first rotating the latter counter-clockwise with the operator $\widehat{J}$ and then scaling the result down by R:

$$|u\rangle = \frac{1}{R}\,\widehat{J}\,|r\rangle\,.$$

Thus,

$$\delta|r\rangle = R\delta\theta\frac{1}{R}\,\widehat{J}\,|r\rangle = \delta\theta\,\widehat{J}\,|r\rangle\,.$$

The rate of change of the vector $|r\rangle$ with respect to time then becomes

$$\frac{\delta|r\rangle}{\delta t} = \frac{\delta\theta}{\delta t}\,\widehat{J}\,|r\rangle \qquad \Longrightarrow \qquad \partial_t|r\rangle = \omega\,\widehat{J}\,|r\rangle\,,$$

where we introduced the angular speed $\omega = \partial_t\theta$. Finally, by applying the $\widehat{J}$ operator to both sides of the last equation, we can cast it into the "Schrödinger" form:

$$\widehat{J}\,\partial_t|r\rangle = -\omega|r\rangle\,.$$

We can make one more step, multiplying both sides by the fundamental physical constant $\hbar$ – known as *reduced Planck's constant*– to obtain an equation very similar to Schrödinger equation:

$$\widehat{J}\,\hbar\partial_t|r\rangle = -\hbar\omega|r\rangle\,, \quad \Big(\text{compare: } i\hbar\partial_t|\Psi\rangle = \widehat{H}\,|\Psi\rangle\Big)\,.$$

3.13.1 Solution

When the angular speed ω is constant, the equation of circular motion can be easily solved. The simple solution will play an important part in quantum theory. To find it, we first recognize that, given constant angular speed ω, the angle swept by the vector $|r\rangle$ grows linearly with time: $\theta_t = \omega t$. Next, to obtain a position vector $|r_t\rangle$ we need to rotate the initial position vector $|r_0\rangle$ by the angle θ_t:

$$|r_t\rangle = \widehat{R}_\theta\,|r_0\rangle\,.$$

As illustrated in Figure 3.18(a), we can split a single rotation by the angle θ into a large number of rotations by a smal angle $\phi = \delta\theta = \theta/N$, where N is the number of steps. Mathematically this is expressed as follows:

$$\widehat{R}_\theta = \widehat{R}_\phi \circ \widehat{R}_\phi \circ \widehat{R}_\phi \circ \cdots \circ \widehat{R}_\phi$$

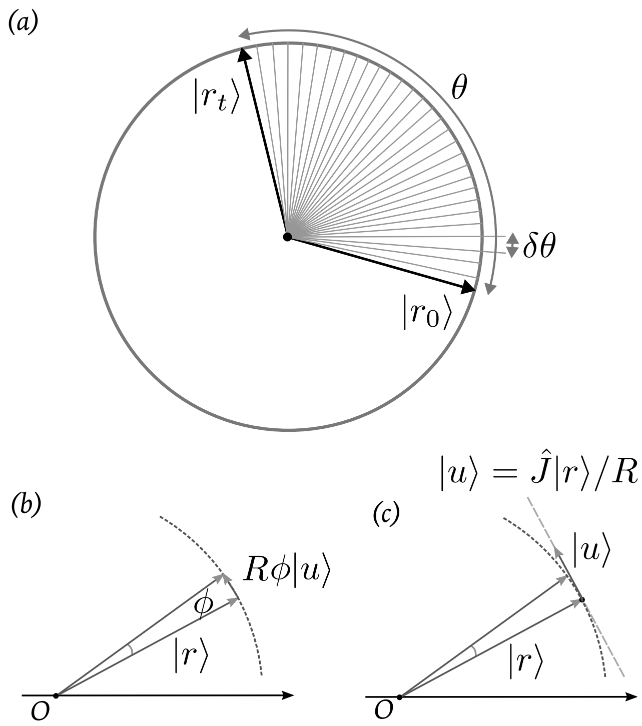


Fig. 3.18: Circular motion.

or more compactly

$$\widehat{R}_\theta = \widehat{R}_\phi^N \; .$$

Each small rotation step can be described by a simple operator which is close to the identity operator:

$$\widehat{R}_\phi = \widehat{1} + \widehat{G}\,\phi \, .$$

To see this, recall that rotating a vector $|r\rangle$ by a small angle ϕ adds to it a tiny vector $\delta|r\rangle$ which perpendicular to $|r\rangle$ and has the length equal to the length of the small arc, as shown in Figure 3.18(b)(c):

$$\widehat{R}_\phi \,|r\rangle = |r\rangle + \phi\,\widehat{J}\,|r\rangle \, .$$

Thus, the operator of small step rotation is given by

$$\widehat{R}_\phi = \widehat{1} + \phi\,\widehat{J} \; .$$

Since $\phi = \delta\theta = \theta/N$, we obtain the rotation operator by the finite angle θ:

$$\widehat{R}_\theta = \left(\widehat{1} + \frac{\theta\,\widehat{J}}{N}\right)^N \; .$$

The expression on the right is the approximation to the exponential function:

$$e^x \approx (1 + \frac{x}{N})^N \, ,$$

and when N grows increasingly larger, the approximation becomes more exact. Finally, substituting $\theta = \omega t$, we arrive at the solution of the circular motion problem. The position vector $|r_t\rangle$ changes in time according to the equation

$$|r_t\rangle = e^{\widehat{J}\omega t}|r_0\rangle \, .$$

Let's remind ourselves that this is the solution to the equation of circular motion

$$\widehat{J}\,\partial_t|r\rangle = -\omega|r\rangle \, .$$

Chapter Highlights

- *Arrows in a plane provide a simple model for vectors.*

4. Classical Physics

> *I think we may ultimately reach the stage when it is possible to set up quantum theory without any reference to classical theory, just as we already have reached the stage where we can set up the Einstein gravitational theory without any reference to the Newtonian theory. But from the point of view of teaching students, I think one would always have to proceed by stages – not expect too much from them, teach them first the elementary theories and gradually develop their minds; and that will always involve working from the classical theory first.*
>
> *P. A. M. Dirac, Lectures on Quantum Field Theory, Belfer Graduate School of Science, Yeshiva University, New York, 1966, p.43.*

GIVEN HOW CLASSICAL AND QUANTUM PHYSICS ARE INTERCONNECTED, it is important to have a solid understanding of the former. In this chapter, we develop a clearer picture of what a system, state, and dynamics are, and also encounter such important concepts as *degrees of freedom*, *mode*, *action*, *Hamiltonian*, *phase space*, *field* and more.

> ***Prerequisite Knowledge***
> To fully understand the material of this chapter, readers should be comfortable with the following concepts:
> - State
> - Dynamical equations

4.1 System

A part of nature that can be clearly isolated and studied is called a *physical system*. An electron, an atom, a molecule, a crystal, a pendulum, a comet,

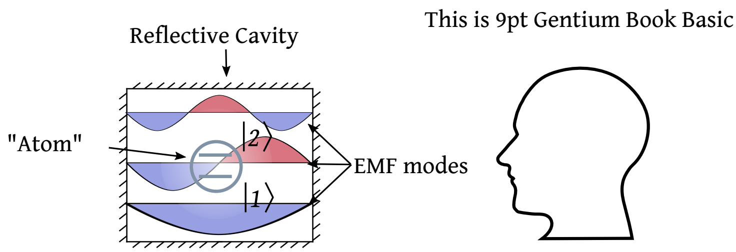


Fig. 4.1: Operators extend the idea of functions. (a) An unary function f can be applied to a number x to produce another number y. (b) An unary operator $\widehat{F}$ can be applied to a vector $\vec{a}$ to yield another vector $\vec{b}$.

a star – these are examples of physical systems of various degrees of complexity.

Often a physical system is a body or several bodies interacting with each other or with some external bodies. Figure 4.2 provides several examples of *mechanical systems*. Let's examine them in more detail.

(a) *Free falling body*: An elastic body falls down vertically under the force of gravity, bounces back, goes up and then down to repeat the bounce again and again. Also, a projectile launched at an angle.
(b) *String pendulum*: A compact body is attached to a string of fixed length. It is allowed to swing back and forth without experiencing air friction.
(c) *Atwood machine*: Two bodies with slightly unequal masses are connected with a non-stretchable string going over a frictionless pulley.
(d) *Inclined plane*: A solid cylinder rolling down an inclinded plane.
(e) *Piston*: A system of three bodies (cylindrical crankshaft, rod, and piston) connected in a way that locks rotation of a cylinder and the vertical motion of the piston.
(f) *Spring oscillator*: A body, attached to a spring, is allowed to slide left and right across a frictionless surface.
(g) *Linear chain of oscillators*: A set of pairwise interconnected identical bodies; the allowed motion happens along the horizontal axis.
(h) *Sun and planet*: A planet circling around the sun.

We will study oscillator and circular motion in great detail.

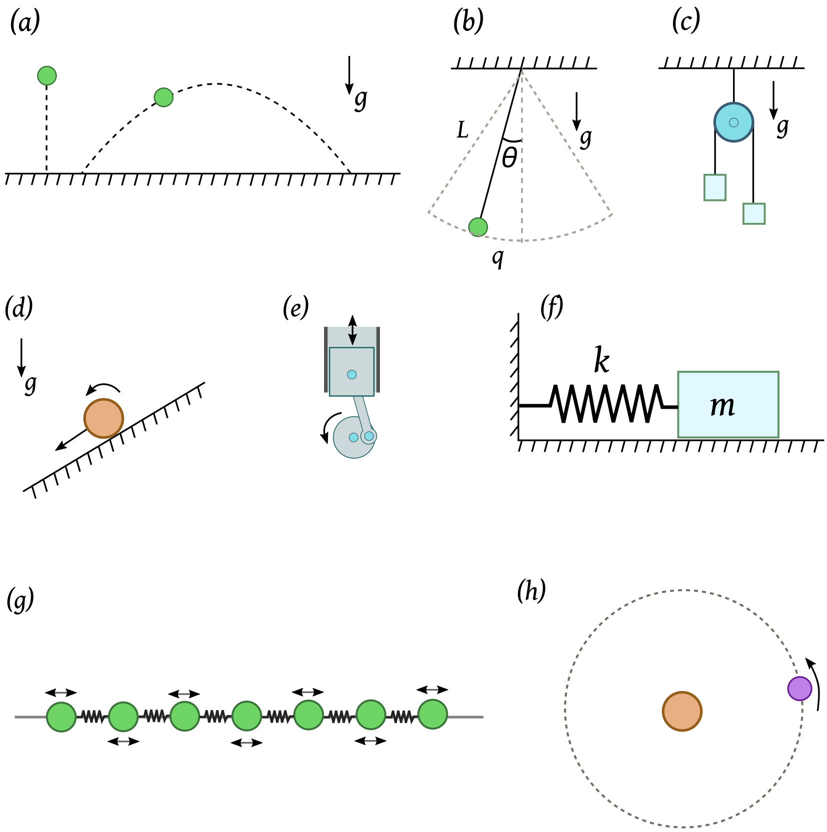


Fig. 4.2: Examples of mechanical systems. See text for explanation..

4.1.1 Oscillator

The model of an oscillator is extremely important. It appears in various guises in almost all physical theories. Let's study it in details.

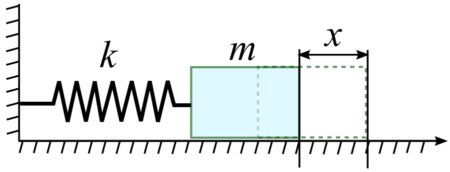


Fig. 4.3: A mechanical model of an oscillator: A body attached to an ideal spring.

Consider a body with the mass m is attached to a spring with the stiffness k. The body is allowed to move across a frictionless surface. The force required to strech a spring by the amount x is given by the Hooke's law

$$F = kx.$$

This is the force applied *to the spring*. The force created *by the spring, and applied to the attached body*, is of equal magnitude but points in the opposite direction.

When the body is displaced from its equilibrium position, by stretching or compressing the spring, and then released, it will undergo periodic motion. During this motion, the position, velocity, kinetic energy of the mody, and the potential energy of the spring will be constantly changing.

To remind, the kinetic energy of a body is

$$E_k = \frac{mv^2}{2}, \text{ or } K = \frac{p^2}{2m}.$$

The potential energy of a spring, stretched or compressed by the amount x is given by

$$E_p = \frac{kx^2}{2}, \text{ or } \Pi = \frac{kq^2}{2}.$$

4.1.2 Configuration

In mechanics, *configuration* means a formal way to describe the arrangement of a system at a given time.

The behavior of a system in time can be described by specifying its *configuration* as the function of time. In relatively simple systems, configuration may consist of a set of coordinates that uniquely determine the arrangement of bodies in the system. For example, the configuration of a pendulum can be given by a single coordinate — the length of the arc q. Of course, as the pendulum swings, both Cartesian coordinates x and y are changing, but not independently, due to the relation

$$x^2 + y^2 = L^2 .$$

Given x, we can find $y > 0$ as $y = +\sqrt{L^2 - x^2}$, thus reducing the number of required coordinates.

Consider another example, shown in the Figure (4.4)(a): a system of two bodies, connected with each other using ideal springs with stiffness k, and each body is connected to a rigid wall.

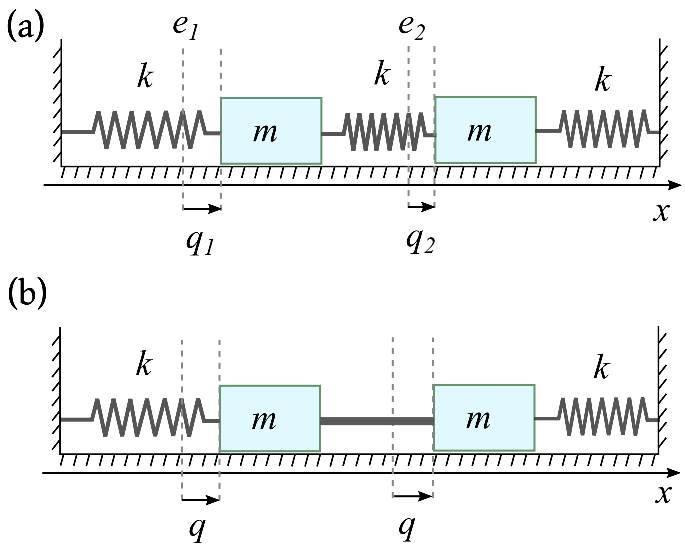


Fig. 4.4: (a); (b).

When the system is in equilibrium, the bodies occupy positions on the horizotal axis denoted as e_1 and e_2. During motion, the position of the first body changes by

$$q_1(t) = x_1(t) - e_1 ,$$

and similarly for the second body: $q_2 = x_2 - e_2$. It is important to realize, that although the two bodies are connected with a spring, they can still move with different velocities, and have different displacements

$q_1 \neq q_2$. Indeed, we can set the system in motion by moving each body independently and then releasing them. Contrast this with the situation, shown in the Figure (4.4)(b), where the bodies are connected with a rigid rod, fusing two masses into essentially a single body. In this case only a single displacement q is required to specify the configuration of the system.

4.1.3 Qoordinates

The coordinates specifying the configuration of a system do not have to be Cartesian. In the example of a pendulum, the configuration can be conveniently given by the length of the arc $q = L\theta$, see Figure (4.2)(b).

Consider another example, shown in the Figure (4.2)(h): Two bodies interact gravitationally. In this problem, it turns out, the equations describing the motion of the system are simpler if, instead of the usual positions x_1 and x_2 we use the relative distance

$$q_1 = x_2 - x_1$$

and the position of the center of mass

$$q_2 = (m_1 x_1 + m_2 x_2)/(m_1 + m_2),$$

where m_1 and m_2 are the masses of the bodies.

We thus come to the idea of *generalized coordinates* – arbitrary coordinates completely specifying the configuration of a system. Generalized coordinates can be based on positions, angles, or some combinations of those.

4.1.4 Modes

Let's use a simple system of two coupled oscillators to explore an important concept of *modes*. The system is illustrated in Figure (4.4)(a). To make the analysis simpler, we will assume identical masses and spring constants. Using the qoordinates q_1 and q_2 we can write the total energy of such a system as follows:

$$E = E_1 + E_2 + E_i\,,$$

where

$$E_1 = \frac{kq_1^2}{2} + \frac{mv_1^2}{2} \text{ and } E_2 = \frac{kq_2^2}{2} + \frac{mv_2^2}{2}$$

are the energies of the "oscillators" without the middle coupling spring. Both E_1 and E_2 have potential energy of stretched spring and the kinetic energy of a moving body. The term E_i represents *interaction* or coupling between the oscillators, and equals to the potential energy of the stretched middle spring:

$$E_i = \frac{k(q_1 - q_2)^2}{2} \,.$$

The velocities v_1 and v_2 are given by

$$v_i = \partial_t q_i = \partial_t x_i \,, \quad i = \overline{1,2} \,.$$

Now if we look at this problem from a different point of view, using generalized coordiates

$$Q_1 = q_1 - q_2 \quad \text{and} \quad Q_2 = q_1 + q_2$$

the situation will be a bit simpler. To demonstrate it, first express q_1 and q_2 in terms of Q_1 and Q_2:

$$q_1 = (Q_1 + Q_2)/2 \,, \quad q_2 = (Q_2 - Q_1)/2 \,.$$

The generalized velocities $U_i = \partial_t Q_i$ are simply

$$U_1 = v_1 - v_2 \quad \text{and} \quad U_2 = v_1 + v_2$$

and, consequently

$$v_1 = (U_1 + U_2)/2 \,, \quad v_2 = (U_2 - U_1)/2 \,.$$

The potential energy of all three springs

$$E_p = \frac{k}{2}\left[q_1^2 + q_2^2 + (q_1 - q_2)^2\right]$$

when expressed in terms of the new qoordinates Q_i becomes:

$$E_p = \frac{k_1 Q_1^2}{2} + \frac{k_2 Q_2^2}{2}$$

where $k_1 = 3k/2$ and $k_2 = k/2$. Note how this expression resembles two disconnected springs. The kinetic energy

$$E_k = \frac{m}{2}\left[v_1^2 + v_2^2\right]$$

is transformed into

$$E_k = \frac{m/2}{2}\left[U_1^2 + U_2^2\right] .$$

The final result is the total mechanical energy, when expressed in Q_i and U_i, looks as follows:

$$E = \frac{k_1 Q_1^2}{2} + \frac{M_1 U_1^2}{2} + \frac{k_2 Q_2^2}{2} + \frac{M_2 U_2^2}{2} .$$

But this looks exactly like two independent oscillators with different spring constants and oscillation frequencies. The frequency of the first oscillator is

$$\omega_1 = \sqrt{\frac{k_1}{M_1}} = \sqrt{3}\omega ,$$

and the frequency of the second oscillator is

$$\omega_2 = \sqrt{\frac{k_1}{M_1}} = \omega ,$$

where $\omega = \sqrt{k/m}$ is the oscillation frequency of the original, unconnected oscillators.

Oscillations with frequencies ω_1 and ω_2 represent distinct *types of motion* or *modes of oscillations.* To understand what they look like in terms of motion of the original bodies, imagine we excite only Q_1-type oscillation, with Q_2 remaining constant in time. In other words, we assume

$$Q_1 = A_1 \cos\omega_1 t \quad \text{and} \quad Q_2 = C .$$

The constant Q_2 means that the "middle point" $Q_2/2 = (q_1 + q_2)/2 = (x_1 + x_2)/2 + (e_1 - e_2)/2$ remains fixed. Oscillating Q_1 means that the distance between the bodies $x_2 - x_1 = (q_2 - q_1) + (e_2 - e_1) = -Q_1 + (e_2 - e_1)$ changes. The bodies periodically move away from each other, then stop, then move towards each other, then stop, and the process repeats.

The second type of motion corresponds to $Q_1 = C$, and $Q_2 = A_2 \cos\omega_2 t$. The constant Q_1 means that the distance between the bodies is fixed and they both move as one whole, with the total mass $M = 2m$. This "whole" system is pushed and pulled by two springs, which combine their forces, effectively acting as the spring with the constant $2k$. This results in the same oscillation frequency.

In general, both modes can be excited and corresponding qoordinates will be changing in time as

$$Q_1 = A_1 \cos\omega_1 t + B_1 \sin\omega_1 t\,,$$

$$Q_2 = A_2 \cos\omega_2 t + B_2 \sin\omega_2 t\,.$$

The original coordinates x_1 and x_2 are then given by

$$x_1 = q_1 + e_1 = (Q_1 + Q_2)/2 + e_1$$

and

$$x_2 = q_2 + e_1 = (Q_2 - Q_1)/2 + e_2\,,$$

which is simply a long sum of oscillations with various frequencies. Thus, in general, the motion of each individual body can not be characterized by any signle frequency Ω.

4.1.5 Degrees of Freedom

Degree of freedom is a separate independent motion of a mechanical system. Each independent motion corresponds to the change in time of a separate generalized coordinate. The number of degrees of freedom is the number of generalized coordinates required to completely specify the configuration of a mechanical system at different moments of time.

Take, for example, a pendulum, shown in the Figure (4.5). In general Cartesian coordinates, all three coordinates x, y, and z will be changing in time. However, only a single generalized coordinate $q(t)$ — the length of the arc – is required to fully describe the configuration, and thus the motion, of this mechanical system. The number of degrees of freedom, in this example, equals 1.

4.2 State

State of a system is the *minimal* collection of observables which is, in certain sense, *complete* and *self-sufficient*. State is "all there is to know" about a system. If the state of a system is known at one moment of time t_0, then we should be able to determine the state at any later moment of

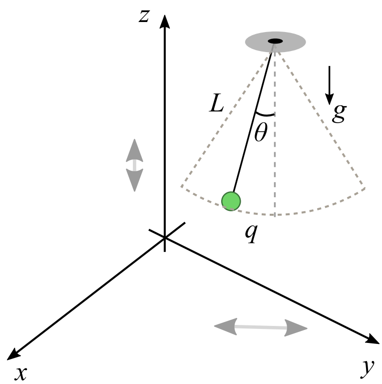


Fig. 4.5: A pendulum has one degree of freedom, despite the fact that all three Cartesian coordinates can be changing during its motion.

time t. In classical mechanics the pair of observables (x, p) defines the state of a mechanical system.

State is the minimal set of quantities describing mechanical system and sufficient to predict their future values from their initial values. State is an important concept not only mechanics, but in other areas of physics. Let's elaborate, using the oscillator as an example.

Suppose that at the moment of time t_0 the position of an oscillator is x_0 and its velocity is v_0. To find their values at some later time $t > t_0$, we can go iteratively in small steps, calculating how much the position and the velocity change after each successive tiny interval of time δt. The first iteration results in the updated value of position

$$x_1 = x_0 + v_0 \delta t.$$

The second, and every other, iteration looks very similar:

$$x_2 = x_1 + v\delta t\,.$$

Now it is important to realize that we can no longer use the same initial velocity v_0 in the second iteration, because the velocity itself changes. Thus, we must update the value of the velocity as well. This is done by using acceleration:

$$v_1 = v_0 + a_0\,.$$

After that, we can find the second iteration of the position: $x_2 = x_1 + v_1 \delta t$. To keep this scheme going, we must be able to update the value of the acceleration, because it is also changing. It appears then, we need some quantity that allows to find the next step:

$$a_1 = a_0 + b_0 \delta t \,.$$

Fortunately, *this is not needed!* At this point we can use the laws of motion. For example, Newton's second law gives the acceleration in terms of the known force acting on the object:

$$a = \frac{F}{m}. \tag{4.1}$$

Forces don't depend on acceleration. (Weber Electrodynamics?)

All known forces in physics depend on positions or distances and– sometimes– velocities of bodies. For example, the force of the spring $F = -kx$ depends only on the coordinate x. The force of gravitational interaction $F = GMm/r^2$ and the Coulomb force between two charges $F = kQq/r^2$ depend on the distance r between the bodies. The force acting on an electron moving through a magnetic field–known as Lorentz force– equals $F = qvB$ and depends on the electron's velocity (and the field's strength B). No known forces depend on acceleration. This fact leads to an important conclusion: It is enough to know position and velocity of an object at time t_0, in order to find their values at any later moment of time $t > t_0$. Obviously, position and velocity at any previous moment of time can be found in the similar way.

Thus, we do not need to advance the acceleration by calculating its small change $\delta a = b\delta t$, we can simply calculate it from the law of motion:

$$a_n = \frac{F(x_n, v_n)}{m} \,. \tag{4.2}$$

This formula says that the acceleration at the iteration step number n is found from the values of the position x_n and the velocity v_n at the same step. Given the velocity, we can advance the postion, and given the acceleration, we can advance the velocity. Then we recalculate the new value for the acceleration and repeat, until we reach the final time t.

The preceding discussion demonstrates that in Newtonian mechanics *the state of a mechanical system* is given by a pair of quantities – (x, v).

There are alternatives to the Newtonian mechanics, and, correspondingly, there are alternatives to the mechanical state. The first such alternative is Hamiltonian dynamics.

4.2.1 State Evolution: Newtonian Approach

We will now apply the ideas and formulas of Newtonian mechanics to an oscillator. We will calculate the motion of the oscillator in time using a simple method of *state evolution*. Specifically, we will setup two simple equations – one for position and one for velocity.

The equation for position is trivial and amounts to the definition:

$$\frac{\delta x}{\delta t} = v \,.$$

The equation for velocity follows from the second law of Newtonian dynamics:

$$\frac{\delta v}{\delta t} = \frac{F}{m} = -\frac{kx}{m} \,.$$

Here we used the expression for the spring force $F = -kx$ acting *on the body* from the side of the spring.

Suppose we know the *initial state* of the oscillator (x_0, v_0) at time $t_0 = 0$. When the clock makes a single tick after a tiny time interval δt the body will move to a new position

$$x_1 = x_0 + v_0 \delta t$$

and the velocity will change due to the action of the spring:

$$v_1 = v_0 - \frac{kx_0}{m} \,.$$

Thus, after a single tick of the clock the state of the oscillator will evolve from $|\xi\rangle_0 = (x_0, v_0)$ to $|\xi\rangle_1 = (x_1, v_1)$. At this point we can keep repeating the steps to calculate the state after any number of ticks, up to the desired time $t = N\delta t$.

We can now formalize the recipe for evolution of the state, writing it as a mathematical relation:

$$|\xi\rangle_{new} = f\,|\xi\rangle_{old} \,,$$

where

$$|\xi\rangle_{new} = (x_{new}, v_{new}) = (x_{old} + v_{old}\delta t, v_{old} - kx_{old}/m) .$$

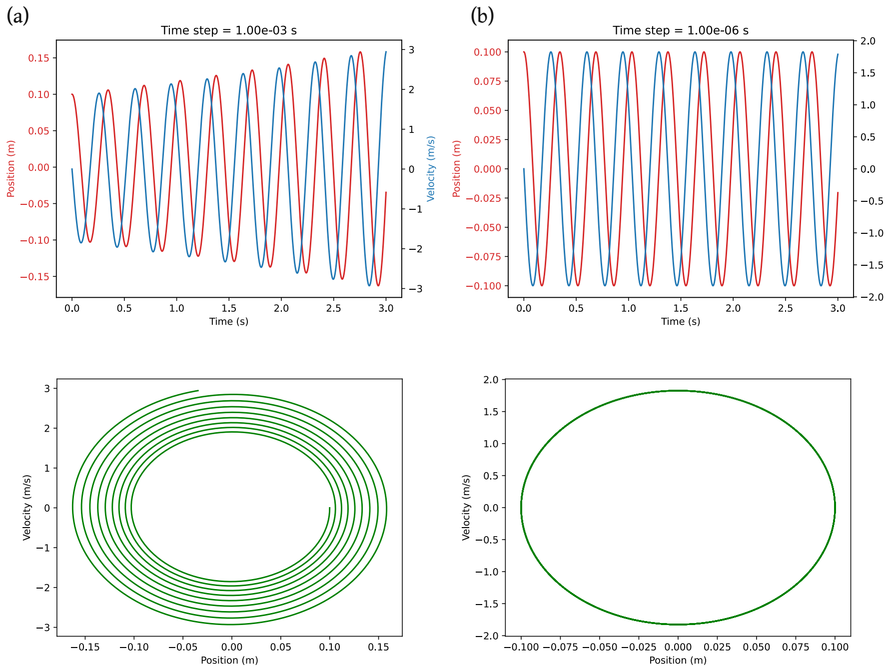


Fig. 4.6: First row: Position (red curve, left axis) and velocity (blue curve, right axis) as functions of time. Second row: Velocity vs position. (a) Calculations with time step of 1 millisecond. (b) Calculations with time step of 1 microsecond.

Using this simple approach, we can calculate the state $|\xi\rangle = (x, v)$ of the oscillator for any moment in the future or past. Figure 4.6 shows two example results. The first result, in the column (a), demonstrates that we must be careful with the step size δt of the time. If it is not sufficiently small, the inherent error of the method accumulates quickly, resulting in wrong behavior, such as the gradual increase of velocity and oscillation amplitude. The column (b) of Figure 4.6 demonstrates the expected behavior of the oscillator – periodic change of position and velocity with constant amplitudes.

4.3 Dynamics

Dynamics is the study of state evolution of various systems subject to known interactions. Physical systems of interest can be mechanical, like the ones given in Section 4.1, or "non-mechanical" like electromagnetic field or even gravitational field. One can also explore dynamics of quantum systems.

The central equation in dynamics describes the state change in time due to known *laws of dynamics*:

$$\partial_t \left|\xi\right\rangle = \widehat{D}_{int} \left|\xi\right\rangle .$$

We will explore three main variants of classical dynamics: Newtonian, Hamiltonian, and Lagrangian. They will prepare us for better understanding *quantum dynamics.*

Newtonian dynamics is based on the notion of *force* as the driver of interaction:

$$(\partial_t x, \partial_t v) = (v, F/m) .$$

Both Hamiltonian and Lagrangian mechanics rely on the notion of energy.

4.4 Hamiltonian

Hamiltonian is the expression for total energy of a system in terms of position and momentum. For example, using the non-relativistic expression for momentum $p = mv$, the Hamiltonian of an oscillator can be written as

$$H = \frac{p^2}{2m} + \frac{kx^2}{2} . \tag{4.3}$$

Hamiltonian, denoted simply as H, still means the total energy. The only difference between H and E is the emphasis on the use of momentum in H instead of velocity.

Momentum is more fundamental than velocity.

Although velocity feels more intuitive and closer related to our visual perception of motion, momentum is a more *fundamental quantity* in physics. There is a *conservation of energy and momentum* law, but there is no law of the conservation of velocity.

4.4.1 Phase Space

Hamiltonian dynamics uses position x and momentum p to study motion. For an isolated system with conserved energy (constant Hamiltonian), the expression for the Hamiltonian establishes the relationship between x and p for all moments of time. For example, using the Hamiltonian for an oscillator (4.3), we can rewrite it as follows:

$$1 = \frac{p^2}{(2mH)} + \frac{x^2}{(2H/k)} \,. \tag{4.4}$$

This has the same form as the equation of an ellipse in Cartesian axes (x, y):

$$1 = \frac{y^2}{b^2} + \frac{x^2}{a^2} \,.$$

The area of such an ellipse with the semi-axes a and b is $A = \pi ab$. For $a = b$ an ellipse becomes a circle with the area $A = \pi R^2$.

The equation (4.4) describes an ellipse in a special xp-plane, every point of which can be specified by a pair of values (x, p). Such a plane is called *phase space*. It plays an important role in Hamiltonian dynamics.

The maximum value of the momentum is $p_{max} = \sqrt{2mH}$ and the maximum value for the displacement of the body is $x_{max} = \sqrt{2H/k}$. It turns out, the *area of the ellipse in the phase space is directly proportional to the total energy of the oscillator*:

$$A = \pi x_{max} p_{max} = 2\pi H \sqrt{m/k} \,,$$

or

$$H = \frac{A}{2\pi} \sqrt{\frac{k}{m}} \,.$$

This important relationship will be used later when we will study quantum properties of the oscillator.

4.4.2 Hamiltonian Equations

Hamiltonian dynamics tracks the time dependence of position x_t and momentum p_t. The pair of values (x, p) represents the *state in Hamiltonian dynamics*. Indeed, since there is a unique relationship between the momentum and velocity (e.g. $p = mv$ for non-relativistic speeds $v \ll c$, c — speed of light), the knowledge of $|\xi\rangle = (x, p)$ is equivalent to the knowledge of $|\xi\rangle = (x, v)$ which is the state in Newtonian dynamics. Put differently, the pair (x, p) provides the same complete description of the system as (x, v).

The main difference between Newtonian and Hamiltonian approaches, is that in the latter the equations for the rate of change of x and p are *explicitly tied to the Hamiltonian*— to the form of the energy expressed in terms of position and momentum. In other words, the equations for $\partial_t x$ and $\partial_t p$ are written in the following form:

$$\partial_t x = \widehat{X}\ H \qquad \text{and} \qquad \partial_t p = \widehat{P}\ H\,.$$

where $\widehat{X}$ and $\widehat{P}$ are some *rules* for calculating velocity $v = \partial_t x$ and force $F = \partial_t p$ from Hamiltonian H. Mathematically, $\widehat{X}$ and $\widehat{P}$ must be *operators*: they take the function $H(x, p)$ and calculate the functions $v(x, p)$ and $F(x, p)$.

Let's use the oscillator model to see how exactly these equations look like.

Exercise 4.1

Show that for an oscillator $v = p/m = \partial_p H$ and $F = -kx = -\partial_x H$.

Using the results of the exercise, we can conclude that for the oscillator the equations of Hamiltonian dynamics are

$$\partial_t x = \partial_p H\,, \tag{4.5}$$

$$\partial_t p = -\partial_x H\,. \tag{4.6}$$

The equations (4.5) and (4.6) are called *Hamiltonian equations of motion*.

Exercise 4.2

In special theory of relativity it is shown that the energy E and momentum p of any particle are related to its mass m as follows:

$$m^2 = E^2 - p^2\,.$$

(Special units must be used for this equation to hold. In these special units all velocities are measured as the fractions of the speed of light, while energy, momentum, and mass are all measured in the same units.)

Starting from the Hamiltonian of a relativistic particle

$$H^2 = p^2 + m^2 \,,$$

show that the momentum depends on velocity as

$$p = \frac{mv}{\sqrt{1 - v^2}} \,.$$

Then show that momentum and energy are related as $p = Ev$.

4.4.3 Solving Oscillator Equations

We will now find the exact time dependence of both position x_t and momentum p_t of the oscillator. To do that, we will need to take another look at the evolution of the state of the oscillator in phase space.

The state of an oscillator corresponds to a point in the phase space. This point can be mathematically represented either as a vector $|\xi\rangle$ connecting the origin and the point, or as a pair of values (x, p) . Two descriptions are equally valid. Moreover, the pair of values (x, p) can be considered as the components of the vector $|\xi\rangle$.

As the time progresses, the position and momentum of the oscillator change, but the tip of the arrow $|\xi\rangle$ remains on the ellipse, corresponding to a constant energy H. The axes of "ordinary" phase space have different physical units, whereas the "usual" Cartesian axes (x, y) have the same units and are equivalent in their meaning. It is convenient to temporarily introduce *normalized* energy, position, and momentum.

First, let us express the energy of the oscillator in terms of the rest-energy of the electron $E_e = m_e c^2$. Then the Hamiltonian of the oscillator (its total energy) can be written as $H = \bar{H} E_e$. If the rest-energy and Hamiltonian are measured in Joules, then $\bar{H}$ is a number without units. (COOKIES?)

Next, we will introduce a "scale" for position of the oscillator. Specifically, we will denote $x_e = \sqrt{2E_e/k}$ as the amplitude of the oscillation when the total energy is equal to E_e. This amplitude is measured in meters. Then the displacement can be written as $x = \bar{x} x_e$, where $\bar{x}$ is a number without units.

Finally, we will introduce a "scale" for momentum of the oscillator: $p_e = \sqrt{2mE_e}$ as the amplitude of the oscillation when the total energy is equal to E_e . (*Note*: the mass m in $\sqrt{2mE_e}$ is the mass of the body attached to the spring, *not* the mass of the electron) The "scale" p_e is measured in kilograms times meters per second – the usual units for momentum. Therefore, the momentum of the oscillator can be written as $p = \bar{p}p_e$, where $\bar{p}$ is a number without units.

Substituting $H = \bar{H}E_e$, $p = \bar{p}p_e$, and $x = \bar{x}x_e$ into the expression for the Hamiltonian, we obtain

$$\bar{H}E_e = \frac{p_e^2}{2m}\bar{p}^2 + \frac{kx_e^2}{2}\bar{x}^2 = E_e\bar{p}^2 + E_e\bar{x}^2 \, ,$$

from which follows a very simple relationship between pure numbers: $\bar{H} = \bar{p}^2 + \bar{x}^2$. This equation describes a circle in *normalized phase space* $(\bar{x}, \bar{p})$. The tip of the *normalized state* vector $|\bar{\xi}\rangle = (\bar{x}, \bar{p})$ performs clock-wise circular motion. If we denote the magnitude of the angular speed of this motion as ω, then we can use the familiar equation for the circular motion:

$$\partial_t |\bar{\xi}\rangle = -\omega \, \widehat{J} \, |\bar{\xi}\rangle \, .$$

The minus sign in the right-hand side is due to the clock-wise rotation of the state vector $|\bar{\xi}\rangle$.

The left side of the last equation is a two-component vector $(\partial_t \bar{x}, \partial_t \bar{p})$. The right hand side is also a vector with components $-\omega \ \widehat{J} \ |\bar{\xi}\rangle = (\omega\bar{p}, -\omega\bar{x})$. The equality of vectors means the equality of their components:

$$\partial_t \bar{x} = \omega\bar{p} \quad \text{and} \quad \partial_t \bar{p} = -\omega\bar{x} \, .$$

Let's examine the equality $\partial_t \bar{x} = \omega\bar{p}$. The left hand side can be expanded as follows:

$$\partial_t\bar{x} = \frac{\delta\bar{x}}{\delta t} = \frac{\delta(x/x_e)}{\delta t} = \frac{1}{x_e}\frac{\delta x}{\delta t} = \frac{v}{x_e} \, .$$

The right hand side can be written as $\omega\bar{p} = \omega\frac{p}{p_e} = \frac{\omega m v}{p_e}$. We showed that

$$\frac{v}{x_e} = \frac{\omega m v}{p_e}$$

and consequently, $\omega = p_e/(m x_e)$. The conclusion is very important: *The angular speed of rotation of normalized state arrow for harmonic oscillator is constant.* The oscillator returns into its initial state with the period $T = 2\pi/\omega$.

When we plug in the expression for x_e and p_e in terms of the energy E_e and the parameters of the oscillator k and m, we will arrive at the following equations for the frequency and period of oscillatory motion:

$$\omega = \sqrt{\frac{k}{m}} \quad \text{and} \quad T = 2\pi\sqrt{\frac{m}{k}}\,.$$

Now we can write the components of the normalized state vector as

$$\bar{x} = \sqrt{\bar{H}}\cos(\omega t) \quad \text{and} \quad \bar{p} = -\sqrt{\bar{H}}\sin(\omega t)\,.$$

The minus sign for momentum comes from the fact that the "circular motion" in normalized phase space is clockwise and $\omega < 0$.

Going back to "normal" energy, position, and momentum is easy. Indeed, we first find $x_t = x_e\sqrt{H/H_e}\cos(\omega t)$ which simplifies to simple harmonic motion:

$$x_t = \sqrt{2H/k}\cos(\omega t)\,.$$

Similarly for momentum: $p_t = -\sqrt{2mH}\sin(\omega t)$.

Exercise 4.3

Show that for harmonic oscillator

$$\Delta x = \sqrt{\langle x^2\rangle - (\langle x\rangle)^2} = \sqrt{H/k}\,,$$

and

$$\Delta p = \sqrt{\langle p^2\rangle - (\langle p\rangle)^2} = \sqrt{mH}\,.$$

Then show that $\Delta x \Delta p = H/\omega$.

4.5 Lagrangian

Lagrangian of a physical system is a special function of its coordinates and velocity that captures some aspect of energy, not covered by the total energy or Hamiltonian. For many systems Lagrangian describes the *imbalance* of kinetic energy over the potential energy. For example, the Lagrangian of the oscillator is

$$L = E_k - E_p = \frac{mv^2}{2} - \frac{kx^2}{2}\,. \tag{4.7}$$

Unlike total energy, Lagrangian is generally not conserved.

Exercise 4.4
Using the results for x_t and p_t from Hamiltonian dynamics for the oscillator, find the explicit form of its Lagrangian as the function of time.

The units for Lagrangian are the same as the units for energy or Hamiltonian (Joules in International System SI), but the meaning and the role of Lagrangian is what sets it apart from other functions of the state of a mechanical system. Lagrangian is used to find the *equations of motion* for the system. For each independent coordinate (e.g. x, y, or z) one finds a separate equation of motion using Lagrangian. Let's see how this works for the oscillator.

The first step is to extract momentum $p = mv$ from the Lagrangian. Looking at the expression (4.7), we find $p = \partial_v L$. Next, we should find the "force" acting on the body. The force will allow us write down Newton's second law in the form $F = \partial_t p$. Again, from the expression (4.7), we find $F = -kx = \partial_x L$.

Finally, rewriting $F = \partial_t p$ in terms of the Lagrangian, we arrive at Euler-Lagrange equation:

$$\partial_t \left(\partial_v L\right) = \partial_x L\,.$$

The dynamical equations for a general system will have the same form.

Lagrangian in Qoordinates
The power of Lagrange approach to mechanics lies in its indifference to the type of coordinates used to describe mechanical system. It is often convenient to use some *generalized coordinate* q instead of Cartesian coordinate x (or y and z). *Generalized velocity* is then $\nu = \partial_t q$. In terms of generalized variables, Euler-Lagrange equations still have the same form:

$$\partial_t \pi = \partial_q L$$

where $\pi = \partial_\nu L$ is *generalized momentum.*

4.5.1 Stationary Action Principle

Euler-Lagrange equations — the equations of motion for a system with specified Lagrangian $L(x, v)$ — are more rigorously derived using an important physical principle known as the *Stationary Action Principle*.

Sometimes it is called *"principle of least action"* and has a special case known as *"principle of least time"*. However, the proper name is the stationary action principle. To understand the idea behind the stationary action principle we must recall the concept of a functional, and in particular the *action functional* .

Although mathematically Lagrange function $L(x, v)$ depends on two inputs – position and velocity – it is usually transformed into a function of time only L_t and then used to calculate an imporant physical quantity called *action*. We will study a specific example of this below.

Simply speaking, action is the *imbalance of energies accumulated during the particular motion of a system*. In practice it works as follows: One chooses a particular way a mechanical system moves by choosing a particular dependence of position on time x_t. This function immediately determines the velocity $v(t) = \partial_t x$. Then we can calculate kinetic and potential energies and corresponding Lagrange function as a function of time

$$L_t = E_k(t) - E_p(t) \, .$$

With this function, we can calculate action – the total accumulated imbalance of energies – during a specific period of time:

$$A = \int \delta t L_t \, .$$

> ***Stationary Time Principle***
> When a system has constant Lagrange function, then the total action becomes proportional to total time. In this case the principle of least action is known as the *principle of extreme (least or maximal) time*.

One can try *any type of physically allowable motion* x_t and calculate the total action A. Different ways of motion will result in different values for the total action. Turns out, there is one unique function x_t^* for which the value of action A is *stationary* or *extreme* – either maximal or minimal and *insensitive to small deviations* of the motion x_t from its "true" form x_t^*. This is the essense of the *stationary action principle*. It is better to demonstrate it using an example.

Vertical Motion

Consider a body moving vertically. We can imagine two different scenarios: 1) Motion with constant acceleration (e.g. free fall) and 2) Motion with constant speed.

If the initial height is h, velocity zero, and the acceleration a, then the position along the vertical axis y and corresponding velocity are known:

$$y_a = h - at^2/2 , \qquad v_a = -at .$$

The body will reach the "ground" ($y = 0$) at time $T = \sqrt{2h/a}$.

Imagine the second body that begins and ends its motion simultaneously with the first body but moving with constant speed $u = h/T$. Then we can write

$$y_c = h - ut , \qquad v_c = -u .$$

Let's denote the first type of motion as $q_1(t)$ and the second as $q_0(t)$. We can consider other types of motion, as long as they start at $y = h$ when $t = 0$ and end at $y = 0$ when $t = h$. For example, we can consider the following motion:

$$q_\alpha(t) = \alpha q_1 + (1 - \alpha) q_0 , \qquad \alpha \in [0, 1] .$$

It is an type of motion somewhere in between the two variants mentioned above. For the value $\alpha = 0$ of the parameter we recover the motion with constant speed $q_0(t)$, while for $\alpha = 1$ we get the uniformly accelerated motion $q_1(t)$ with zero initial velocity.

It is not difficult to see that the velocity for this general motion is given by similar combination of velocities:

$$v_\alpha(t) = \alpha v_a + (1 - \alpha) v_c .$$

Let's calculate the Lagrange function for a body moving freely, without any forces acting on it. Then the Lagrange function is simply its kinetic energy:

$$L(t) = \frac{m v_\alpha^2}{2} = \frac{m}{2} \left[\alpha^2 v_a^2 + (1 - \alpha)^2 v_c^2 + 2\alpha(1 - \alpha) v_a v_c \right] .$$

Plugging in the expressions for velocity, we arrive at

$$L(t) = \frac{m}{2} \left[\alpha^2 a^2 t^2 + (1 - \alpha)^2 u^2 + 2\alpha(1 - \alpha) a u t \right] .$$

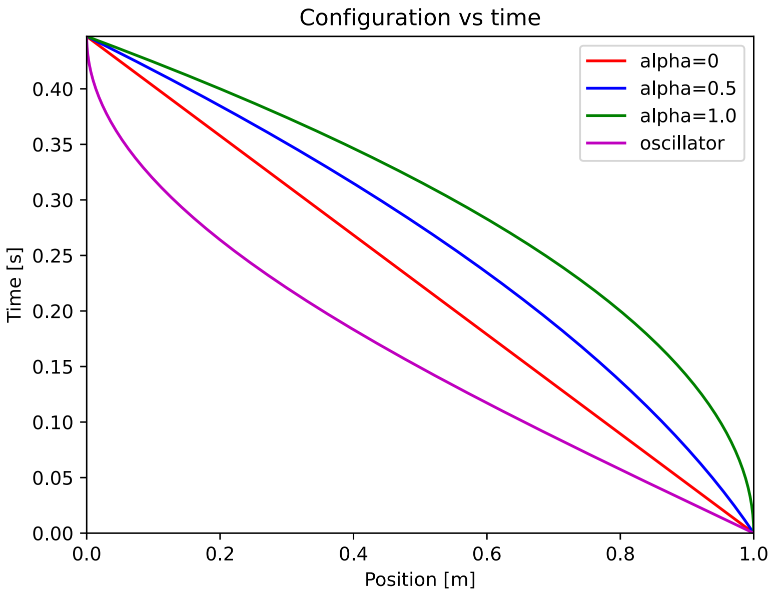


Fig. 4.7: Different types of vertical motion in Lagrange picture. Particles start and stop at the same times and in the same places, moving between initial and final configurations according to different equation $q(t)$.

Now we can calculate the total action accumulated during the motion from $t = 0$ to $t = T$:

$$A = \int L(t)\delta t = \alpha^2 S_1 + (1-\alpha)^2 S_2 + \alpha(1-\alpha)S_3 \,.$$

Here we denoted

$$S_1 = \frac{ma^2}{2}\int t^2\delta t\,, \qquad S_2 = mu^2\int \delta t\,, \qquad S_3 = 2mau\int t\delta t\,.$$

Examining the expression for the total action, we see that A depends on the "mixing" parameter α in a quadratic way:

$$A(\alpha) = (S_1 + S_2 - S_3)\alpha^2 + (S_3 - 2S_2)\alpha + S_2\,.$$

The sums can be evaluated:

$$S_1 = \frac{ma^2}{2}\frac{T^3}{3}\,, \qquad S_2 = mu^2 T = \frac{mh^2}{T}\,, \qquad S_3 = mauT^2 = mahT\,.$$

Recalling that $T = \sqrt{2h/a}$ and therefore $h = aT^2/2$, we find

$$S_3 - 2S_2 = (ma^2T^3/2) - 2ma^2T^4/(4T) = 0\,.$$

The coefficient in front of α^2 is

$$S_1 + S_2 - S_3 = \frac{ma^2T^3}{6} + \frac{ma^2T^3}{4} - \frac{ma^2T^3}{2} = -\frac{ma^2T^3}{12}\,.$$

Finally, the total action depends in the parameter α as

$$A = \frac{ma^2T^3}{12}(3-\alpha^2)\,.$$

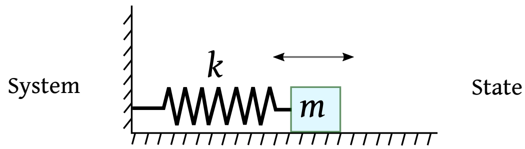
To find the condition for extremal/stationary action, we need to find α that maximizes or minimizes the action. Since $A(\alpha)$ is the inverted parabola with maximum at $\alpha = 0$, the maximal action is achieved for the motion $q_0(t)$ with constant speed without acceleration and forces.

Exercise 4.5

Consider the Lagrange function for a body moving under the force of gravity:

$$L = \frac{mv^2}{2} - mgq\,.$$

Calculate $L(t)$ for $q_\alpha(t)$ and then calculate total action accumulated between $t = 0$ and $t = T$. Find α for which the action $A(\alpha)$ becomes stationary.



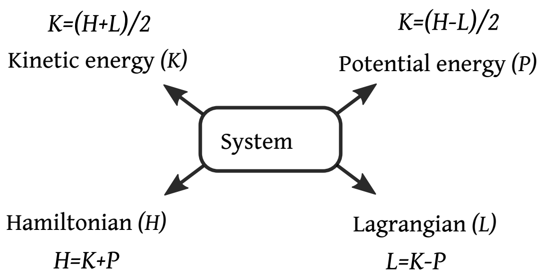


Fig. 4.8: System can be described either by kinetic and potential energies, or Hamiltonian, or Lagrangian.

Exercise 4.6

Consider the following ways of traveling from the height $y = h$ to the ground $y = 0$ in time T:

$$q_1(t) = h - gt^2/2\,,$$

$$q_2(t) = h(1 - t/T)\,,$$

$$q_3(t) = h\left(1 - \sin\frac{\pi t}{2T}\right)\,,$$

$$q_4(t) = h\frac{ee^{-t/T} - 1}{e - 1}\,,$$

and

$$q_5(t) = h(1 - t/T)^2 \,.$$

4.5.2 Summary of Three Mechanics

Let's review what we've learned about three different approaches to study dynamics.

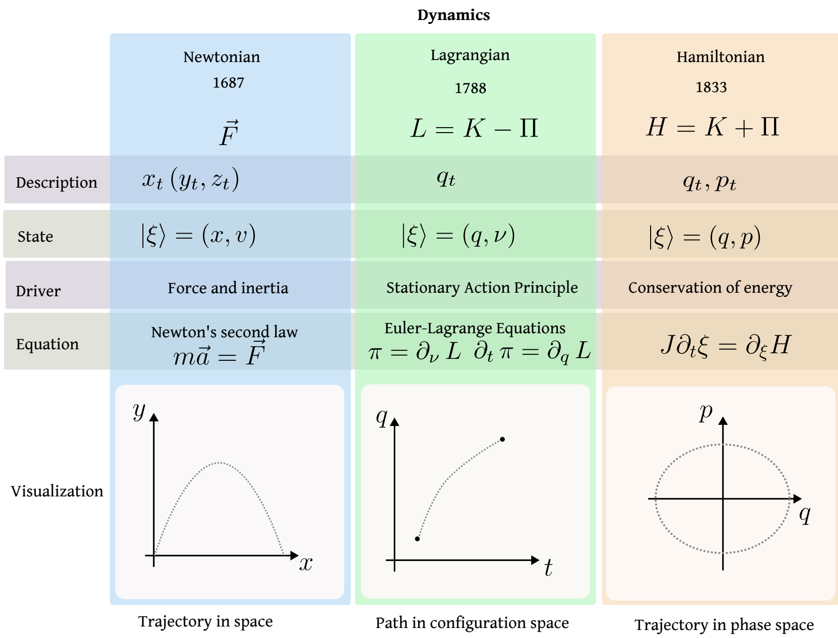


Fig. 4.9: Summary of three different approaches to problems of motion.

State is $|\xi\rangle = (q, u)$ where q is the generalized coordinate and $u = \partial_t q$ is *generalized velocity*. Generalized momentum:

$$\pi = \partial_u L \,,$$

Euler-Lagrange equation:

$$\partial_t \pi = \partial_q L \,.$$

4.6 Field

The *field* concept is extremely important for modern physics, both classical and quantum. Without proper understanding of fields, it is impossible to appreciate the concept of a particle in current theories.

In order to understand fields, we will consider their mathematical aspect first, starting with the simplest type – *scalar field*. Then we will see how the most fundamental physical systems can be described within the framework of fields.

4.6.1 Scalar Field

Imagine we want to describe how temperature varies inside a room, as shown in Figure 4.10. For simplicity, we assume we are interested in the area near the wall with a heater and a window. In this case at each point specified by a pair of coordinates (x, y) we can assign a measured value of the temperature T. This results in a map

$$(x, y) \quad \leftrightarrow \quad T$$

or in a function $T(x, y)$. Since temperature is a simple number (a scalar), the function $T(x, y)$ is called *scalar field*.

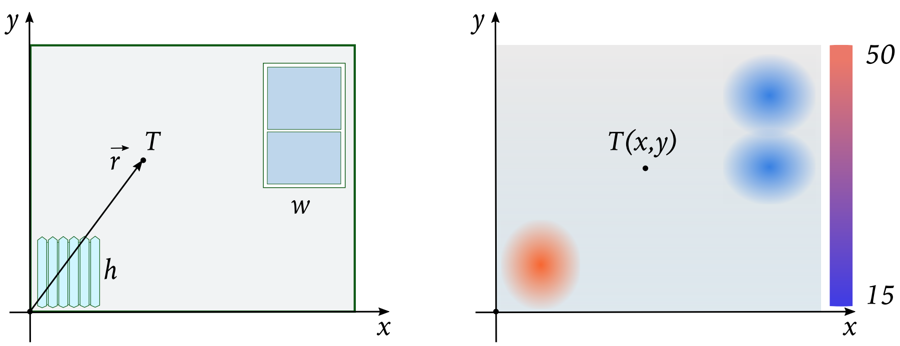


Fig. 4.10: Temperature inside a room can be described using *scalar field* $T(x, y)$.

4.6.2 Vector Field

Next, suppose we study the motion of air in the vicinity of some surface. We can specify how fast and in what direction each "particle of air" or a tiny volume of air is moving at every point of interest, as shown in

Figure 4.11. This way we obtain a map

$$(x, y) \quad \leftrightarrow \quad \vec{v}$$

or a *vector-valued function* $\vec{v}(x, y)$. This is a typical example of a *vector field.*

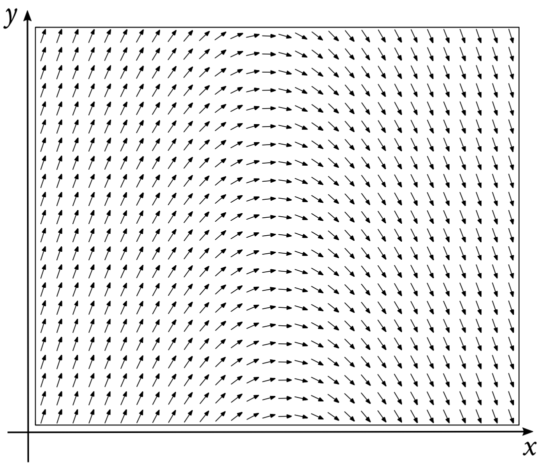


Fig. 4.11: The state of flow of air or water can be described using *vector field* $\vec{v}(x, y)$ which assigns a velocity to each point.

4.6.3 Tensor and Spinor Fields*

The examples of scalar and vector fields can be generalized to include any kind of fields. The core idea is that we are dealing with an *extended system,* some "object occupying region of space" and we want to describe its properties at each point. This way we establish a map

$$\text{point} \quad \leftrightarrow \quad \text{value}\,.$$

The type of value determines the type of field. And what determines the type of value? It is the nature of the system and the specific properties of the physical object we want to study.

For EMF one needs *tensors* to describe the properties correctly. In other words, EMF is mathematically described as a *tensor field.*

For EPF one needs *spinors* – special "vector-like" objects that capture the properties of electrons related to spin. Put differently, EPF is mathematically described as a *spinor field.*

4.6.4 Physical Fields

In physics, *field* is a *dynamical system* with infinite number of degrees of freedom. It's dynamics can be studied using either Lagrangian or Hamiltonian approach.

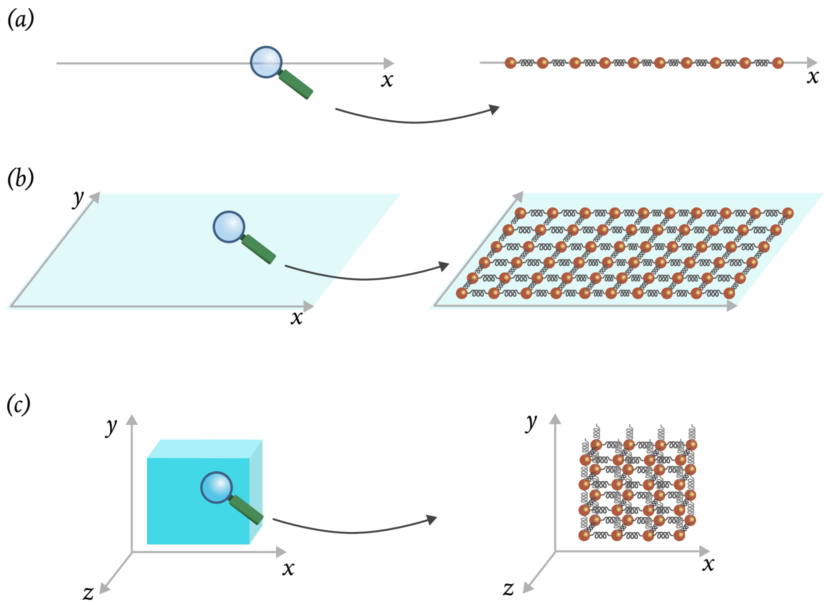


Fig. 4.12: Mechanical model of coupled oscillators can be used to describe physical fields, such as electromagnetic field or electron-positron field.

4.7 Ideal Versus Real

So far when discussing motion of various systems, we assumed that positions, velocities, or momenta and other quantities can be specified with arbitrary precision and, in the limit, absolutely accurately.

Chapter Highlights

- *Operators extends the idea of functions.*
- *Numeric functions (e.g.,* $\sin x$*) act on numbers and yield other numbers. Operators may act on vectors to yield other vectors or numbers.*

- *Linear operators represent the simplest and yet powerful class of operators on vectors.*
- *Linear operators can be represented graphically or symbolically.*

5. Quantum Physics

WE ARE NOW READY TO TAKE THE PREVIOUSLY DEVELOPED CONCEPTS and tools into the realm of quantum phenomena. Firstly, we will upgrade our understanding of *system* and *state* to the quantum level. Then we will see how the concepts of dynamics can be applied to quantum systems in general. Finally, we will do a detailed analysis of quantum dynamics of simplest and yet most important quantum systems: *qubit* and *quantum oscillator*. The latter will lead to the concepts of *quantum field*. We will also discuss the core quantum concept of *entanglement*.

Prerequisite Knowledge
To fully understand the material of this chapter, readers should be comfortable with the following concepts:

- State
- Dynamical equations

5.1 Quantum System

In quantum theory a *system* is understood more broadly than in classical physics. Take, for instance, a "two-body" system consisting of a star and a planet orbiting it. Neglecting the gravitational field, we can say that this is a two-part system which can be at any time described by a state

$$|\xi\rangle = |star\rangle|planet\rangle\ ,$$

where $|star\rangle$ and $|planet\rangle$ are the states of the star and the planet, respectively. In addition to orbital motion, each part of this system can be rotating, and this rotation constitutes additional degree of freedom for each system. Therefore, the state of, for example, the planet has two components:

$$|planet\rangle = |orbital\rangle|spin\rangle\,.$$

The orbital part of the state is described by the position and momentum of the planet: $|orbital\rangle = (x, p)$. The degree of freedom related to the spinning is called *intrinsic degree of freedom* as it is not directly responsible for the (outward) movement of the planet as a whole. In quantum physics such degree of freedom can be the focus of an experiment and can be considered a quantum system of its own right.

As another example consider an hydrogen atom. Viewed as a classical system it has such parts as an electron, a positron, and an electromagnetic field. The electron, however, has an *intrinsic degree of freedom – spin* – which is fundamentally quantum and is unrelated to "spinning". When working with atoms we can affect both "orbital" and intrinsic states of the electron. However, the energy required to do that differs greatly for spin and for the "atom" (electron+positron+field). Thus, we can choose to manipulate spin only and treat this intrinsic degree of freedom as a quantum system in its own right.

Finally, consider a light beam – a special mode of electromagnetic field. Apart from spatial distribution of the power in the beam we must specify its polarization.

These examples are not exhaustive, but hopefully sufficient to convey the idea of quantum system. We thus arrive at the following definition:

Definition 5.1 ☞ ***Quantum System***
A degree of freedom of a physical system can be considered a *quantum system* if that degree of freedom exhibits quantum behavior.

5.2 Fundamental Randomness

5.3 Quantum State

State is the fundamental concept of quantum theory. It is the source of all *predictions* about the results of *any* possible measurement. Classical state of a system tells us what the values of various quantities *certainly are*, while quantum state tells us what the results of an arbitrary measurement *will be* and *with what probability*.

We are looking for a binary operator $\widehat{\sigma}$ that yields a number based on two vectors:

$$|\sigma\rangle\langle\sigma|\vec{a}\,\vec{b} = x\,.$$

5.3.1 Superposition

Suppose we measure the energy of a quantum system (e.g. a hydrogen atom) and find that it takes on discrete values E_1, E_2, E_3, and so on. A distinct quantum state $|e_1\rangle$, $|e_2\rangle$, $|e_3\rangle$, and so on, is associated with each energy value.

If a long series of measurements always yields the same energy value E_k, we conclude that the system was prepared in the state of *definite* energy $|e_k\rangle$ (compare with EPR criterion of reality). If the results of a long series of measurements are random, then the initial state of atoms can not be the state of a definite energy. In practice, a long series of measurements of *identically prepared* atoms reveals certain *statistical* pattern. Specifically, for a number of measurement N, the energy E_1 results in N_1 random outcomes, the energy E_2 results in N_2 random outcomes and so on. One can speak of the *probability* of obtaining the result E_1, defined as the *relative frequency* $p_1 = N_1/N$ of occurrences of E_1; similarly for other results. Generally, the probability of obtaining the result E_k is $p_k = N_k/N$.

If the probabilities $(p_1, p_2, p_3, \ldots, p_k)$ are *reproducible* (the same set of probabilities in different repeated long series of measurements), then the state of system can be characterized by the combination of energy values and their probabilities:

$$|\xi\rangle \;\longrightarrow\; \{(E_1, p_1), (E_2, p_2), (E_3, p_3), \ldots, (E_k, p_k)\}\,.$$

If there is a one-to-one correspondence $E_k \leftrightarrow |e_k\rangle$ between the energy and the state, then the state $|\xi\rangle$ can be said to be *determined* by (or is a function of) the set of pairs $(p_i, |e_i\rangle)$:

$$|\xi\rangle \;\longleftrightarrow\; \{(p_1, |e_1\rangle), (p_2, |e_2\rangle), (p_3, |e_3\rangle), \ldots, (p_k, |e_k\rangle)\}\,.$$

The *simplest* function that constructs one vector from other vectors is a *linear combination*. We can try using such a function to express the state $|\xi\rangle$ in terms of the states with definite energy $|e_i\rangle$ as follows:

$$|\xi\rangle = p_1|e_1\rangle + p_2|e_2\rangle + p_3|e_3\rangle + \ldots p_k|e_k\rangle\,.$$

There is no guarantee that this approach will be effective. In fact, the expression given above is *not the one* used in quantum theory, but it is not far from the "proper" quantum form.

Mixed State
The expression for the state as a linear combination of other states with coefficients equal to corresponding probabilities *can* be applied in systems in so-called *mixed state.*

$$\widehat{\rho} = p_1 \ \widehat{\rho_1} \ + p_2 \ \widehat{\rho_2} \ + \ldots + p_k \ \widehat{\rho_k} \ .$$

Every state can be viewed as a linear combination – called *superposition* – of some other states:

$$|\psi\rangle = c_1|\phi_1\rangle + c_2|\phi_2\rangle + \ldots + c_k|\phi_k\rangle \, .$$

This is analogous to the representation of arrow-vectors in terms of some basis.

Born Rule

5.3.2 States Overlap

$$\langle\psi|\phi\rangle.$$

5.3.3 Pure and Mixed States

State Purification

It is possible to "upgrade" a mixed state of a quantum system S given by a density operator $\widehat{\rho}$ to a pure state $|\Psi\rangle$. The mathematical procedure is called *purification*. Physically, the operation means finding a larger system S_L which contains S as its "part" and can be described using a pure state vector $|\Psi\rangle$.

Purification can always be achieved. One natural choice of a "larger" quantum system S_L can be a system consisting of a pair of S-like systems. As an example, a mixed state of a qubit can be purified to a state of qubit pair.

5.3.4 Fidelity

When the state of a quantum system cannot be described using a state vector $|\psi\rangle$, we need to use a more general approach – *state operator* or *density operator* $\widehat{\rho}$.

How do we compare two states $\widehat{\rho}$ and $\widehat{\sigma}$ for similarity? A useful measure of similarity in this case is called *fidelity* and has the following definition:

$$\mathcal{F}(\widehat{\rho}, \widehat{\sigma}) = \mathtt{Tr}\sqrt{\sqrt{\widehat{\rho}}\ \widehat{\sigma}\ \sqrt{\widehat{\rho}}}\,.$$

This complicated definition captures almost the same idea as the scalar product. In fact, for pure states $\widehat{\rho}= |\psi\rangle\langle\psi|$ and $\widehat{\sigma}= |\phi\rangle\langle\phi|$ it returns the absolute value of the scalar product $|\langle\psi|\phi\rangle|$.

Using fidelity allows comparing pure states and mixed states. Indeed, for a pure state $\widehat{\rho}= |\psi\rangle\langle\psi|$ (a projector) the square root is $\widehat{\rho}$, and

$$\widehat{\rho}\widehat{\sigma}\widehat{\rho}= \langle\psi|\ \widehat{\sigma}\ |\psi\rangle\ \widehat{\rho}\ \ .$$

Taking the square root of the last expression gives

$$\alpha\,\widehat{\rho}\,,$$

where $\alpha = \sqrt{\langle\psi|\ \widehat{\sigma}\ |\psi\rangle}$ is a number. Finally, using the fact that $\mathtt{Tr}\ \widehat{\rho}= 1$, we obtain

$$\mathcal{F}(\widehat{\rho}, \widehat{\sigma}) = \sqrt{\langle\psi|\ \widehat{\sigma}\ |\psi\rangle}\,.$$

Recall that $\langle\psi|\ \widehat{\sigma}\ |\psi\rangle$ is the expectation value of the operator $\widehat{\sigma}$ in the state $|\psi\rangle$.

Exercise 5.1

Prove that the definition of fidelity implies that it is symmetric in its arguments.

5.4 Correspondence Principle

5.5 Quantum Dynamics

We are looking for a binary operator $\widehat{\sigma}$ that yields a number based on two vectors:

$$|\sigma\rangle\langle\sigma|\,\vec{a}\,\vec{b} = x\,.$$

5.6 Quantum Hamiltonian

We are looking for a binary operator $\widehat{\sigma}$ that yields a number based on two vectors:

$$|\sigma\rangle\langle\sigma|\,\vec{a}\,\vec{b} = x\,.$$

5.7 Quantum Bit

Any quantum system with two active states is called a *qubit*. The state with lower energy is usually called *ground state* and denoted as $|g\rangle$ or $|0\rangle$ (zero). The state with higher energy is usually called *excited state* and denoted as $|e\rangle$ or $|1\rangle$ (one). The notation $|0\rangle, |1\rangle$ is used in the field of quantum information and computation.

If the energy of the ground and excited states are E_g and E_e, respectively, then the Hamiltonian of a qubit can be written using projectors

$$\widehat{H} = E_g|g\rangle\langle g| + E_e|e\rangle\langle e| .$$

It requires an energy $\Delta E = E_e - E_g$ to excite the qubit from the lower energy state to the higher energy state. This energy may come from a quantum of electromagnetic field oscillating with frequency $\omega = \Delta E/\hbar$.

5.7.1 Flipping Operator

Transition between the states of a qubit can be described mathematically using operators that map one state into another. For example, an operator $\widehat{F}$ that *flips* states must do the following:

$$\widehat{F}\ |0\rangle = |1\rangle , \quad \widehat{F}\ |1\rangle = |0\rangle .$$

Such operator can be easily built from the tensor products:

$$\widehat{F} = |1\rangle\langle 0| + |0\rangle\langle 1| .$$

Each term in this sum is useful in quantum theory. The first term is called *raising operator* and is denoted as $\widehat{\sigma}_+ = |1\rangle\langle 0|$. The second term is called *lowering operator* and is denoted as $\widehat{\sigma}_- = |0\rangle\langle 1|$. Apparently, the raising operator excites the qubit from the ground state, while the lowering operator brings the qubit down from the excited state.

Exercise 5.2
Calculate $\widehat{F}^2$.

Exercise 5.3
Calculate (a) $\widehat{\sigma}_+\widehat{\sigma}_+$; (b) $\widehat{\sigma}_-\widehat{\sigma}_-$; (c) $\widehat{\sigma}_+\widehat{\sigma}_-$; (d) $\widehat{\sigma}_-\widehat{\sigma}_+$.

Exercise 5.4
Show that $\widehat{\sigma}_+\widehat{\sigma}_- + \widehat{\sigma}_-\widehat{\sigma}_+ = \widehat{I}$, where $\widehat{I}$ is the identity operator.

Exercise 5.5
Show that the qubit Hamiltonian can be written in terms of the raising and lowering operators as follows:

$$\widehat{H} = \hbar\omega \left(\widehat{\sigma}_+\widehat{\sigma}_- + \epsilon\,\widehat{I}\right) ,$$

where $\epsilon = E_g/\Delta E$.

5.7.2 Number Operator

The operator $\widehat{n} = \widehat{\sigma}_+\widehat{\sigma}_-$ is called *number operator* for the following reason. First, note that $\widehat{\sigma}_+\widehat{\sigma}_- = |1\rangle\langle 1|$ is the projector on the excited state of qubit.

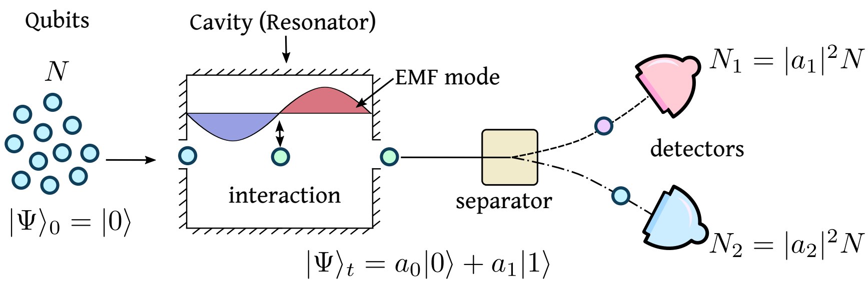


Fig. 5.1: Qubits initially in the ground state $|0\rangle$ travel through a cavity resonator with an electromagnetic field mode attuned to the qubits transition energy $\Delta E = \hbar\omega$. After interaction, the state of each qubit changes to $|\Psi_t\rangle$. Then the qubits are "measured"/separated into two groups, based on their final state.

Next, consider an experiment schematically shown in Figure 5.1. In this experiment a large number N of qubits start in the ground state $|0\rangle$ and go through a region (cavity) where they can interact with an oscillating electromagnetic field (mode). After leaving the cavity, each qubit goes through a special device which splits the beam into two parts, depending on the measured state of a qubit. The top detector detects N_1 qubits in state $|1\rangle$, the bottom detector detects N_2 qubits in ground state. The N_1 qubits correspond to the number of energy quanta absorbed from cavity. This number can be expressed as

$$N_1 = N|a_1|^2 = N\langle\Psi_t|\,\widehat{n}\,|\Psi_t\rangle\,.$$

In other words, the expectation value of the operator $\widehat{n}$ determines the number of energy quanta absorbed from the mode of electromagnetic field inside the cavity.

5.8 Quantum Oscillator

The *principle of the quantization of action* can be applied to harmonic oscillator. The result is the quantization of energy levels.

The energy of a harmonic oscillator can be expressed in terms of the maximum momentum p_m or in terms of the maximum displacement x_m:

$$H = \frac{p_m^2}{2m} \quad \text{or} \quad H = \frac{kx_m^2}{2}\,.$$

Multiplying these two equalities and recalling that $\omega^2 = k/m$, we obtain

$$H = \frac{\omega x_m p_m}{2}\,.$$

The path which the state vector $|\xi\rangle = (x, p)$ follows in phase space is an ellipsis with the major semi-axes x_m and p_m. The area of this ellipsis is $A = \pi x_m p_m$. Therefore, the connection between the energy of harmonic oscillator and the area is given by

$$H = \frac{\omega}{2\pi}A\,.$$

The area A is a physical quantity with the units of *action*.

As shown in Figure 5.2(a), areas in phase space have the smallest size limited by the elementary quantum of action h – known as Planck constant. The quantization of action and, consequently, the quantization of phase-space area, has two important implications for harmonic oscillator.

First, every time an oscillator absorbs some energy ΔE, the maximum deviation and the maximum momentum increase. The ellipsis in phase space increases its area. But since the area in phase space can't grow continuously–changes in discrete quanta $\delta A = h$–we must have discrete changes in energy. Second, the existence of the elementary quantum

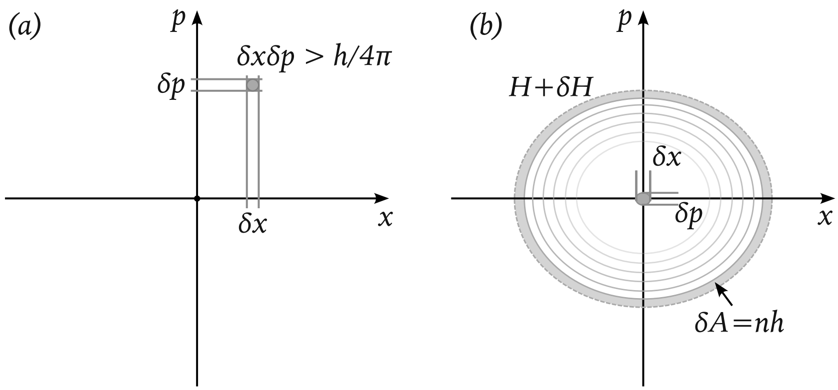


Fig. 5.2: Areas of phase-space regions have the units of action. Quantization of action implies quantization of phase-space area. (a) The smallest area in phase space is limited by the fundamental quantum of action h – Planck's constant. (b) Area of the ellipsis inside the path of harmonic oscillator is proportional to its energy. Quantization of area leads to the quantization of energy of harmonic oscillator.

of action and the smallest are in phase space, require that the lowest energy state of harmonic oscillator is described not by a point in phase space, but by an elementary ellipsis such that $A_0 = \delta x \delta p \propto h$. Putting these two ideas together, we conclude that the area of the ellipsis can be written as

$$A_n = A_0 + nh\,.$$

The energy of the oscillator then takes the form

$$H = n\hbar\omega + E_0\,,$$

where $\hbar = h/2\pi$ is called *reduced Planck's constant*, and E_0 is the lowest energy of the harmonic oscillator. From the expression for H follows that harmonic oscillator can be in a countable set of states, growing in energy from E_0 by a fixed step $\hbar\omega$.

The energy of the lowest state can be written in terms of the step size $\hbar\omega$: $E_0 = e_0\hbar\omega$, where e_0 is some number (it will be found later). Finally, we can write the energy of harmonic oscillator as follows:

$$H = \hbar\omega(n + e_0)\,.$$

5.8.1 Hamiltonian Operator

For any quantum system with discrete energy states $E_0, E_1, E_2, \ldots, E_n \ldots$ the Hamiltonian operator can be written in terms of projectors:

$$\widehat{H} = \int E_k |k\rangle\langle k| \,, \quad k = 0, 1, 2, \ldots, n \ldots$$

For harmonic oscillator $E_k = E_0 + k\hbar\omega$ for $n > 1$ and the Hamiltonian operator can be written as follows

$$\widehat{H} = \int (E_0 + k\hbar\omega)|k\rangle\langle k| \,.$$

The number k tells how many excitations quantum oscillator absorbed to reach the energy state $|k\rangle$. Opening the parentheses and recalling that

$$\int |k\rangle\langle k| = \widehat{I} \,,$$

we obtain

$$\widehat{H} = E_0 \, \widehat{I} + \hbar\omega \int k|k\rangle\langle k| \,.$$

This expression is very similar to the Hamiltonian of a qubit

$$\widehat{H}_{qb} = E_g \, \widehat{I} + \hbar\omega \, \widehat{n}$$

where $\widehat{n} = \widehat{\sigma}_+ \widehat{\sigma}_-$ is a number operator. The similarity is not accidental, as the operator $\widehat{n} = \int k|k\rangle\langle k|$ plays the role of the number operator. Indeed, it is easy to check by direction application that:

$$\widehat{n} \; |n\rangle = n|n\rangle \,.$$

In other words, the energy states $|n\rangle$ of harmonic oscillator, are the eigen-states of the number operator $\widehat{n}$ with the eigen-value n corresponding to the number of excitation level.

Exercise 5.6
Prove that $\widehat{n} \; |n\rangle = n|n\rangle$ by direction application of the operator $\widehat{n} = \int k|k\rangle\langle k|$.

The expression for the number operator $\widehat{n}$ can be obtained in a different way. First note that

$$\widehat{H} \; |n\rangle = E_n|n\rangle \,, \quad \text{where } E_n = E_0 + n\hbar\omega \,.$$

From this follows

$$(\widehat{H} - E_0\,\widehat{I})\,|n\rangle = n\hbar\omega\,|n\rangle\,,$$

and, consequently,

$$\frac{(\widehat{H} - E_0\,\widehat{I})}{\hbar\omega}\,|n\rangle = n\,|n\rangle\,.$$

The operator on the left hand side of this equation is the number operator $\widehat{n}$. It can be simplified once we recall that

$$\widehat{H} = \int E_k|k\rangle\langle k| \quad \text{and} \quad \widehat{I} = \int |k\rangle\langle k|\,.$$

Using these relations, we first write

$$\widehat{H} - E_0\,\widehat{I} = \int (E_k - E_0)|k\rangle\langle k|\,.$$

Then, remembering that $E_k = E_0 + k\hbar\omega$, we immediately arrive at

$$\widehat{n} = \frac{(\widehat{H} - E_0\,\widehat{I})}{\hbar\omega} = \int k|k\rangle\langle k|\,.$$

Thus, the operator of quantum harmonic oscillator can be written in the following form:

$$\widehat{H}_{osc} = E_0\,\widehat{I} + \hbar\omega\,\widehat{n}\;\;.$$

5.8.2 Ladder Operators

The number operator for qubit could be expressed as the product of two simple operators that raised or lowered qubit states:

$$\widehat{n} = \widehat{\sigma}_+ \widehat{\sigma}_-\;\;.$$

The idea of raising and lowering states is also applicable to harmonic oscillator. Similar to qubit, we can write such operators as tensor products:

$$\widehat{a}_+ = |n+1\rangle\langle n| \text{ and } \quad \widehat{a}_- = |n-1\rangle\langle n|\,.$$

Unfortunately, these operators will act properly only on the state $|n\rangle$.

Exercise 5.7
Evaluate (a) $\widehat{a}_+\;|n\rangle$; (b) $\widehat{a}_-\;|n\rangle$; (c) $\widehat{a}_+\;|n+m\rangle$; (d) $\widehat{a}_+\;|n+m\rangle$.

It is easy to fix this problem by summing over all states:

$$\widehat{a}_+ = \int |k+1\rangle\langle k| \quad \text{and} \quad \widehat{a}_- = |0\rangle\langle 0| + \int |m-1\rangle\langle m| \,, \quad m > 0 \,.$$

The first term in the expression for $\widehat{a}_-$ ensures that the vacuum state remains unchanged: $\widehat{a}_- \; |0\rangle = |0\rangle$.

Exercise 5.8
Evaluate (a) $\widehat{a}_+ \; |n\rangle$; (b) $\widehat{a}_- \; |n\rangle$.

Susskind-Glogower Operator
The operator $\widehat{a}_+$ is known as *Susskind-Glogower operator.*

Let's check whether $\widehat{a}_+\widehat{a}_-$ yields the number operator $\widehat{n} = \int k|k\rangle\langle k|$. Even without explicitely evaluating the composition $\widehat{a}_+\widehat{a}_-$ we can see that it is unlikely to contain the required factor k.

Exercise 5.9
Show that $\widehat{a}_+\widehat{a}_- = |1\rangle\langle 0| - |0\rangle\langle 0| + \widehat{I}$.

To find better operators for raising and lowering states of harmonic oscillator, we can taken a closer look at the qubit case. There we had $\widehat{\sigma}_+ \; |0\rangle = 1|1\rangle$ and $\widehat{\sigma}_- \; |1\rangle = 1|0\rangle$. We explicitly added "1" in front of the final states, to highlight the following property of the $\widehat{\sigma}$-operators:

$$\widehat{\sigma}_+ \; |k\rangle = \sqrt{k+1}|k+1\rangle \quad \text{and} \;\; \widehat{\sigma}_- \; |k\rangle = \sqrt{k}|k-1\rangle \,.$$

Thus, we can "upgrade" the raising and lowering operators $\widehat{a}_+$ and $\widehat{a}_-$ to include the information about the state they act on. We want them to behave as follows:

$$\widehat{a}_+ \; |k\rangle = \sqrt{k+1}|k\rangle \quad \text{and} \quad \widehat{a}_- \; |m\rangle = \sqrt{m}|m-1\rangle \,.$$

Exercise 5.10
Evaluate $(\widehat{a}_+)^p \; |0\rangle$.

Exercise 5.11
(a) Show that the upgraded operators have the property

$$\widehat{a}_{+}\widehat{a}_{-}\ |m\rangle = m|m\rangle \quad m > 0\,.$$

(b) Evaluate $\widehat{a}_{-}\widehat{a}_{+}\ |m\rangle$.

Exercise 5.12
(a) Show that

$$\widehat{a}_{-}\widehat{a}_{+} - \widehat{a}_{+}\widehat{a}_{-}=\widehat{I}\ \ .$$

Such raising and lowering operators (also called *ladder operators*) are very useful when working with quantum harmonic oscillators. In terms of the ladder operators, the Hamiltonian of quantum oscillator is written as

$$\widehat{H}_{osc}= \hbar\omega\ \widehat{n}\ + E_0\ \widehat{I}\,,$$

where the number operator $\widehat{n}=\widehat{a}_{+}\widehat{a}_{-}$.

5.8.3 Conjugation

The raising operator $\widehat{a}_{+}$ can be written in terms of the tensor products:

$$\widehat{a}_{+}= \int_0 \sqrt{k+1}|k+1\rangle\langle k|\,.$$

If we limit the lowering operator to states $|m\rangle$ with $m > 0$, then it also allows a simple representation

$$\widehat{a}_{-}= \int_1 \sqrt{m}|m-1\rangle\langle m|\,.$$

By changing the summation variable $m-1 = k$ (and, therefore, $m = k+1$), we can re-write the summation over $k = 0, 1, 2\ldots$:

$$\widehat{a}_{-}= \int_0 \sqrt{k+1}|k\rangle\langle k+1|\,.$$

Now the expression for $\widehat{a}_{-}$ became similar to the expression for $\widehat{a}_{+}$, with the exception that the order of states in the tensor product is flipped:

$$|k+1\rangle\langle k| \leftrightarrow |k\rangle\langle k+1|\,.$$

This change of order of factors in a tensor product is called *conjugation*. The operators $\widehat{a}_-$ and $\widehat{a}_+$ are therefore related to each other via the *conjugation operation*. These operators are said to be *conjugates* of each other.

The relation of conjugation gives some insight into what the lowering operator $\widehat{a}_-$ does to the vacuum state:

$$\widehat{a}_- \; |0\rangle = \int_0 \sqrt{k+1}|k\rangle\langle k+1|0\rangle = 0 \int_0 \sqrt{k+1}|k\rangle = 0|\infty\rangle \,,$$

where we introduced a vector

$$|\infty\rangle = |0\rangle + \sqrt{2}|1\rangle + \sqrt{3}|2\rangle + \ldots + \sqrt{n+1}|n\rangle + \ldots$$

Obviously, $|\infty\rangle \neq |0\rangle$. The overall factor of zero negates any possible contributions of $|\infty\rangle$, making the product $0|\infty\rangle$ a special "zero vector" $|z_0\rangle$, with the natural property

$$|k\rangle + |z_0\rangle = |k\rangle \,.$$

The vector $|z_0\rangle$ does not correspond to any physical state, but represents a mathematical "zero vector". Since for all mathematical manipulations the vectors $0|\infty\rangle$ and $0|0\rangle$ are equivalent, we can express the action of the lowering operator $\widehat{a}_-$ on the vacuum state as follows:

$$\widehat{a}_- \; |0\rangle = 0|0\rangle \,.$$

Finally, the action of the number operator $\widehat{n}$=$\widehat{a}_+\widehat{a}_-$ on the vacuum state can be evaluated:

$$\widehat{a}_+\widehat{a}_- \; |0\rangle = \widehat{a}_+ \; (\widehat{a}_- \; |0\rangle) = 0(\widehat{a}_+ \; |0\rangle) = 0|1\rangle = 0|0\rangle \,,$$

here we used the mathematical equivalence of states $0|1\rangle$ and $0|0\rangle$.

Dagger Notation

The relation of conjugation between operators is denoted using a special notation. For example, if we denote the lowering operator $\widehat{a}_-$ simply as $\widehat{a}$, then its conjugate operator– raising operator– is denoted using a special "dagger" symbol as the superscript:

$$\widehat{a}_+ = \widehat{a}^\dagger \;\; .$$

The use of dagger notation is standard in quantum theory.

Let's use the dagger notation to summarize the basis facts about the ladder operators, the number operator, and the Hamiltonian of quantum oscillator. First, raising and lowering properties:

$$\widehat{a}\ |n\rangle = \sqrt{n}|n-1\rangle\,, \qquad \widehat{a}^{\dagger}\ |n\rangle = \sqrt{n+1}|n+1\rangle\,.$$

Second, number operator and commutator:

$$\widehat{a}^{\dagger}\,\widehat{a}\ |n\rangle = n|n\rangle\,, \qquad \widehat{a}\,\widehat{a}^{\dagger} - \widehat{a}^{\dagger}\,\widehat{a} = \widehat{I}\ .$$

Finally, conjugation relation between the ladder operators:

$$\widehat{a} \xrightarrow{\dagger} \widehat{a}^{\dagger}\ .$$

Normal Order

Exercise 5.13

Ladder operators are used many important applications of quantum theory. Often one encounters expressions with several operators in no particular order, for example $\widehat{X} = \widehat{a}^{\dagger}\,\widehat{a}^{2}\,\widehat{a}^{\dagger}\,\widehat{a}$. For calculations it is necessary to rearrange these operators into a *normal order* where all raising operators appear on the left, before the lowering operators.

Use the commutation relation $\widehat{a}\,\widehat{a}^{\dagger} - \widehat{a}^{\dagger}\,\widehat{a} = \widehat{I}$ to put $\widehat{X}$ into a normal order.

5.8.4 Canonical Commutation

The Hamiltonian operator for quantum harmonic oscillator can be written in different ways. One way relies on energy eigen-values E_k:

$$\widehat{H} = \int_0 E_k|k\rangle\langle k|\,.$$

Another way utilizes raising and lowering operators:

$$\widehat{H} = \hbar\omega\,\widehat{a}^{\dagger}\,\widehat{a} + E_0\,\widehat{I}\ .$$

However, the starting point was the expression in terms of position and momentum. The question then becomes whether we can introduce *position and momentum operators* such that for harmonic oscillator we get

$$\widehat{H} = \frac{\widehat{p}^2}{2m} + \frac{m\omega^2\,\widehat{x}^2}{2}\,.$$

We already saw in Exercise X the hint that some relationship must exist between the operators $\widehat{a}$, $\widehat{a}^\dagger$ and $\widehat{x}$, $\widehat{p}$. Such relationship must be linear in order to transform the expression for $\widehat{H}$ quadratic in terms of raising and lowering operator

$$\widehat{H} = E_0\,\widehat{a}\,\widehat{a}^\dagger + (\hbar\omega - E_0)\,\widehat{a}^\dagger\,\widehat{a} = \square\,\widehat{a}\,\widehat{a}^\dagger + \square\,\widehat{a}^\dagger\,\widehat{a}$$

into the expression quadratic in terms of position and momentum

$$\widehat{H} = \square\,\widehat{p}^2 + \square\,\widehat{x}^2\,.$$

We are thus looking for a linear transformation

$$\widehat{x} = A\,\widehat{a} + B\,\widehat{a}^\dagger \qquad \text{and} \qquad \widehat{p} = C\,\widehat{a} + D\,\widehat{a}^\dagger$$

which will lead to the Hamiltonian operator $\widehat{H} = E_0\,\widehat{a}\,\widehat{a}^\dagger + (\hbar\omega - E_0)\,\widehat{a}^\dagger$ $\widehat{a}$.

Exercise 5.14

Show that the Hamiltonian operator for harmonic oscillator in terms of the unknown coefficients A, B, C and D has the form:

$$\widehat{H} = \left(\frac{C^2}{2m} + \frac{m\omega^2 A^2}{2}\right)\widehat{a}^2 + \left(\frac{D^2}{2m} + \frac{m\omega^2 B^2}{2}\right)\widehat{a}^\dagger + \\ + \left(\frac{CD}{2m} + \frac{m\omega^2 AB}{2}\right)\widehat{a}\,\widehat{a}^\dagger + \left(\frac{CD}{2m} + \frac{m\omega^2 AB}{2}\right)\widehat{a}^\dagger\,\widehat{a}\,.$$

Exercise 5.15

Using the result of the previous exercise, show that it implies that $E_0 = \hbar\omega/2$.

Exercise 5.16

Using the results of the two previous exercises, show that the four unknown coefficients A, B, C and D satisfy the following equations:

$$C^2 = -(m\omega A)^2\,,$$

$$D^2 = -(m\omega B)^2\,,$$

and

$$CD + (m\omega)^2 AB = m\hbar\omega\,.$$

Exercise 5.17

Using the result of the previous exercise, show that $CD = (m\omega)^2 AB$ (convince yourself that CD can't be $CD = -(m\omega)^2 AB$!). Then show that one possible solution is the set of coefficients:

$$A = B = \sqrt{\frac{\hbar}{2m\omega}}\,,$$

and

$$C = -D = -\,\widehat{J}\,\sqrt{\frac{\hbar m\omega}{2}}\,.$$

With the steps outlined above, we obtain the following expressions for the operators of position and momentum:

$$\widehat{x} = \sqrt{\frac{\hbar}{2m\omega}}\left(\widehat{a}^\dagger + \widehat{a}\right) = x_\omega \left(\widehat{a}^\dagger + \widehat{a}\right)$$

and

$$\widehat{p} = \widehat{J}\,\sqrt{\frac{\hbar m\omega}{2}}\left(\widehat{a}^\dagger - \widehat{a}\right) = \widehat{J}\; p_\omega \left(\widehat{a}^\dagger - \widehat{a}\right)\,.$$

Using these relations, it is now easy to find so called *canonical commutation relation* for the basic physical operators of position and momentum. First, we find

$$\widehat{x}\widehat{p} = \widehat{J}\;\frac{\hbar}{2}\left(\widehat{a}^\dagger\widehat{a}^\dagger - \widehat{a}\widehat{a} + \widehat{a}\widehat{a}^\dagger - \widehat{a}^\dagger\widehat{a}\right),$$

then

$$\widehat{p}\widehat{x} = \widehat{J}\;\frac{\hbar}{2}\left(\widehat{a}^\dagger\widehat{a}^\dagger - \widehat{a}\widehat{a} + \widehat{a}^\dagger\widehat{a} - \widehat{a}\widehat{a}^\dagger\right).$$

Subtracting the latter equation from the former, we arrive at

$$\widehat{x}\widehat{p} - \widehat{p}\widehat{x} = [\widehat{x}, \widehat{p}] = \widehat{J}\,\hbar[\widehat{a}, \widehat{a}^{\dagger}] = \widehat{J}\,\hbar\,.$$

5.9 Physical Realization of Qubits

Recall that harmonic oscillator is *any* physical system with the Hamiltonian in the form

$$H = \frac{p^2}{2m} + \frac{kx^2}{2}\,.$$

Many physical systems can be described using this Hamiltonian and thus provide specific *realizations* of the oscillator model. Similarly, many concrete physical systems realize the idea of a qubit.

5.10 Interacting Qubits

Isolated quantum systems are idealizations. Interactions between systems are not only hard to avoid, but are often desirable for practical purposes. We now turn our attention to the interaction between a pair of simplest quantum systems − quantum qubits.

5.10.1 Joint State

Treating a pair of qubits as a single quantum system implies describing the state of two qubits with a single state vector $|\Psi\rangle$. It is called a *joint state* vector and is used to describe the results of measurements performed on a pair of qubits.

To measure two separate parts of a single quantum system one may use two detectors. Each detector measures a single qubit and finds it in either state $|0\rangle$ or $|1\rangle$ with different probabilities. The results of a *joint measurement* can be expressed as a linear combination of all possibilities:

$$|\Psi\rangle = c_{00}|0\rangle|0\rangle + c_{01}|0\rangle|1\rangle + c_{10}|1\rangle|0\rangle + c_{11}|1\rangle|1\rangle\,.$$

For example, if all results are equally probable, the state of the qubit-pair system can be written as follows:

$$|\Psi_0\rangle = (|0\rangle|0\rangle + |0\rangle|1\rangle + |1\rangle|0\rangle + |1\rangle|1\rangle)\,/2\,.$$

There are no *correlations* between the measurements for this state. Two qubits are not only spatially separated, but also *statistically unrelated.* In this case each qubit can be assigned its own state vector and the joint state $|\Psi\rangle$ is a product of two states:

$$|\Psi_0\rangle = |\psi\rangle|\phi\rangle$$

where $|\psi\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and $|\phi\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$.

Whenever the results from two detectors are correlated, the separation of a qubit pair into independent parts becomes impossible. The state $|\Psi\rangle$ then can't be factorized: $|\Psi\rangle \neq |\psi\rangle|\phi\rangle$. Indeed, suppose the state is

$$|\Psi_1\rangle = (|0\rangle|0\rangle + |1\rangle|1\rangle)/\sqrt{2}\,.$$

Assuming that $|\Psi\rangle = |\psi\rangle|\phi\rangle$ and $|\psi\rangle = a|0\rangle + b|1\rangle$ and $|\phi\rangle = c|0\rangle + d|1\rangle$, we get four requirements for the coefficients a, b, c and d:

$$ac = 1/\sqrt{2}\,, \qquad ad = 0$$
$$bd = 1/\sqrt{2}\,, \qquad bc = 0\,.$$

These requirements can't be satisfied, because a and d can't be zero, while their product must be zero (similarly b and c).

The state $|\Psi_1\rangle = (|0\rangle|0\rangle + |1\rangle|1\rangle)/\sqrt{2}$ is an example of *entangled* state of a qubit pair. It illustrates a peculiar quantum situation when the complete knowledge about the joint system does not mean complete knowledge of its "parts". In other words, in a strongly *correlated system* there are no independent parts which can be described with their own individual state vectors. Qubit pair acts as *one inseparable whole*.

Exercise 5.18

Suppose a qubit pair is in the state

$$|\Psi_2\rangle = (|0\rangle|1\rangle + |1\rangle|0\rangle)/\sqrt{2}\,.$$

Show that this state can't be written as the product state:

$$|\Psi_2\rangle \neq |\psi\rangle|\phi\rangle\,.$$

5.10.2 Computational Basis

The four joint states of a qubit pair

$$|\Upsilon_1\rangle = |0\rangle|0\rangle, \ |\Upsilon_2\rangle = |0\rangle|1\rangle, \ |\Upsilon_3\rangle = |1\rangle|0\rangle, \ |\Upsilon_4\rangle = |1\rangle|1\rangle$$

are called *computational basis*. It is often used in the field of quantum information processing. Each state of the computational basis has the form of a product of states of separate qubits. State vectors of the computational basis are *not entangled*.

A general state of a qubit pair can be expanded in the computational basis:

$$|\Psi\rangle = \int c_i|\Upsilon_i\rangle \, .$$

A general operator representing processes involving qubit pair can also be written using computational basis:

$$\widehat{A} = \int a_{ij}|\Upsilon_i\rangle\langle\Upsilon_j| \, .$$

For example, transition operator describing the transfer of energy from the first qubit to the second takes the form $\widehat{T}_{12} = |\Upsilon_2\rangle\langle\Upsilon_3|$.

Exercise 5.19

Write general form of Hamiltonian of a qubit pair in the computational basis.

$$\widehat{H} = \int h_{ij}|\Upsilon_i\rangle\langle\Upsilon_j| \, .$$

Q: Are there other states, which are also basis and product? Smth like

$$|\Xi\rangle = |+\rangle|+\rangle \, .$$

5.10.3 Creating Entanglement

How do qubits become *entangled*? Let's show that an interaction between qubits, responsible for the transfer of energy from the first qubit to the second, can, in certain scenarios, lead to an entangled state. The mechanism described below is universal in the sense that it works for any two interacting quantum systems, not just a pair of qubits.

Energy exchange between two qubits implies the following transition:

$$|1\rangle|0\rangle \quad \longrightarrow \quad |0\rangle|1\rangle\,.$$

This is a process in which the first qubit is initially excited, and the second qubit is in the ground state, but in the end the first qubit drops into the ground state, while the second qubit becomes excited. Mathematically, this transition is described by a product of operators:

$$\widehat{T}_{12}{=}\widehat{\sigma}_1 \otimes \widehat{\sigma}_2^+{=}\widehat{\sigma}_1\widehat{\sigma}_2^+ \ .$$

The qubits are identical and there is no fundamental reason for the energy to flow only in one direction. Therefore, the exchange must be reversible and described not only by $\widehat{T}_{12}$, but also by the opposite process:

$$\widehat{T}_{21}{=}\widehat{\sigma}_1^+ \otimes \widehat{\sigma}_2{=}\widehat{\sigma}_1^+\widehat{\sigma}_2 \ .$$

Hamiltonian with interaction between two qubits can then be written as follows:

$$\widehat{H}_\epsilon{=}\widehat{H}_1\widehat{I}_2 + \widehat{I}_1\widehat{H}_2 + \epsilon\left(\widehat{T}_{12} + \widehat{T}_{21}\right)\,, \tag{5.1}$$

where the parameter ϵ determines the strength of the interaction. Obviously, for $\epsilon = 0$ there is no interaction between the qubits.

Exercise 5.20
Show that the state $|\Psi_2\rangle = (|0\rangle|1\rangle + |1\rangle|0\rangle)/\sqrt{2}$ is the eigen-state of the Hamiltonian (5.1). Find the eigen-value (energy) for this state.
Suggestion: Use the fact that $E_1 = E_0 + \hbar\omega$.

Exercise 5.21
Check whether the state $|\Psi_3\rangle = (|0\rangle|1\rangle - |1\rangle|0\rangle)/\sqrt{2}$ is the eigen-state of the Hamiltonian (5.1).

Exercise 5.22
Find the result of the action of the Hamiltonian (5.1) on the state $|1\rangle|0\rangle$.

Now we can find how the state of the qubit pair changes from the initial $|1\rangle|0\rangle$ at time $t = 0$ to an entangled state a moment later. Schrödinger equation states

$$\partial_t |\Psi\rangle = -\frac{\widehat{J}}{\hbar}\ \widehat{H}_\epsilon\ |\Psi\rangle\,.$$

Therefore, after a time interval δt, the state evolves as

$$|\Psi\rangle \quad \longrightarrow \quad |\Psi\rangle + \delta|\Psi\rangle$$

where

$$\delta\,|\Psi\rangle = -\frac{\widehat{J}\ \delta t}{\hbar}\ \widehat{H}_\epsilon\ |\Psi\rangle\,.$$

For $|\Psi\rangle = |1\rangle|0\rangle$ the right-hand side becomes

$$\delta\,|\Psi\rangle = a|1\rangle|0\rangle + b|0\rangle|1\rangle\,,$$

where

$$a = -\ \widehat{J}\ (2E_0/\hbar + \omega)\delta t\,, \quad b = -\ \widehat{J}\ \epsilon/\hbar\delta t\,.$$

The key point is that due to the interaction term $\widehat{T}_{12}$ the initial state evolves into the superposition

$$|\Psi\rangle = (1+a)|1\rangle|0\rangle + b|0\rangle|1\rangle$$

which clearly has the form of an entangled state. Continuing this process, we will arrive, step by step, to a fully entangle state $|\Psi_2\rangle = (|0\rangle|1\rangle + |1\rangle|0\rangle)\,/\sqrt{2}$.

Where is Energy?
When a qubit pair is entangled, neither qubit has definite energy. However, the system as a whole does have a certain energy. How is this energy "divided" between the qubits?

5.10.4 Bell States

Among the bases for a qubit pair some stand out due to their special properties. One such basis is named after John Steward Bell. The basis vectors can be expressed in terms of the computational basis as follows:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\left(|\Upsilon_1\rangle + |\Upsilon_4\rangle\right)\,, \tag{5.2}$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}\left(|\Upsilon_1\rangle - |\Upsilon_4\rangle\right)\,, \tag{5.3}$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}\left(|\Upsilon_2\rangle + |\Upsilon_3\rangle\right) , \tag{5.4}$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}\left(|\Upsilon_2\rangle - |\Upsilon_3\rangle\right) , \tag{5.5}$$

Each state vector of the Bell states basis corresponds to an entangled state. In fact, such states are *maximally entangled*, in contrast with the computational basis.

Exercise 5.23
Express computational basis in terms of the Bell states.

Exercise 5.24
Express qubit pair Hamiltonian in terms of the operators built from Bell states.

5.10.5 GHZ State

The idea of entangled quantum systems can be naturally extended beyond a simple pair. If we are working with three qubits, their joint states can be written using a computational basis

$$|000\rangle , |001\rangle , |010\rangle , |011\rangle , |100\rangle , |101\rangle , |110\rangle , |111\rangle .$$

Each state from this basis is a product state and corresponds to a statistically independent (not entangled) states. Fully entangled states are possible, as one such state–GHZ state– is named after Daniel Greenberger, Michael Horne, and Anton Zeilinger. It has a simple form:

$$|GHZ\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}} .$$

5.10.6 Qat States

Macroscopical systems, such as crystals, rocks, tables, animals, planets, and so on, contain an enormous number of "parts": molecules, atoms, protons and electrons. Such systems are extremely complex and their quantum description is challenging. However, researchers are constantly improving the techniques to create *macroscopic quantum states* – quantum states of systems with macroscopically large number of parts. Such quantum systems are sometimes called *macroscopic Schrödinger's cat* *states*. We will call them *qat states*.

Although a single biological cell does not exhibit quantum properties, let's model it as a qubit, with the states $|0\rangle$ and $|1\rangle$ representing a "dead" and "alive" cell, respectively. A body of a cat will have about $M = 10^{10}$ cells, and the state of M qubits can be written as a superposition of 2^M computational basis vectors. Two extreme states have simple structure:

$$|A\rangle = |111\dots1\rangle = |1\rangle^{\otimes M} \quad \text{– all cells are alive (alive cat),}$$

$$|D\rangle = |000\dots0\rangle = |0\rangle^{\otimes M} \quad \text{– all cells are dead (dead cat).}$$

Exercise 5.25
What does the following state correspond to:

$$|B\rangle = |0111\dots1\rangle\,?$$

How many cells need to be "alive" for the cat to be considered alive?

The following qat state is very interesting:

$$|S\rangle = \frac{|A\rangle + |D\rangle}{\sqrt{2}}\,.$$

Dead or Alive?

Erwin Schrödinger first discussed such a state in his 1935 paper XXX, using a cat as an example. Today the term "Schrödinger's cat" is one of the most recognized expressions from quantum physics. What is not usually mentioned, is the purpose of Schrödinger's example.

5.11 Quantum Field

The state of electromagnetic field is the linear superposition of modes. Electric part of it can be written as

$$E = \iint E_{mp}\,,$$

where each mode E_{mp} has certain polarization $\vec{e}_{mp}$ and spatial distribution $u_{mp}(\vec{r})$:

$$E_{mp} = \vec{e}_{mp}\,\widehat{a}_{mp}\,u_{mp}(\vec{r})\,.$$

5.12 Quantum States of Light Chapter Highlights

- *Two vectors can be compared for similarity by calculating the "degree of overlap". The longer two vectors are and the closer their mutual direction – the greater the overlap is.*
- *Degree of overlap can be described by a binary linear operator $\widehat{\sigma}$. This operator is closely related to the concept of scalar product of two vectors.*
- *When scalar product (or, equivalently, degree of overlap) is defined for vectors, each vector receives a "special relative" – conjugate vector – that lives in different vector space, called conjugate or dual space.*
- *When the degree-of-overlap operator $\widehat{\sigma}$ is partially applied, the result is a unary linear operator that yields a number for every input vector. Importantly, such an operator is also a vector, albeit not an arrow-like vector.*

6. Applications

QUANTUM PHYSICS CAN BE SUCCESSFULLY APPLIED TO A WIDE VARIETY of problems in science and technology. In this chapter we will study several examples demonstrating how quantum effects can be explained using the concepts and tools discussed in the previous chapters. We will first start with principle of action quantization, applying to the spectra of atoms and atom-like structures. Then we will learn how atomic systems can interact with electromagnetic field, leading to such important effects as spontaneous and stimulated emissions which are at the core of the laser operation. At the end of the chapter we will examine the *entanglement* as a *resource* for quantum information processing tasks.

☑ ***Prerequisite Knowledge***

To fully understand the material of this chapter, readers should be comfortable with the following concepts:

- State
- Dynamical equations

6.1 Hydrogen-like Atoms

We will start with the simplest atomic system: Hydrogen atom.

6.1.1 Hydrogen Atom

The most abundant element in the observable universe is hydrogen. A hydrogen atom has two components: heavy nucleus (proton) and light "shell" (electron). The two components can be separated in the process of *ionization*. A significant energy is required to tear-off the shell:

$$E_{ion} = 2.18 \times 10^{-18}\,(J) = 13.6\,(eV)\,.$$

These numbers do not look big; however, they correspond to an extremely high temperature:

$$T_{ion} = \frac{E_{ion}}{k} = \frac{2.18 \times 10^{-18}}{1.38 \times 10^{-23}} = 157\,887\,(K)\,.$$

This is almost 30 times higher than the temperature near the "surface" of the sun; the estimated temperature in the core of the sun is about 100 times higher than the hydrogen ionization temperature.

The bottomline is that in normal conditions the electron is reliably *trapped/localized/confined* in a small region near the nucleus. The diameter of hydrogen atom provides a convenient "scale" for distances in atomic world:

$$d_H \approx 1\,\text{Å} \qquad \text{Ångstrom} \; = 10^{-10}\,(m)\,.$$

R If every person on earth had a hydrogen atom and placed it in a single row next to each other, then the resulting chain would be about a meter long.

Ångstrom

Angstrom is a convenient unit for measuring distances between atoms inside solid bodies. For example, copper atoms are about 2.8 Å in diameter and are arranged in a crystal with the distance between the nearest neighbors $d \approx 3.6$ Å.

To see how those values for the ionization energy and atomic radius appear, let us consider the following model of the atom, shown in the Figure 6.1.

The circular motion of the electron around the nucleus is mathematically similar to the problem of a planet circling a star. Using the second law of Newton and the expression for the Coulomb's force, we can write

$$\frac{m_e v^2}{r} = k\frac{q_e^2}{r^2}\,,$$

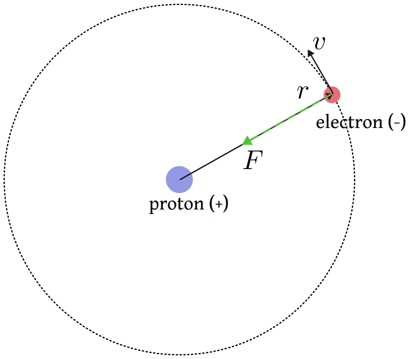


Fig. 6.1: Planetary model of hydrogen atom.

where

$$k = \frac{1}{4\pi\varepsilon_0} \,.$$

Newtonian momentum is given by $p = mv$; from the motion equation above, we find

$$p^2 = k\frac{m_e q_e^2}{r} \,.$$

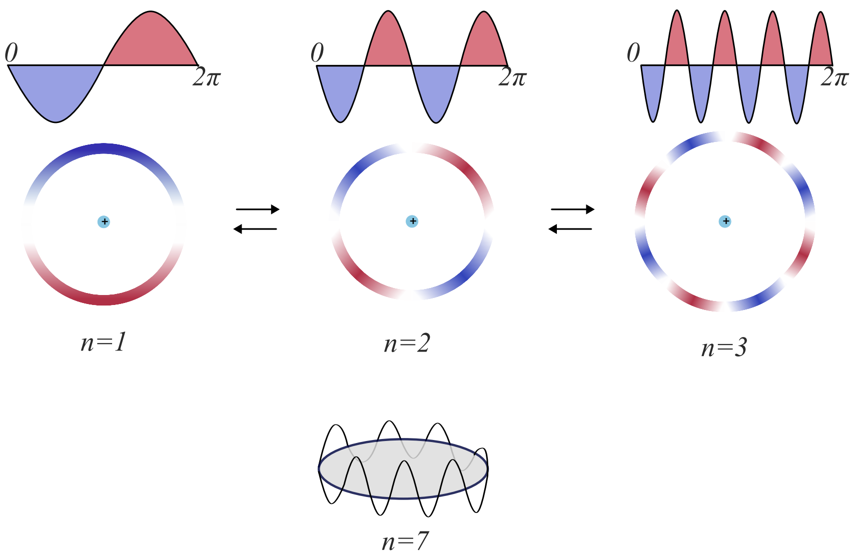


Fig. 6.2: Orbits allowed by the semi-quantum model.

Using de Broglie hypothesis about the relation between momentum and wavelength

$$p\lambda = h\,,$$

and combining it with *quantization hypothesis*:

$$n\lambda = 2\pi r$$

or, equivalently,

$$pr = n\frac{h}{2\pi} = n\hbar$$

we find possible solutions for the radius of the “orbit”:

$$r_n = \frac{\hbar^2 n^2}{k m_e q_e^2} \qquad n = 1, 2, 3, \dots$$

The smallest value of the radius is

$$a_0 = r_1 = \frac{4\pi\varepsilon_0 \hbar^2}{m_e q_e^2} = 0.523\,\text{Å}\,.$$

It is called *Bohr radius*. Thus, the smallest diameter of electron’s orbit is about 1 Ångstrom.

Exercise 6.1
Find the numbers for which the radius of electron “orbit” equals: 1) 1 mm; 2) 1 cm; 3) 1 m.

How fast does the electron move around the nucleus? The velocity of the electron is

$$v = \sqrt{\frac{k q_e^2}{m_e r}}\,.$$

Substituting $r_n = a_0 n^2$ we get

$$v_n = \frac{1}{n}\sqrt{\frac{k q_e^2}{m_e a_0}} = \frac{v_1}{n}\,.$$

The value of v_1 is

$$v_1 = 2.19 \times 10^6\ (m/s)$$

Although this is a big number, it is much smaller than the speed of light; we are therefore justified in using Newtonian mechanics and not taking relativity into account.

Quantization in Astronomy

The "quantization" of orbit radius can be found even for planets in solar system. If we denote the distance between the earth and the sun as A (called *astronomical unit*), then the distances to planets from the sun are given by the formula

$$r_n = \frac{A}{10}\left(4 + 3 \times 2^n\right).$$

This relation is known as *Titius-Bode* law. It works remarkably well for all planets from Venus ($n = 0$) to Uranus ($n = 6$).

How strong is the electron attracted to the nucleus? The Coulomb force between the charges is

$$F = \frac{kq_e^2}{r^2}.$$

Substituting $r_n = a_0 n^2$ we get

$$F_n = \frac{kq_e^2}{a_0^2 n^4} = \frac{F_1}{n^4}.$$

The value of F_1 is

$$F_1 = 8.24 \times 10^{-8}\,(N)$$

This is a tiny force on the human scale of forces. Even a baby ant has enough force to tear a hydrogen atom apart with its "bare hands"!

Although the force acting on the electron is small on a human scale, the acceleration ($a = v^2/r$) it produces is very large (on the human scale, compare to g). This is due to extreme lightness of the electron:

$$m_e = 9.1 \times 10^{-31}\,(kg).$$

What is the total energy of the hydrogen atom? It can be found as follows:

$$E_n = \frac{m_e v_n^2}{2} - k\frac{q_e^2}{r_n} = -\frac{E_1}{n^2},$$

where

$$E_1 = k\frac{q_e^2}{2m_e a_0}.$$

The value of E_1 is

$$E_1 = 2.18 \times 10^{-18}\,(J)\,.$$

In atomic world a special unit of energy is used. It is the energy of electon accelerated by a voltage drop of 1 Volt:

$$E_{ev} = 1\,(V) \times q_e = 1.6 \times 10^{-19}\,(J)\,.$$

Using this atomic unit, the hydrogen atom has energy

$$E_1 = 13.6\,(eV)\,.$$

Exercise 6.2
What is the frequency of revolution of electron around the nucleus for an orbit number n?

Atoms Can't Exist?

The model described above leads to the conclusion that atoms must not exist for long. According to the theory of classical electrodynamics, a charge moving in a circle will emit electromagnetic waves with the frequency of revolution. As the atom loses its energy, the electron spirals ever closer to the nucleus. Taking classical electrodynamics into account, the life-time of a hydrogen atoms should be about nanosecond. This contradicts the observation of stability of atoms.

Niels Bohr suggested that the "orbits" represent so called *stationary states*, where electrons can "move" without radiating away electromagnetic waves. Radiation only happens when electron *transitions* between stationary levels, for example between 2 and 4. The energy carried away by the light is related to the frequency of the electromagnetic wave ν as follows:

$$\Delta E = E_m - E_n = h\nu_{nm} \qquad m > n\,.$$

Exercise 6.3
How much energy is required to "move" electron from the"orbit" of 1 mm to 1 cm? From 1 cm to 1m?

Line Spectra Explained

Using the results obtained above, we can now describe the spectra of light emitted or absorbed by hydrogen.

The stationary states are discrete, therefore there are discrete energies of transition between a pair of levels. When atom absorbs electromagnetic radiation, electron "jumps up" to higher energy level and farther distance. In the reverse process, electron "jumps down" from higher level to lower one, resulting in emission.

The wavelength of the radiation is

$$\lambda = cT = \frac{c}{\nu} = \frac{ch}{E_m - E_n} .$$

Plugging in the expression for the energy, we get

$$\lambda = \frac{ch}{E_1} \frac{n^2 m^2}{m^2 - n^2} = 91.127(nm) \times \frac{n^2 m^2}{m^2 - n^2} . \tag{6.1}$$

A special case of this formula was discovered in 1885 by a Swiss mathematician Johan Balmer. Analyzing the visible lines in the spectra of hydrogen, he found some regularity in the wavelengths. Balmer expressed it as follows:

$$\lambda = B \frac{m^2}{m^2 - 2^2} ,$$

where $m > 2$ and $B = 364.51$ nm. Looking at (6.1), we can see that it can be written for $n = 2$ as

$$\lambda = 91.127(nm) \times \frac{4m^2}{m^2 - 2^2} = 364.51(nm) \times \frac{m^2}{m^2 - 2^2} .$$

Exercise 6.4
According to the the formula (6.1), how many hydrogen lines will be in the visible part of the spectrum (from 400nm to 700nm)?

Three Body Problem
The problem of hydrogen atom involves *three* physical entities interacting with each other: Proton, electron, and electromagnetic field. Electron does not interact directly with the proton, it does so via the electric field of the nucleus. This is important to keep in mind, especially if we want to understand the phenomenon of *spontaneous*

emission.

6.1.2 Landau Levels and Magnetic Flux Quantization*

The simple semi-classical approach to quantization used for hydrogen atom can be applied to another interesting quantum effect appearing when electrons are moving in a strong magnetic field.

$$qvB = \frac{mv^2}{r} \rightarrow qBr = mv$$

$$qBr^2 = mvr = n\hbar\,.$$

Thus, the radius can only be $r_n = r_1\sqrt{n}$ where $r_1 = \sqrt{\hbar/qB}$. Kinetic energy is $E_n = E_1 n$ where $E_1 = q\hbar B/2m$.

The quantity $\Phi = \pi r^2 B$ is called *magnetic flux* – a measure of the "flow" of magnetic field B throught the area πr^2. In terms of magnetic flux, the quantization condition becomes

$$\Phi = n\Phi_0\,,$$

where $\Phi_0 = h/2e$ is known as *elementary magnetic flux quantum.* It is the smallest step magnetic flux can change by.

Spontaneous Emission and Cavity QED

According to Schrodinger equation, an hydrogen atom with electron in any statioary state $|\Psi_n\rangle$ with energy E_n will remain in this state *forever.* In reality, every atom will randomly transition into a state with lower energy, all the way to the lowest energy state, emitting radiation as the result. This is called *spontaneous emission.*

To describe spontaneous emission, one must account for the fact that atom is not truly isolated and electron and proton are not the only quantum systems in picture. There is an electromagnetic field with its many modes-oscillators.

Interaction of atom-like quantum systems (qubits, quantum dots, atoms) with quantum states of electromagnetic field is the focus of an exctiting area of research called *Cavity Quantum Electrodynamics* or *cavity QED* (CQED).

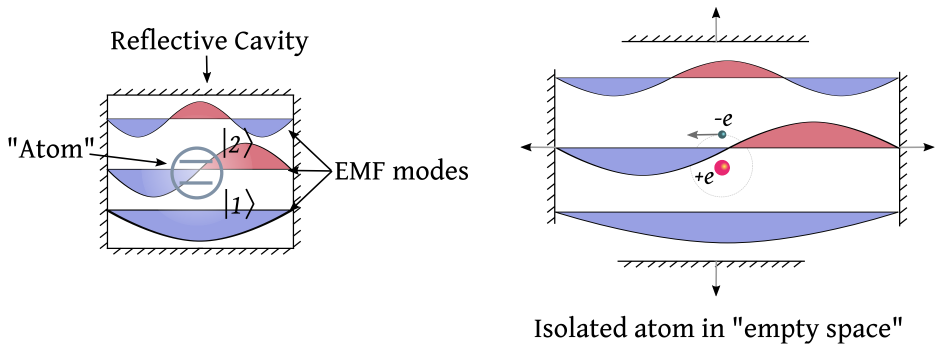


Fig. 6.3: Atom in "empty space" is a quantum system in a very large cavity, coupled to the modes-oscillators of electromagnetic field.

6.1.3 Franck-Hertz Experiment

Atoms can transition between different states by absorbing energy from electromagnetic radiation ("photons"). But this is not the only mechanism. Other particles, like electrons, can transfer their kinetic energy to atoms by bumping into the latter. Such a phenomenon was studied by James Franck and Gustav Hertz in 1914 using collisions of electrons with mercury atoms enclosed in a special vacuum tube. They showed that whenever electrons had a certain kinetic energy – specific to mercury atoms – they transfered this energy very efficiently (*resonantly*) to mercury atoms. When the energies of electrons and mercury atoms were not matched, electrons scattered from the atoms *elastically* – without the loss of their kinetic energy. The work of Franck and Hertz was recognized with 1925 Nobel Prize in Physics. Interestingly, according to James Franck, neither he nor Gustav Hertz had any knowledge of Bohr's theory of atomic states when they performed their experiment.

Idea

The idea of Franck-Hertz experiment is illustrated in figure 6.4. Its central part consists of a closed vacuum tube with mercury gas and several electric elements used as the source and sink of electrons.

Numbers

At room temperature mercury is a liquid metal which can be easily turned into a gas by heating it up to 200°C. Unlike molecular gases such as hydrogen H_2, oxygen O_2, or nitrogen N_2, *mercury gas is monoatomic* – consisting of single Hg atoms.

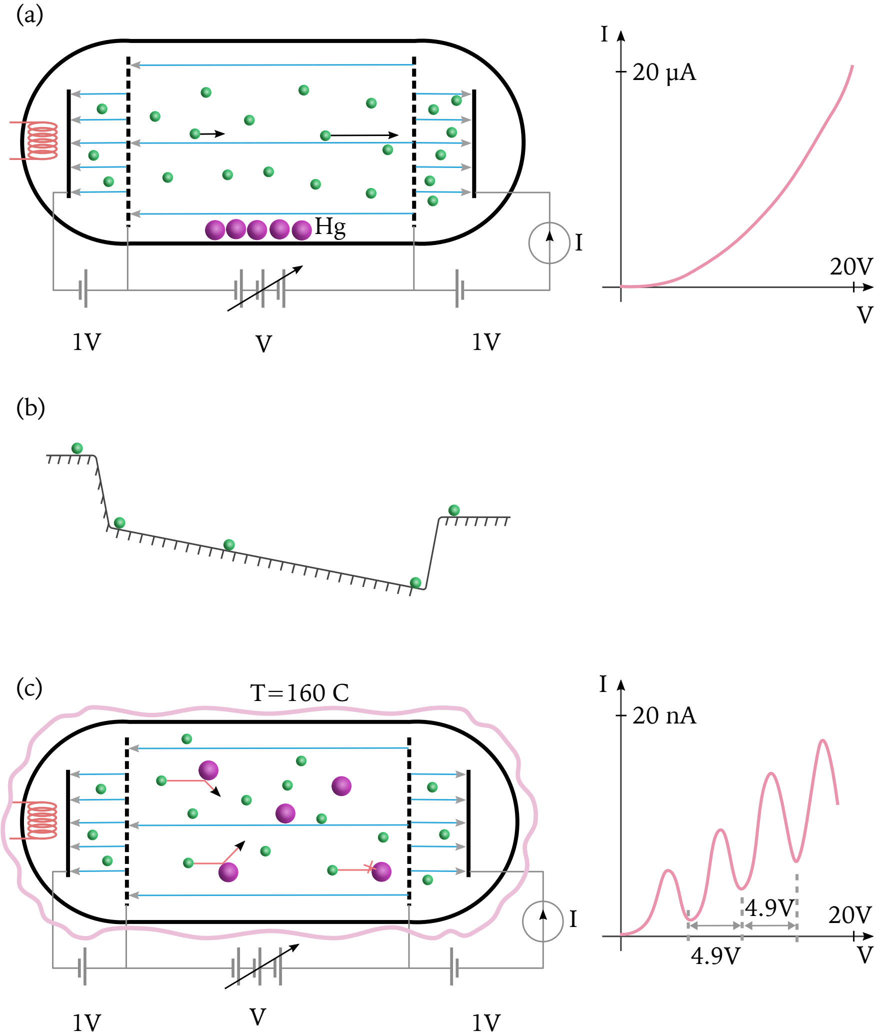


Fig. 6.4: (a) A vacuum tube is used to study the flow of electrons between two metal electrods under various conditions.; (b); (c).

Each mercury atom contains 80 protons and 120 neutrons and is significantly heavier than an electron: $M_{Hg} \approx 4 \times 10^5\, m_e$. Consequently, an electron colliding with a mercury atom won't be able to impart any appreciable recoil speed. In other words, mercury atoms are like immovable objects to electrons.

Exercise 6.5
Show that for the head-on collision the speed of an initially stationary mercury atom after the collision becomes $v = 2v_0 \frac{m_e}{M_{Hg}}$.

6.1.4 Stoke's Rule

Stoke's rule.

6.2 Rydberg Atoms

When an atom has outer electron(s) excited into states with high excitation number n, it is called a *Rydberg atom*(CHECK!). Such atoms are of great importance for experiments of quantum physics.

6.3 Quantum Dots

Quantum dots are physical structures designed to confine electrons in tiny volumes. They are human-made and can be viewed as *artificial atoms*. As the result of confinement, quantum states of the trapped electrons form a discrete set not unlike the states of a hydrogen atom. *Examples:???*

To illustrate the idea, consider an idealized case of an electron confined in one dimension in a region between $x = 0$ and $x = L$.

$$|\alpha\rangle\langle\beta|$$

$$E_n = -\frac{E_i}{n^2}\,.$$

6.4 Spontaneous Emission

When an excited atom is left alone, it must remain in the excited state *forever*. This follows from the Schrodinger's equation:

$$i\hbar\partial_t\,|\psi\rangle = \widehat{H}\,|\psi\rangle$$

If $\widehat{H}\,|\psi\rangle = E_n|\psi\rangle$ we get

$$\partial_t\,|\psi\rangle = \beta|\psi\rangle\,, \text{ where } \beta = \frac{-iE_n}{\hbar}\,.$$

The solution to this equation is

$$|\psi_t\rangle = e^{\beta t}|\psi_0\rangle = e^{-iE_n/\hbar}|n\rangle\,.$$

However, atoms placed in the best isolation allowed by an experiment, inevitably transition to the lowest (ground) energy state, emitting energy in the form of EMF excitation. The process is random and resembles a radioactive decay.

6.5 Stimulated Emission

$$|\alpha\rangle\langle\beta|$$

$$E_n = -\frac{E_i}{n^2}\,.$$

6.6 Lasers

Laser is a device that produces a special kind of light −*monochromatic* and *coherent* electromagnetic radiation. Monochromatic means "single color" or single wavelength. Coherent means capable of interference, which implies stable *phase*.

Laser can be described on several levels: classical, semiclassical, and quantum.

Any laser has three main components:

(a) *Cavity* to create mode.
(b) *Gain medium* to amplify signal.
(c) *Pump* to keep energy flowing into the gain medium.

From the standpoint of quantum physics, lasers are special *sources of excitation of electromagnetic field.* Laser's operation is based on the effect of **L**ight **A**mplification by the **S**timulated **E**mission of **R**adiation (LASER).

$$|\alpha\rangle\langle\beta|$$

$$E_n = -\frac{E_i}{n^2}\,.$$

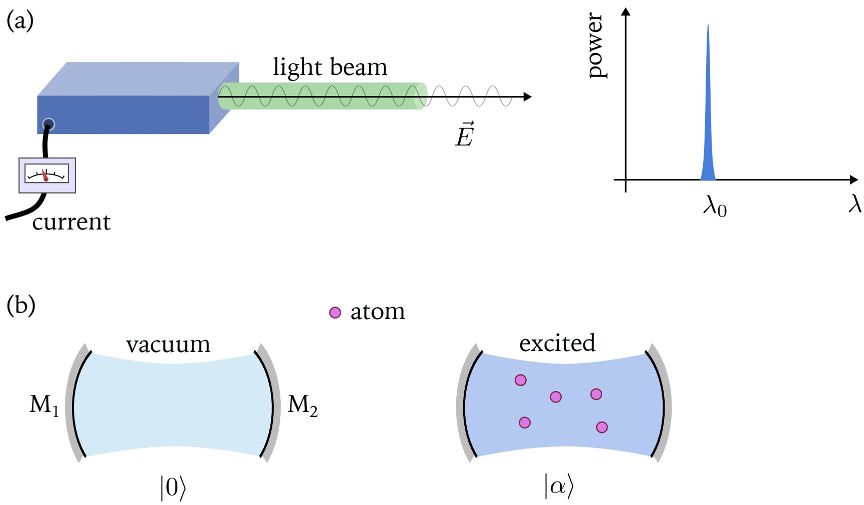


Fig. 6.5: (a); (b)

6.7 Photoeffect

The phenomena related to photoelectric effect, or photoeffect, are so important that three separate Nobel prizes have been awarded to people who contributed to their discovery, explanation, and confirmation.

> ***Noberl Prizes for Photoeffect***
> The first Nobel prize was given to Philipp Eduard Anton von Lenard in 1905 for *"for his work on cathode rays"* (CHECK DETAILS!). The second Nobel prize was awarded to Albert Einstein in 1921 *"for his services to Theoretical Physics, and especially for his discovery of the law of the photoelectric effect"*. The third Nobel prize was awarded to Robert Millikan in 1923 for *"for his work on the elementary charge of electricity and on the photoelectric effect"*.

An EMF excitation – a photon – with frequency ν and energy $E = h\nu$ can impart its energy to an electron bound inside a material (e.g. metal). If the energy is sufficient to free the electron, it can escape with the kinetic energy

$$E = h\nu - W\,,$$

where W is the energy which a *free electron* needs to move outside of a material. W is called a *work function* and it is a number specific to a material.

$$|\alpha\rangle\langle\beta|$$

$$E_n = -\frac{E_i}{n^2}\,.$$

6.8 Black Body Radiation

$$\rho_\nu = \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/kT}-1}\,.$$

$$E_n = -\frac{E_i}{n^2}\,.$$

6.9 Conductors

$$|\alpha\rangle\langle\beta|$$

$$E_n = -\frac{E_i}{n^2}\,.$$

6.9.1 Heat Capacity

Einstein's model.

6.10 Entanglement

Entanglement can be viewed as a *physical resource* of purely quantum nature. As a resource, entanglement facilitates several important processes related to processing of information.

$$|\alpha\rangle\langle\beta|$$

$$E_n = -\frac{E_i}{n^2}\,.$$

6.10.1 Teleportation

In quantum physics *teleportation* means a "transfer" of a quantum state to a distant location. In the process of teleportation the original quantum state is "erased" in one place and "reconstructed" at a target location.

It important to understand that quantum teleportation does not imply sending *objects*, *energy*, or a *signal* instanteneously from one place to another. Quantum teleportation has nothing to do with the popular sci-fi notion of object teleportation.

6.10.2 Entanglement And Measurement

Entanglement – a purely quantum effect – plays the central role in the measurement process.

Chapter Highlights

- *Tensors find application in various areas of science and math.*
- *Geometrical properties of surfaces and spaces can be described using metric tensor.*
- *Physical properties of solids are often anisotropic – depend on the direction of applied "force". Such properties are best described by various tensors: stress tensor, mobility tensor, piezoelectric tensor, and others.*
- *At the fundamental level electric and magnetic fields are united in a single physical object – electromagnetic field. Electromagnetic field is described by an antisymmetric tensor of the second rank.*

7. Implications

IN THIS CHAPTER WE'LL EXAMINE HOW QUANTUM PHYSICS AFFECTS OUR understanding of the world at the fundamental level. We will analyze two big questions: What are the ultimate "building blocks" of reality, and what we can know about them.

> ***Prerequisite Knowledge***
> To fully understand the material of this chapter, readers should be comfortable with the following concepts:
> - State
> - Dynamical equations

7.1 Two Ologies

Ontology and epistemology.

7.2 Apriori Values

Elements of reality.

7.3 Complimentarity

Bohr's ideas.

7.4 Phenomena

Bohr's ideas.

7.5 Wholeness

David Bohm's ideas.

7.6 It From Bit

Wheeler's ideas.

7.7 Non-Kolmogorov Probability

Chapter Highlights

- *Tensors find application in various areas of science and math.*
- *Geometrical properties of surfaces and spaces can be described using metric tensor.*
- *Physical properties of solids are often anisotropic – depend on the direction of applied "force". Such properties are best described by various tensors: stress tensor, mobility tensor, piezoelectric tensor, and others.*
- *At the fundamental level electric and magnetic fields are united in a single physical object – electromagnetic field. Electromagnetic field is described by an antisymmetric tensor of the second rank.*

8. Appendix

We are now ready to appreciate the implications of quantum physics.

8.1 Physics

When a quantity x changes by a tiny amount, we will denote the change using small Greek letter δ (delta) as follows:

$$\boxed{\delta x \text{ - tiny change of } x.}$$

A convenient way to write all components of a second rank tensor is to use table-like structure called *matrix*.

8.1.1 Black Body Radiation

8.1.2 Notation

K and E_k – Kinetic energy of a system.
Π and E_p – Potential energy of a system.
E – Total mechanical energy ($E = E_K + E_P$) written in terms of velocity v and position x.
H – Hamiltonian of a system: $H = K + \Pi$. Differs from E because kinetic energy written in terms of *momentum* p instead of velocity.
L – Lagrangian (Lagrange function) of a system: $L = E_K - E_p$. It is the "imbalance" of energies.
Δx – Change of a value of a variable x.
δx – "Tiny" change of a value of a variable x.
∂ – Rate of change.
∂_t – Rate of change with respect to time.
∂_x – Rate of change with respect to variable x (e.g. position).
$\partial_t f$ – Rate of change of f with respect to t.

It means exactly the following

$$\partial_t f = \frac{\delta f}{\delta t} = \frac{f(t+\delta t) - f(t)}{\delta t} \ .$$

$\vec{\xi}$ – State of a system in Hamiltonian dynamics. It is a vector with components $\vec{\xi} = (x, p)$.
$\hat{J}$– Operation (operator) of rotation by 90 degrees.
$\hat{R}(\theta)$ – Operation (operator) of rotation by θ.
h – Quantum of action (Planck's constant). In SI units its numerical value is $h = 6.626 \times 10^{-34} (J \cdot s)$.
$\hbar$ – "Reduced Planck's constant". A convenience notation for often used combination $\hbar = h/(2\pi)$.
A – Action.
Ψ – Quantum state.
$|\Psi\rangle$ – Quantum state vector.
ϕ, θ – Angle variables.
ω – Angular speed (also angular velocity). Often it has the following meaning: $\omega = \partial_t \theta$.
$\vec{e}_1, \vec{e}_2$ – Basis vectors. Usually they have unit length and point in mutually perpendicular directions.
z – Arbitrary *numeric* variable, $\vec{z}$ – arbitrary *vector* variable, $\hat{z}$ – arbitrary *operator*.
$\overset{\circ}{A}$ – Angstrom, a unit of length in the world of atoms. $1\overset{\circ}{A} = 10^{-9}(m)$. Hydrogen atom is about $1\overset{\circ}{A}$ in diameter.
c – Speed of light in vacuum.
ν – Frequency of oscillations measured as the number of oscillations per second, in Hz.

8.1.3 Physical Constants

Below is the list of various physical constants used in these notes.
$q_e = 1.6 \times 10^{-19}\,(C)$ – Charge quantum (charge of an electron).
$m_e = 9.1 \times 10^{-31}\,(kg)$ – rest-energy (aka mass) of an electron.
$k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9\,(N \cdot m^2/C^2)$ – Coulomb constant – force between two unit charges 1 meter apart.
10^{-9} s = 1 nanosecond – the unit of time in atomic world. It is a "heartbeat of atoms".
$1\,(eV) = q_e\,(J)$ – 1 electron-volt. It is the kinetic energy an electron

would acquire when accelerated by a 1V battery. A tiny value.
$m_e c^2/q_e = 0.5\,MeV$ – rest-energy of an electron measured in electron-volts. Roughly speaking, we will need half a million 1-volt batteries to accelerate an electron to make its kinetic energy comparable to its rest-energy.
$k = 100\,(N/m)$ is a spring constant of a spring that stretches by 0.1 of a meter when 1 kilogram mass is attached to it.

8.2 Mathematics

When a quantity x changes by a tiny amount, we will denote the change using small Greek letter δ (delta) as follows:

$$\boxed{\delta x \text{ - tiny change of } x.}$$

A convenient way to write all components of a second rank tensor is to use table-like structure called *matrix*.

8.2.1 Poisson Series

A British mathematician, statistician, biologist, and geneticist Sir Ronald Aylmer Fisher, once said: *"Among discontinuous distributions, the Poisson series is of first importance."* Indeed, it is a simple and ubiquitous random series. It is used to describe occurrences of failures and defects in industrial processes, distributions of plants and animals in different areas and periods of time, the number of photons received by the retina, number of birth in a hospital per day, a number of clicks in a radioactivity detectors, and many more phenomena.

We will examine this process in the case of Rutherford-Geiger experiment with radioactive decay of radium, as discussed in subsection 2.10.1.

8.2.2 Greek Alphabet

In mathematics most often we use θ and ϕ for angles. Sometimes α and β are also used. Occasionally ψ is used to denote angle.

In physics λ is used to denote the wavelength of light, ν – frequency in Hertz (periods of oscillations per second), ω – angular speed (number of radians of rotation per second).

$\mathrm{A}\,\alpha$	alpha	$\mathrm{B}\,\beta$	beta
$\Gamma\,\gamma$	gamma	$\Delta\,\delta$	delta
$\mathrm{E}\,\epsilon$	epsilon	$\mathrm{Z}\,\zeta$	zeta
$\mathrm{H}\,\eta$	eta	$\Theta\,\theta$	theta
$\mathrm{I}\,\iota$	iota	$\mathrm{K}\,\kappa$	kappa
$\Lambda\,\lambda$	lambda	$\mathrm{M}\,\mu$	mu
$\mathrm{N}\,\nu$	nu	$\Xi\,\xi$	xi
$\mathrm{O}\,o$	omicron	$\Pi\,\pi$	pi
$\mathrm{P}\,\rho$	rho	$\Sigma\,\sigma$	sigma
$\mathrm{T}\,\tau$	tau	$\Upsilon\,\upsilon$	upsilon
$\Phi\,\phi$	phi	$\mathrm{X}\,\chi$	chi
$\Psi\,\psi$	psi	$\Omega\,\omega$	omega

Table 8.1: Greek Alphabet

The symbols Ψ and Φ are usually used to denote quantum state vectors.

8.2.3 Available Environments

Coloredboxed environment, with \coloredboxed{color}{ text }:

$$\boxed{E = mc^2}.$$

Bold text command inside math mode is \btc{ txt}:

$$\textbf{max}\, x\, y$$

Bold text with emphasis (italic) is done with \bem{text}: ***example of a very important piece of text***.

Grey bullet: \tus: • • • •

Quantum state is \qs: $|\psi\rangle$ or better use ket and bra commands: $|\phi\rangle$ and $\langle\psi|$.

Bracket and ketbra combinations using a single command: $\langle\psi|\phi\rangle$ and $|\psi\rangle\langle\phi|$.

Operators are set with \op{name}: $\widehat{\rho}= |\psi\rangle\langle\psi|$.

$$\hat{\rho} \neq \widehat{\rho}$$

Environments

Prerequisite environment \myprereq{text}:

Prerequisites
One, two, and three.

Example environment \myExample{text}:

Example
This is a perfect example.

Analogy environment \analogy{text}:

Analogy
Consider the following analogy: A and B.

Definition environment \mydef{text}:

Definition 8.1 ***Kinetic Energy***
Kinetic energy is the energy due to motion.

Reminder environment \myrem{text}:

Kinetic Energy
Recall that $E_k = mv^2/2$.

Bio environment \mybio{text}:

Max Planck
The full name of Max Planck is Max Karl Ernst Ludwig Planck.

8.3 Temporary Stuff

Work in progress sections will be kep here.

8.3.1 Facts About Light

Our understanding of the physical nature of light is the result of several centuries of studies. Visible, infra-red or ultra-violet light, and microwave or gamma radiation are all various manifestations of *electro-magnetic* *field* (EMF). It is one of the most technologically important and well-understood fields.

Omnipresence

The first important fact about EMF is that it is truly ubiqutous: There is no place in the universe free from EMF. Yes, there are regions of space (and time) where EMF is not excited, but EMF is still present there in its "unexcited" form, called *EMF-vacuum*. This vacuum state is an important quantum state of EMF. It is discussed in more details in Section XXX.

Localizability

The second important fact about EMF is its ability to be localized (concentrated) in finite regions of space, simetimes in small volumes. Put differently, EMF can be excited ("emitted") by relatively small "objects" (sources of EMF excitations) or absorbed by similar objects. Emission and absorption of light by atoms is a perfect example of this principle.

> ❗ ***Localized Excitations***
> We must clarify: It is the *excitation of EMF* which is localizable, *not* EMF. The latter is everywhere, always.

> ❗ ***Localizable ≠ Localized***
> A person can be excitable but not excited, irritable but not irritated. Analogously, a field excitation is localizable: It *can be localized* but it does not mean that it is necesserily localized at all moments of time.

Energy-Momentum

Excitations of EMF propagate through space and possess the basic mechanical aspect – *energy-momentum*. When an atom emits a pulse of light it loses certain energy when and it also experiences a recoil, like a gun that fires a bullet. An EMF excitation carrying energy E also carries momentum p. The two are related simply:

$$E = pc\,,$$

where c is the speed of light – the speed of propagation of EMF through "empy" space. This formula is a special case of a more general relation between energy and momentum, derived in the special theory of relativity:

$$E^2 = p^2c^2 + m^2c^4\,.$$

This is the relation between *total* energy (kinetic and non-kinetic) of an object with mass m and momentum p. It shows that even for an object at rest, when $p = 0$, it still possesses non-kinetic energy – called rest-energy – equal to $E_0 = mc^2$. For EMF excitations the total energy E is *purely due to motion* or, equivalently, EMF excitations are massless ($m = 0$).

> **❗ *Photons***
> In common language, propagating EMF excitations are called *photons*. It is sometimes said that photons with energy E carry momentum $p = E/c$ and that photons are massless particles. It must be emphasized again that such language is not always helpful.

Colors and Periodicity

Q?: Maybe switch colors and energy-momentum?

Besides energy-momentum, EMF excitations exhibit various "colors", with some "colors" visible and and some invisible to the human eye. The "colors" vary smoothly and continuously, forming a so called *spectrum* of electromagnetic radiation.

$$E = h\nu = \frac{hc}{\lambda}\,.$$

It is not yet known whether there are upper or lower *physical limits* to the wavelength. There are mathematical limits: $0 < \lambda < \infty$.

Timeless Law

The laws that describe the behavior of electromagnetic radition in space and time are expressed by the equations which show no preference to the direction of time flow.

$$\partial_t F = G\,.$$

Lorentz force ($q \to -q$?)

$$\partial_t v = qE + qvB\,.$$

Modes

When discussing the excitations of various fields, including EMF, the concept of *field modes* plays an extremely important role. There are modes related to orthogonal polarizations, non-overlapping propagation paths (spatial modes), frequency modes, temporal bin, temporal modes.

Familar picture of *light beams* is actually based on modes.

9. Solutions

Exercise ??

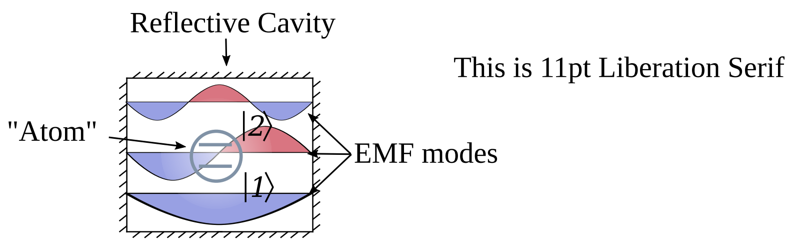


Fig. 9.1: The set M contains all possible makes of cars: Ford, Toyota, etc.

The diagram in the Figure 9.1 shows the set M – the set of all possible makes of cars. A mapping **trk** returns $true$ if a given car maker produces trucks.

Exercise ??

Any binary function can be viewed as a unary function if two inputs are replaced by a single input of a *pair of numbers*. Similarly for a function with two outputs. This idea is illustrated in the Figure 9.2(a): The function **swp** is viewed as a unary function which swaps the numbers in an *ordered pair*:

$$\textbf{swp}\ \ (n, m) = (m, n)\,.$$

Given the set $\mathbb{Z}$ of whole numbers, we can create the set of all possible *ordered pairs* (n, m). This set can be denoted as follows:

$$(\mathbb{Z}, \mathbb{Z}) \ \text{ or } \ \mathbb{Z} \times \mathbb{Z}\,.$$

The latter notation is standard in mathematics, but the former way of writing is

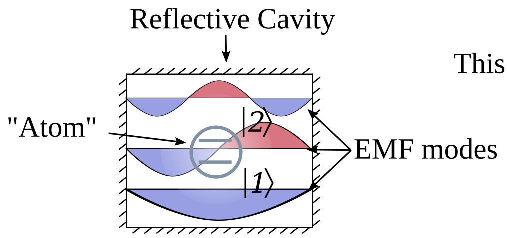


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Fig. 9.2: (a) Two inputs (outputs) of a function can be replaced with a single input of a ***pair*** of numbers, turning a binary function into a unary one. (b) That.

also acceptable. We can similarly denote the set of all *ordered triples*:

$$(\mathbb{Z}, \mathbb{Z}, \mathbb{Z}) \;\text{ or }\; \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}\,.$$

With the notation introduced above, the action of functions with multiple inputs or outputs can be depicted on the level of sets. The Figure 9.2(b) shows how this works for the functions **swp** and **max** .

Index